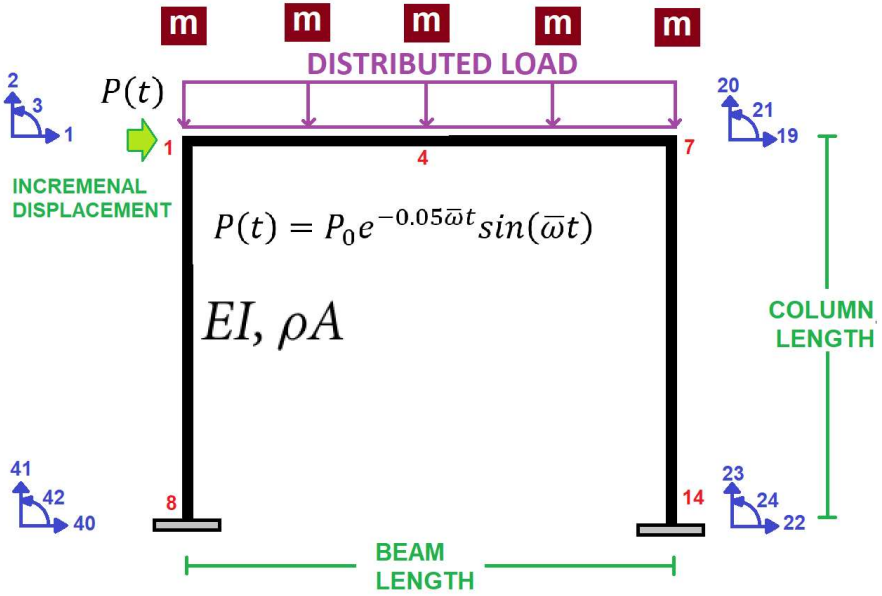
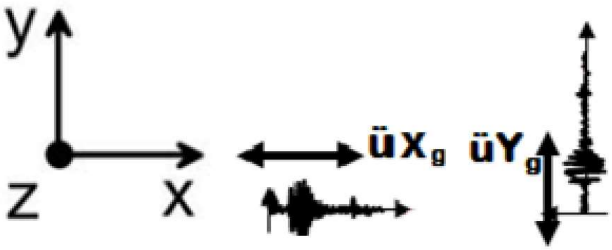
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

# COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A STEEL FRAME: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

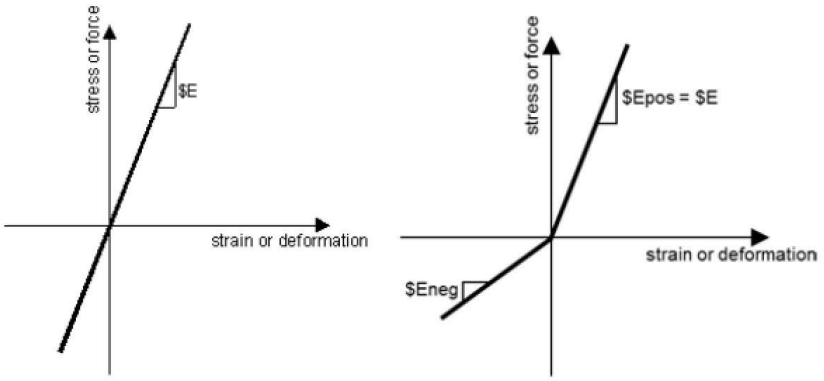


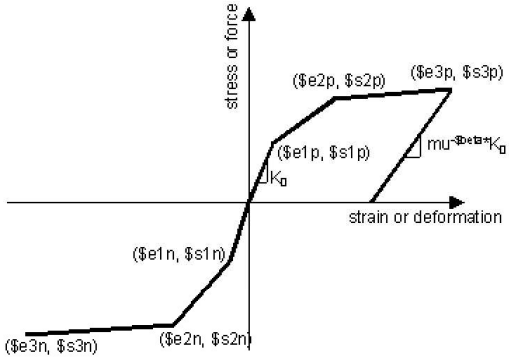

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

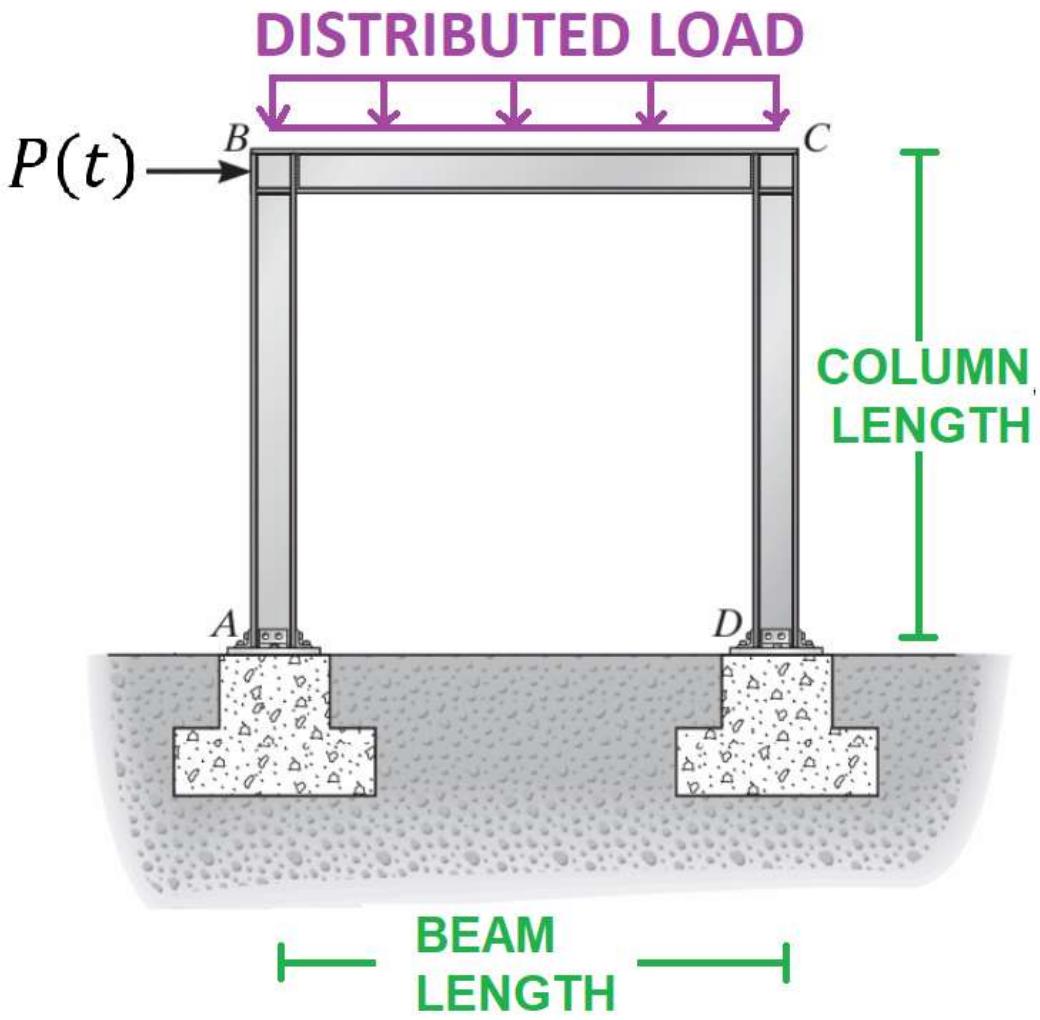



## ELEMENT ELASTIC MATERIAL

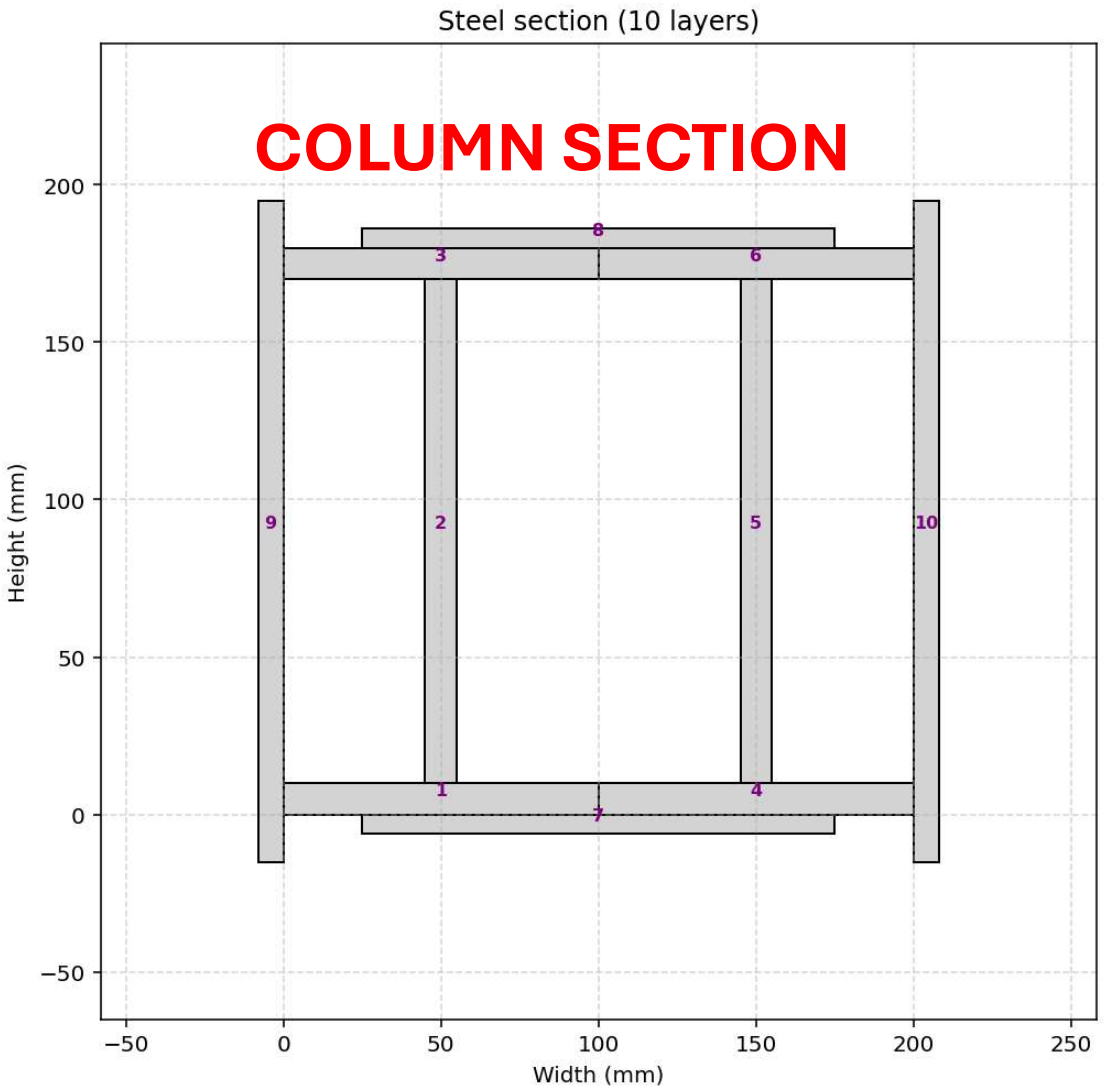


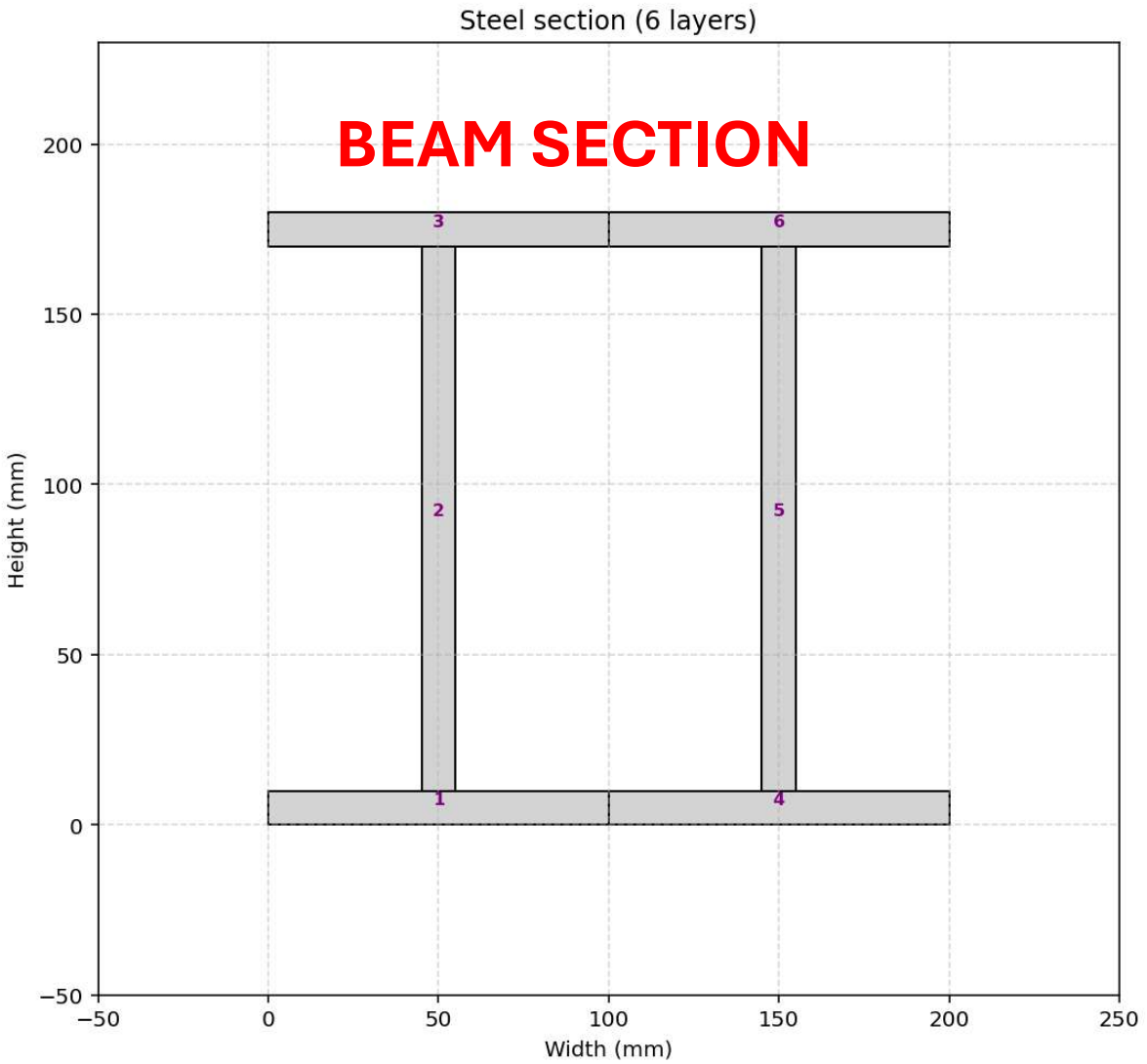

## ELEMENT INELASTIC MATERIAL

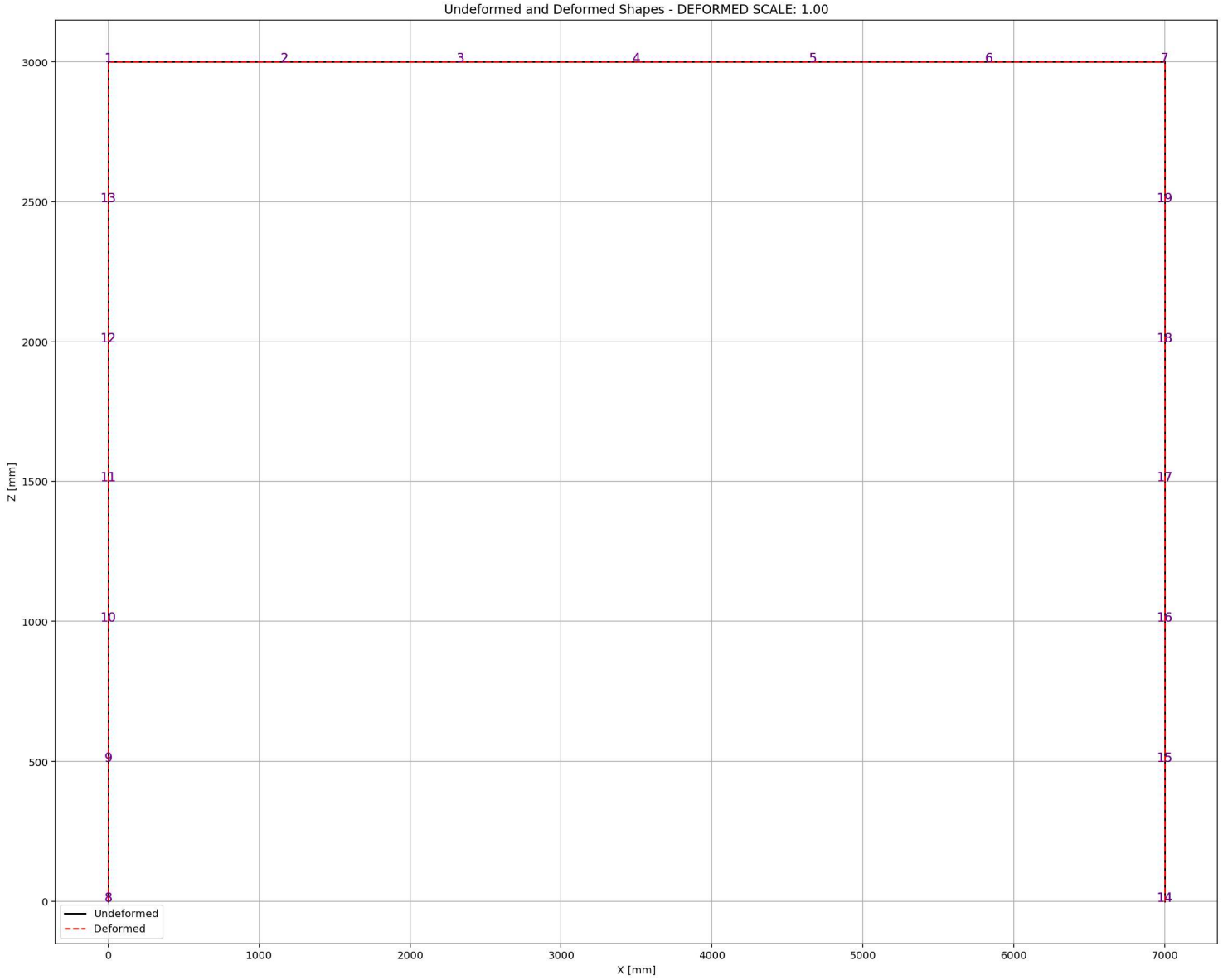
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$



$$\rho A \frac{\partial^2 v(x,t)}{\partial t^2} + EI \frac{\partial^4 v(x,t)}{\partial x^4} = P(t)$$


Steel section (10 layers)
COLUMN SECTION
Height (mm)
Width (mm)
200
150
100
50
0
−50
−50
0
50
100
150
200
250
1
2
3
4
5
6
7
8
9
10


Steel section (6 layers)
BEAM SECTION
Height (mm)
Width (mm)
200
150
100
50
0
−50
−50
0
50
100
150
200
250
1
2
3
4
5
6


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

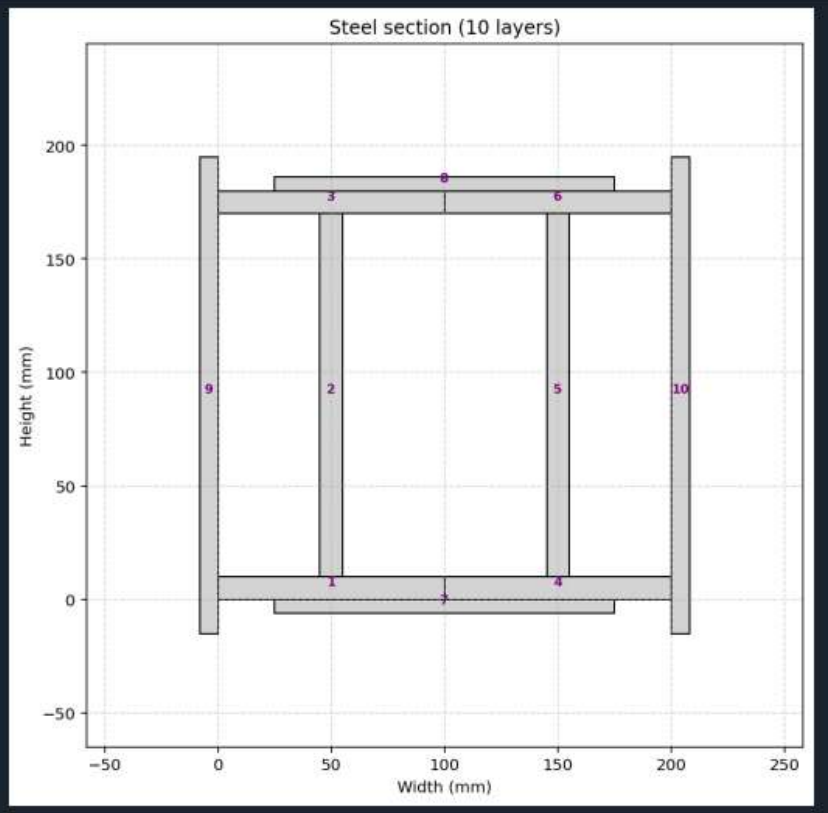
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

P(t)_STEEL_FRAME_8_ANA.py

```
##########################################################################
#                  >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
#          COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A STEEL FRAME : A
#FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND
#--------------------------------------------------------------------------
#                                              P(t) = P0 sin(wt)
#                                       P(t) = P0 exp(-0.05wt) sin(wt)
#--------------------------------------------------------------------------
#        ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
#                                      OF ENERGY DISSIPATION CAPACITY IND
#--------------------------------------------------------------------------
#                    THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAE
#                                  EMAIL: salar.d.ghashghaei@gmail.com
##########################################################################
"""
Nonlinear Seismic Performance Assessment of a steel frame:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati

This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
 for performance-based earthquake engineering. The 2D model incorporates bo
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity to capture P-delta effects
 and large displacements—essential for collapse assessment.

Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
 and plastic mechanism identification
(4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturin
 pinching behavior, and energy dissipation degradation
(5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External
(6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of Externa
(7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting da
```




IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Save file | Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 12, Col 105 | UTF-8 | CRLF | RW | Mem 45%

# STATIC ANALYSIS

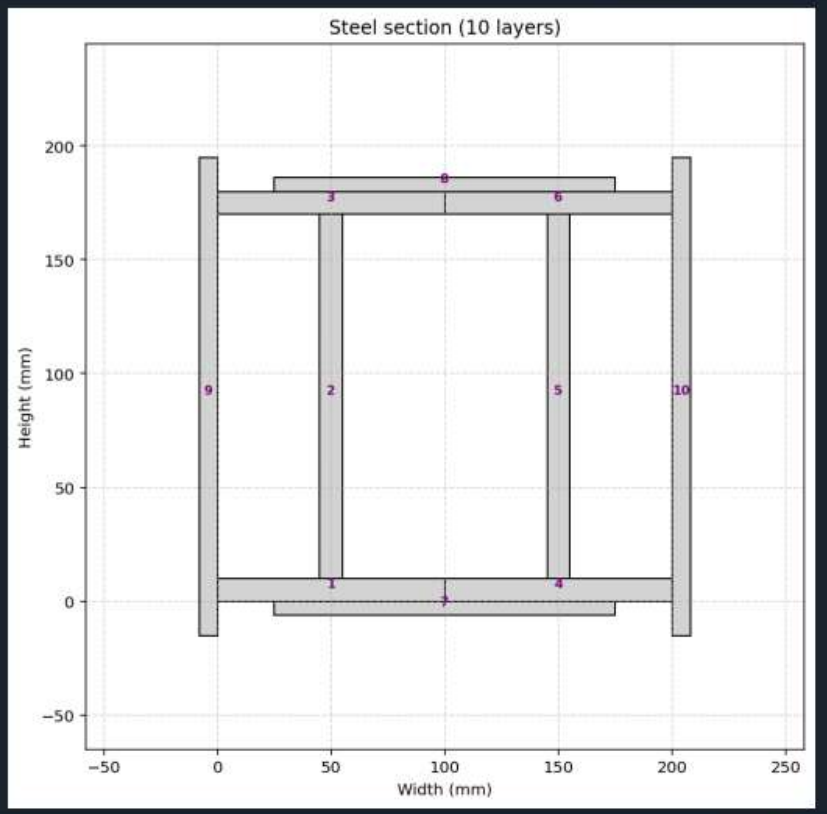
Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

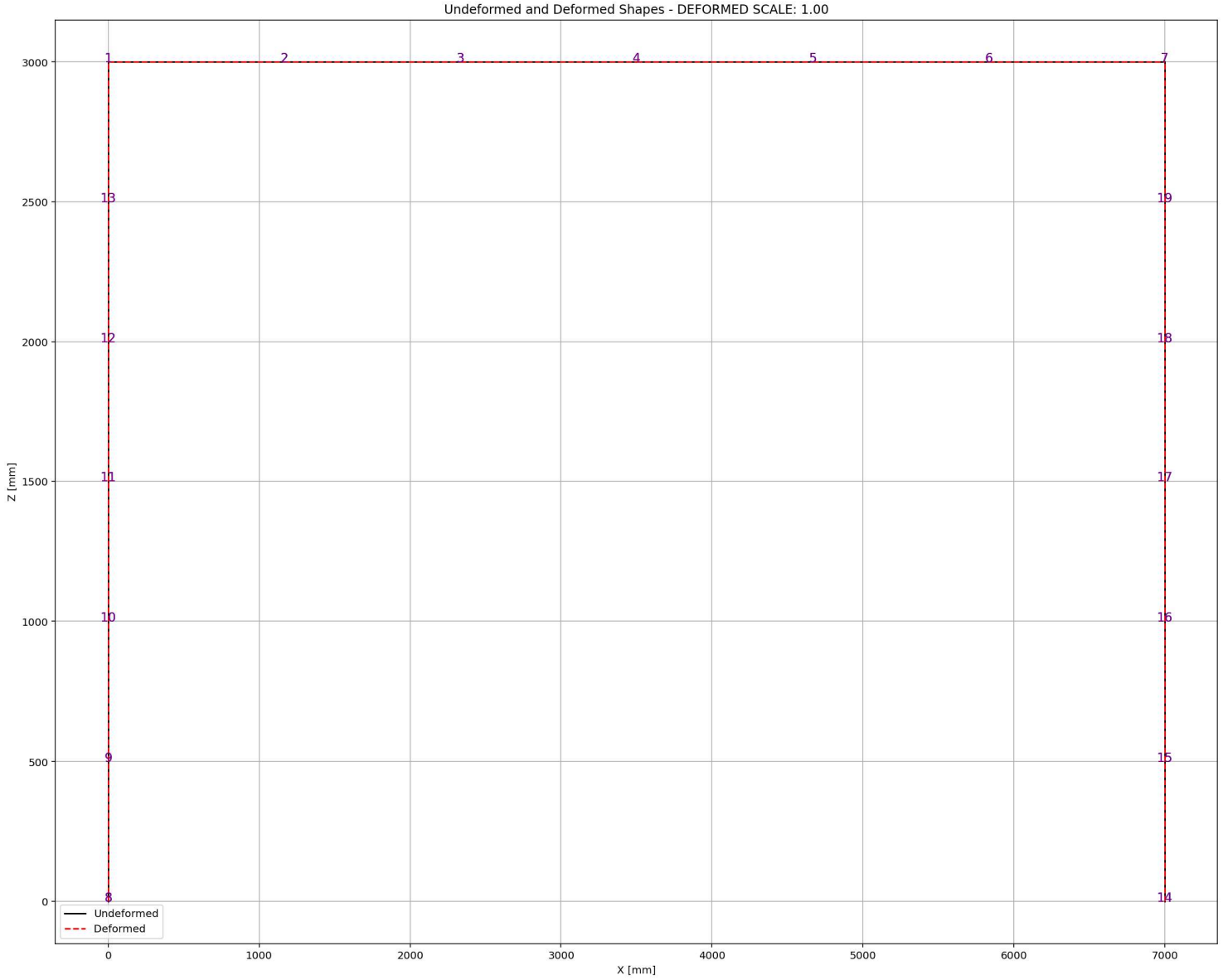
P(t)_STEEL_FRAME_8_ANA.py

```
1171  # PERIOD ANALYSIS
1172  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1173  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1174  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1175  ANAL_TYPE = 'PERIOD'
1176  BRACE = False                        # True: WITH BRACE - False: WITHOUT B
1177
1178  DATA = STEEL_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
1179  (PERIOD_MIN_X, PERIOD_MAX_X) = DATA
1180  print('Structure First Period:  ', PERIOD_MIN_X)
1181  print('Structure Second Period: ', PERIOD_MAX_X)
1182
1183  S01.PLOT_2D_FRAME(deformed_scale=1.0)
1184  #%%-------------------------------------------------
1185  # STATIC ANALYSIS
1186  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1187  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1188  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1189  ANAL_TYPE = 'STATIC'
1190  BRACE = False                        # True: WITH BRACE - False: WITHOUT B
1191
1192  DATA = STEEL_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
1193  (reactionX, reactionY, disp_mid,
1194   ele_axialforce, ele_shearforce, ele_momentforce,
1195   node_displacementsX, node_displacementsY,
1196   dispX, dispY) = DATA
1197
1198  S01.PLOT_2D_FRAME(deformed_scale=1.0)
1199  #%%-------------------------------------------------
1200  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1201  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1202  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1203  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1204  ANAL_TYPE = 'PUSHOVER'
```




IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 46%


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

# PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

...X_&_ENERGY_DISSIPATION_CAPACITY_INDEX\09_STEEL_FRAME_8_ANA

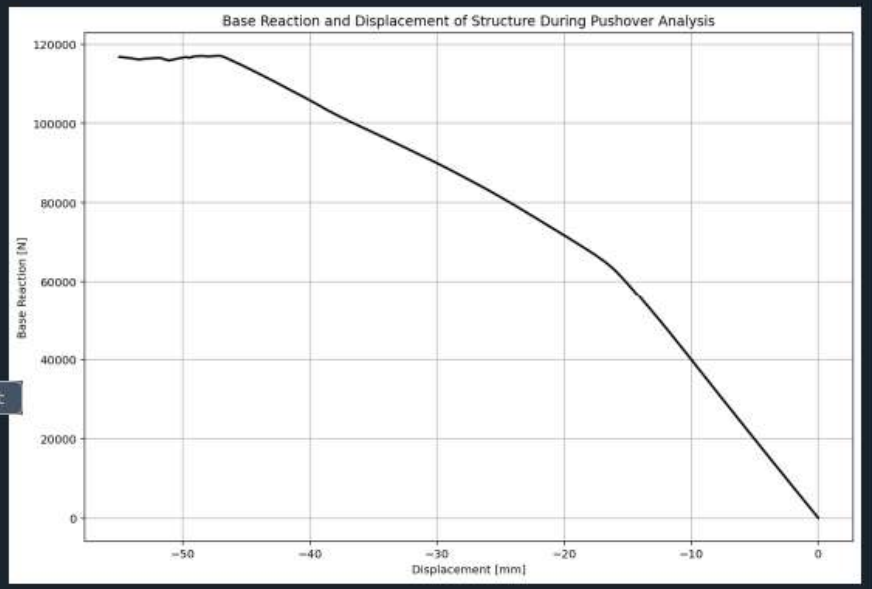
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

P(t)_STEEL_FRAME_8_ANA.py

```
1200  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1201  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1202  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1203  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1204  ANAL_TYPE = 'PUSHOVER'
1205  BRACE = False                        # True: WITH BRACE - False: WITHOUT B
1206
1207  DATA = STEEL_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
1208  (reactionX_PUSH, reactionY_PUSH, disp_mid_PUSH,
1209   ele_axialforce_PUSH, ele_shearforce_PUSH, ele_momentforce_PUSH,
1210   node_displacementsX_PUSH, node_displacementsY_PUSH,
1211    dispX_PUSH, dispY_PUSH,
1212   PERIOD_MIN_PUSH, PERIOD_MAX_PUSH,
1213   STRESSb_PUSH, STRAINb_PUSH, STRESSt_PUSH, STRAINt_PUSH, CURVATURE_PUSH) =
1214
1215
1216  XDATA = disp_mid_PUSH
1217  YDATA = reactionX_PUSH
1218  XLABEL = 'Displacement [mm]'
1219  YLABEL = 'Base Reaction [N]'
1220  TITLE = 'Base Reaction and Displacement of Structure During Pushov
1221  COLOR = 'black'
1222  SEMILOGY = False
1223  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1224
1225  DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reactionY_PUSH), SLO
1226  (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_S
1227
1228  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1229  plt.figure(0, figsize=(12, 8))
1230  plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1231  plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1232  plt.title('Period of Structure During Pushover Analysis')
1233  plt.ylabel('Structural Period [s]')
```

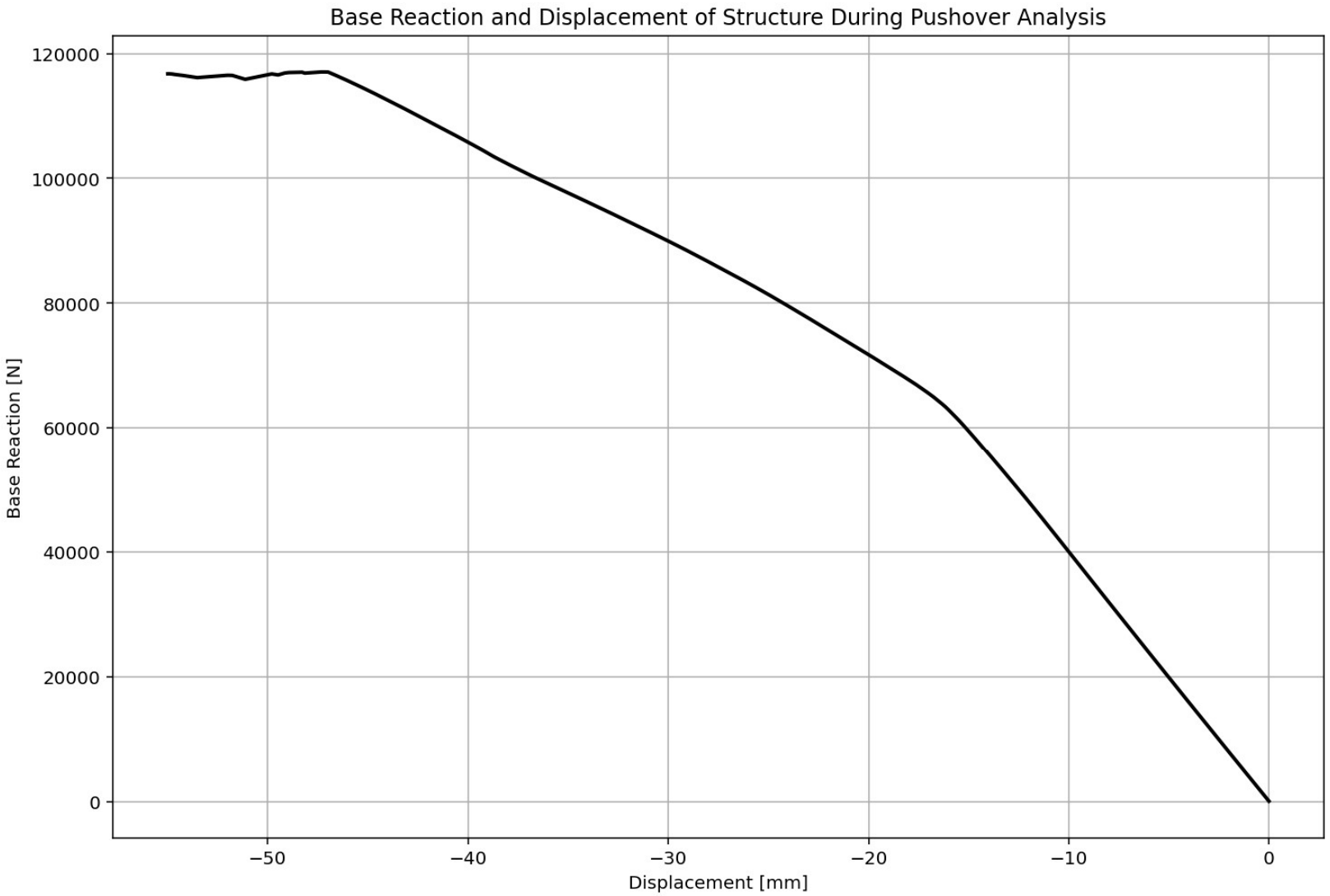
This is a float

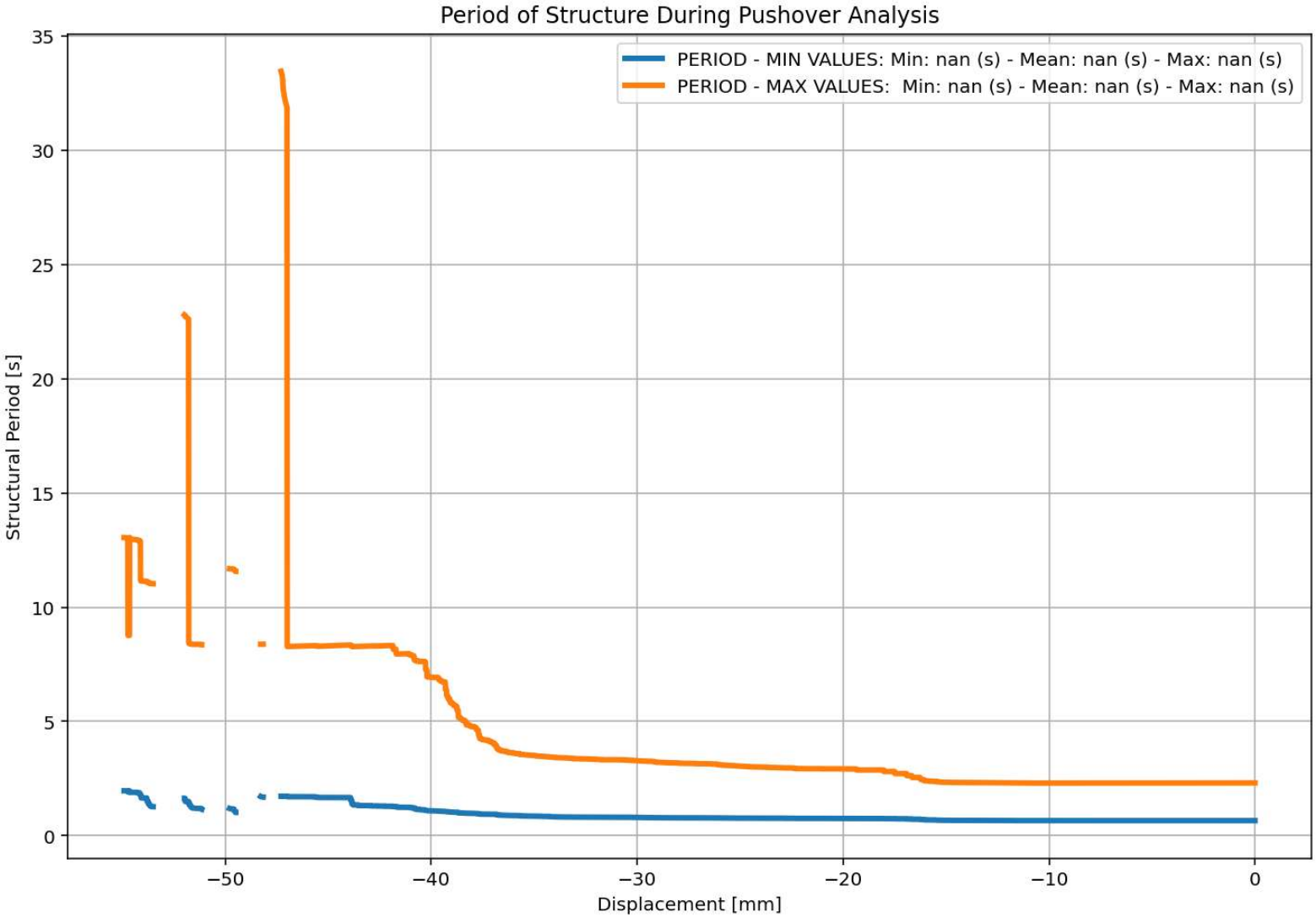
37 %

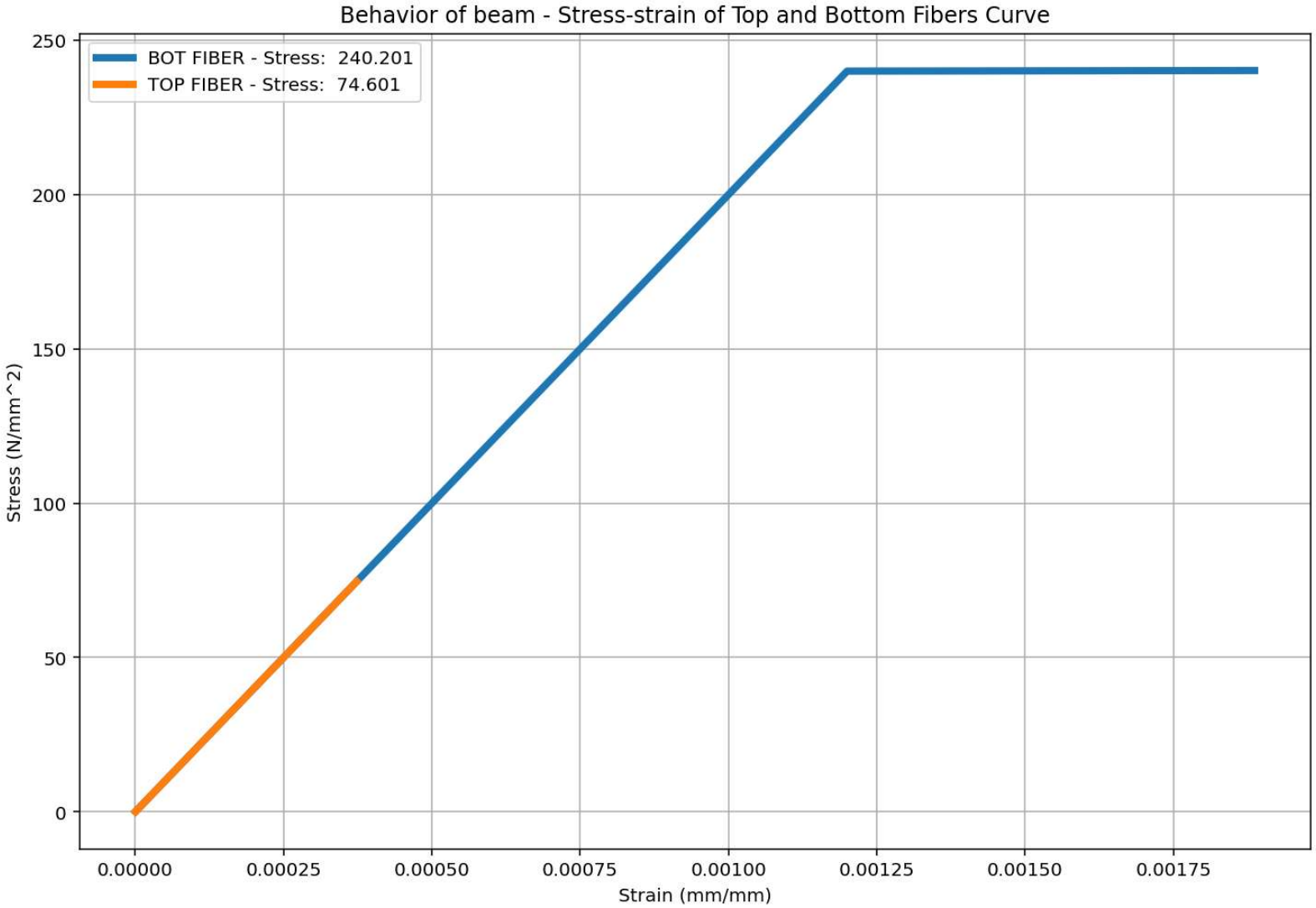


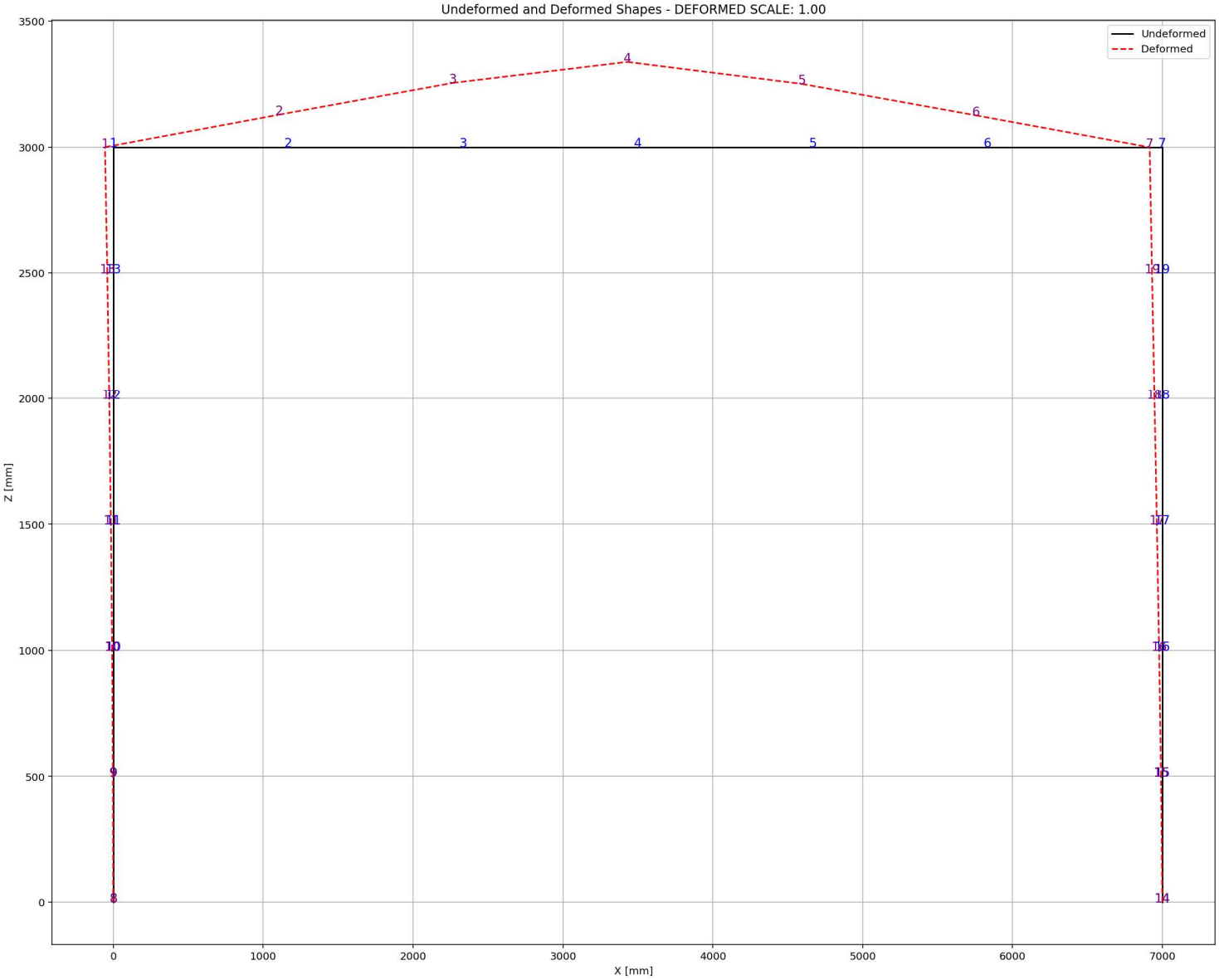

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 46%


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
120000
100000
80000
60000
40000
20000
0
−50
−40
−30
−20
−10
0
Displacement [mm]


Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 240.201
TOP FIBER - Stress: 74.601
Stress (N/mm^2)
250
200
150
100
50
0
0.00000
0.00025
0.00050
0.00075
0.00100
0.00125
0.00150
0.00175
Strain (mm/mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
3500
3000
2500
2000
1500
1000
500
0
0
1000
2000
3000
4000
5000
6000
7000

# CYCLIC DISPLACEMENT ANALYSIS

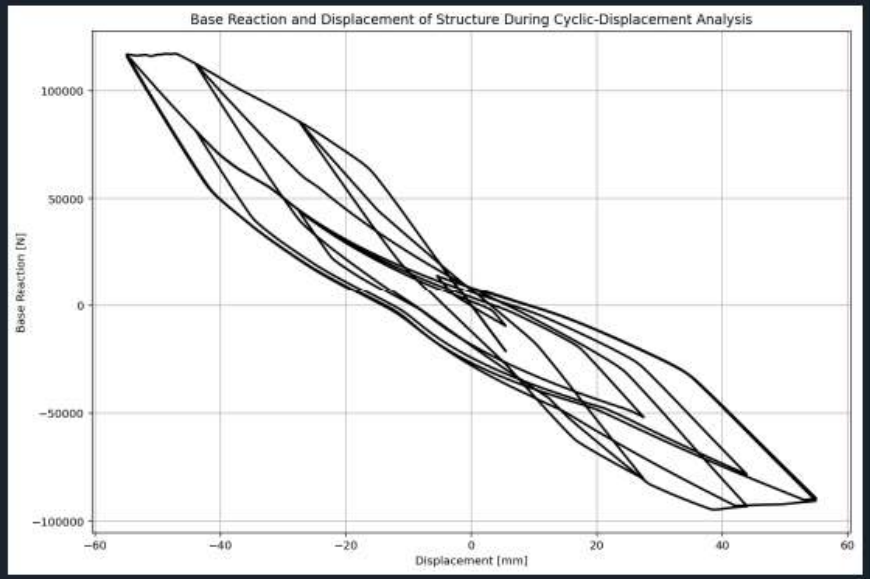
Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

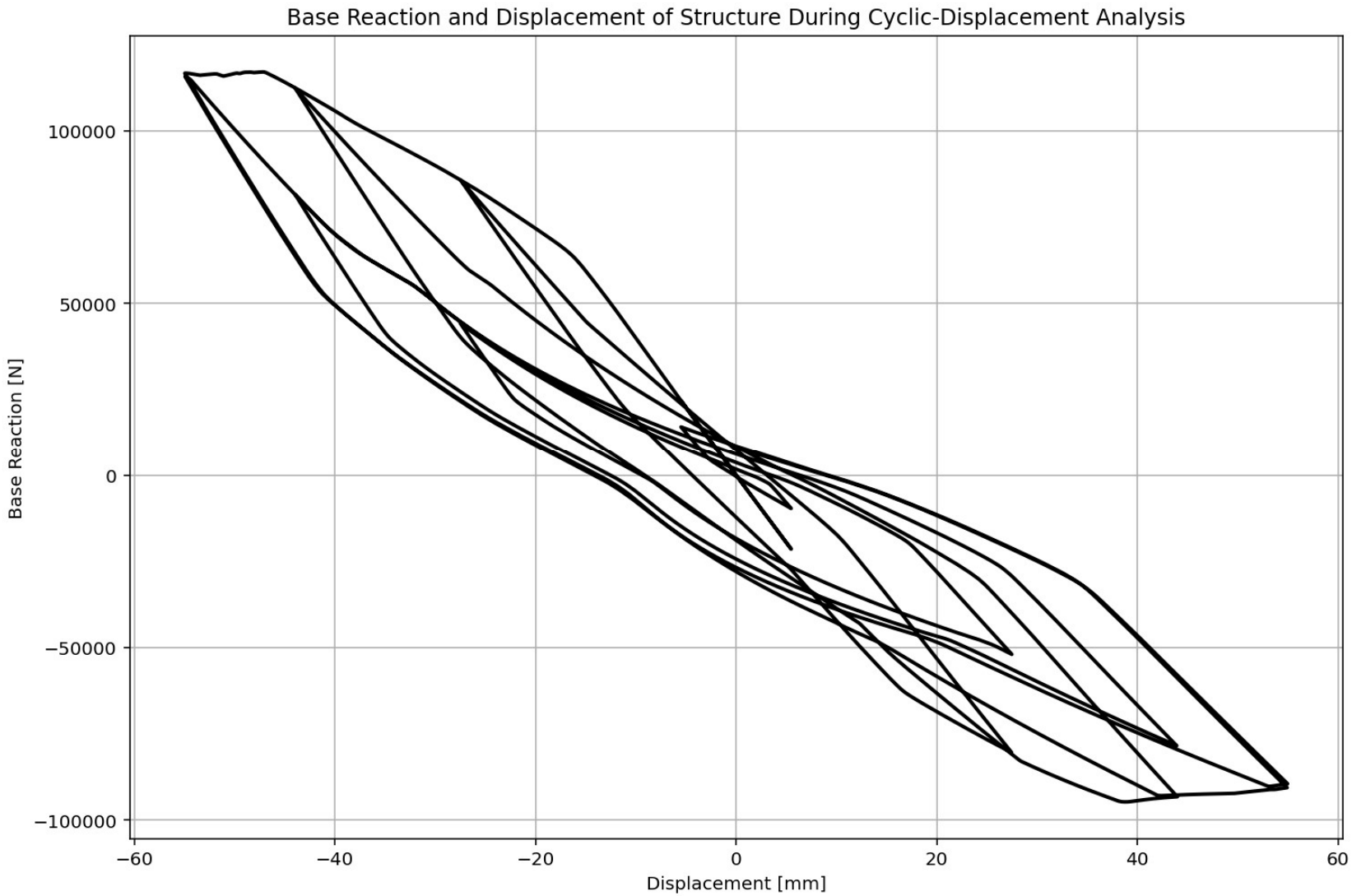
P(t)_STEEL_FRAME_8_ANA.py

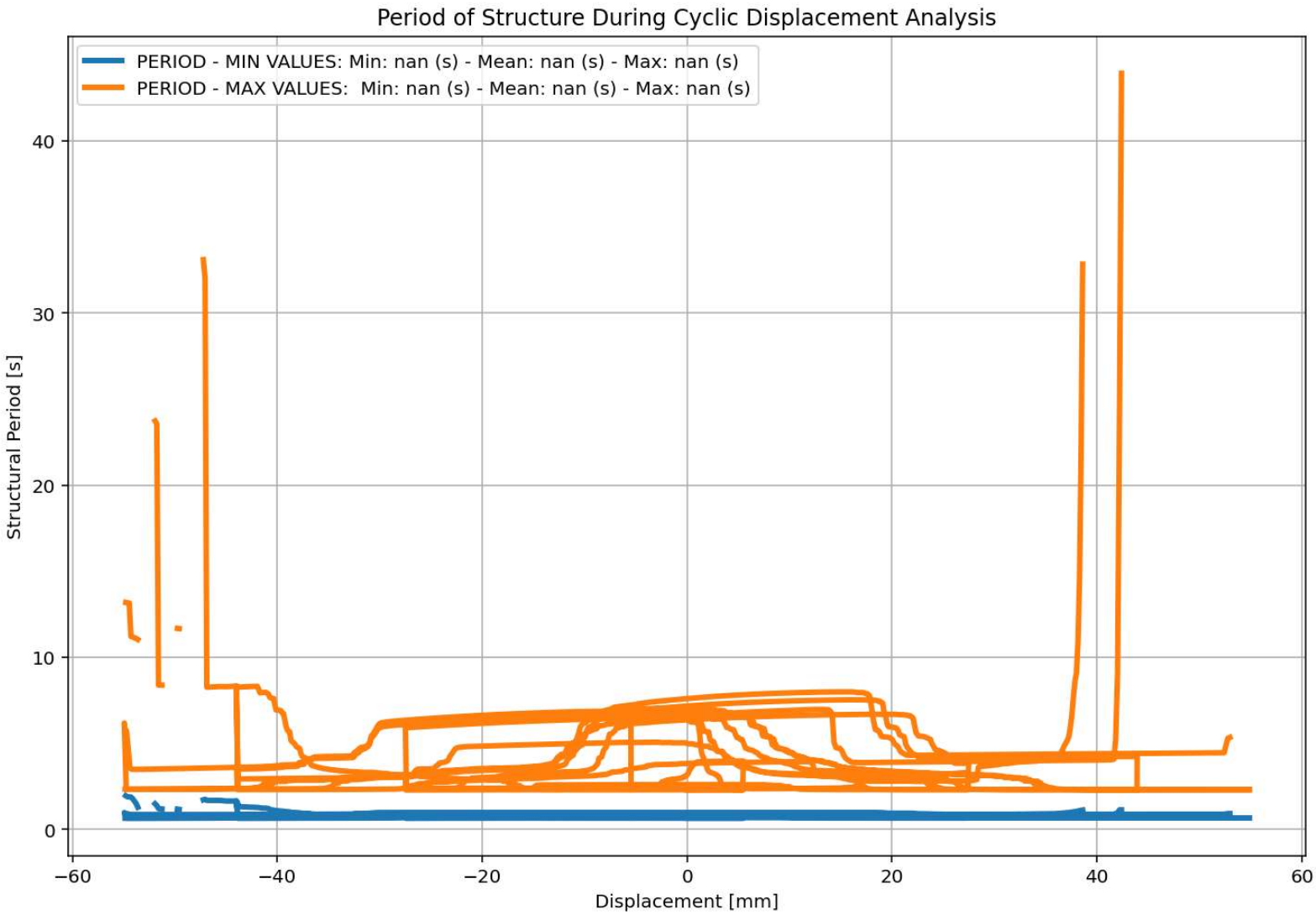
```
1255  # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1256  #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1257  ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1258  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
1259  ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1260  BRACE = False                        # True: WITH BRACE - False: WITHOUT B
1261
1262  DATA = STEEL_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
1263  (reactionX_CP, reactionY_CP, disp_mid_CP,
1264   ele_axialforce_CP, ele_shearforce_CP, ele_momentforce_CP,
1265   node_displacementsX_CP, node_displacementsY_CP,
1266    dispX_CP, dispY_CP,
1267   PERIOD_MIN_CP, PERIOD_MAX_CP,
1268   STRESSb_CP, STRAINb_CP, STRESSt_CP, STRAINt_CP, CURVATURE_CP) = DATA
1269
1270
1271  XDATA = disp_mid_CP
1272  YDATA = reactionX_CP
1273  XLABEL = 'Displacement [mm]'
1274  YLABEL = 'Base Reaction [N]'
1275  TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displace
1276  COLOR = 'black'
1277  SEMILOGY = False
1278  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1279
1280  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1281  plt.figure(0, figsize=(12, 8))
1282  plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
1283  plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
1284  plt.title('Period of Structure During Cyclic Displacement Analysis')
1285  plt.ylabel('Structural Period [s]')
1286  plt.xlabel('Displacement [mm]')
1287  #plt.semilogy()
1288  plt.grid()
```

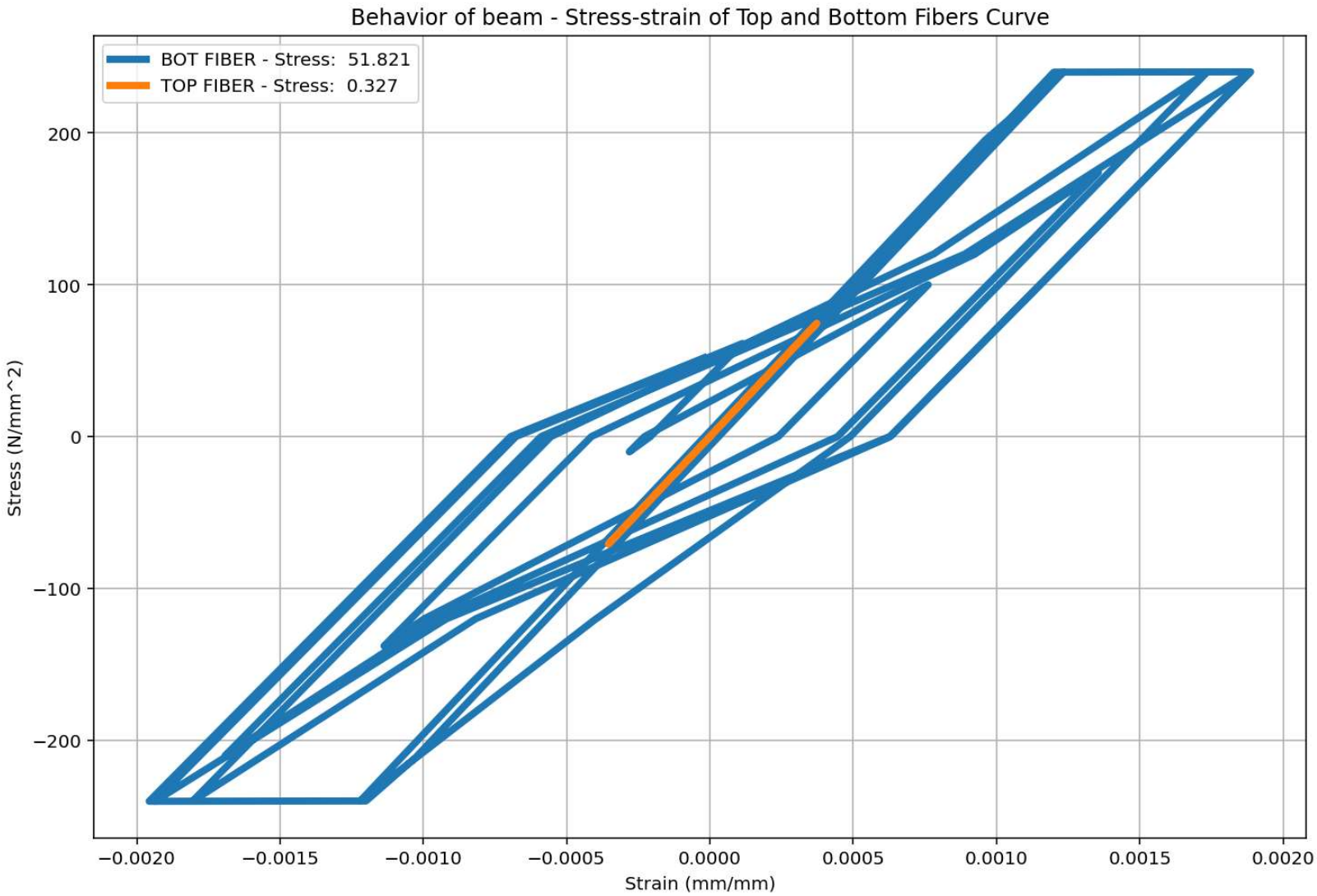


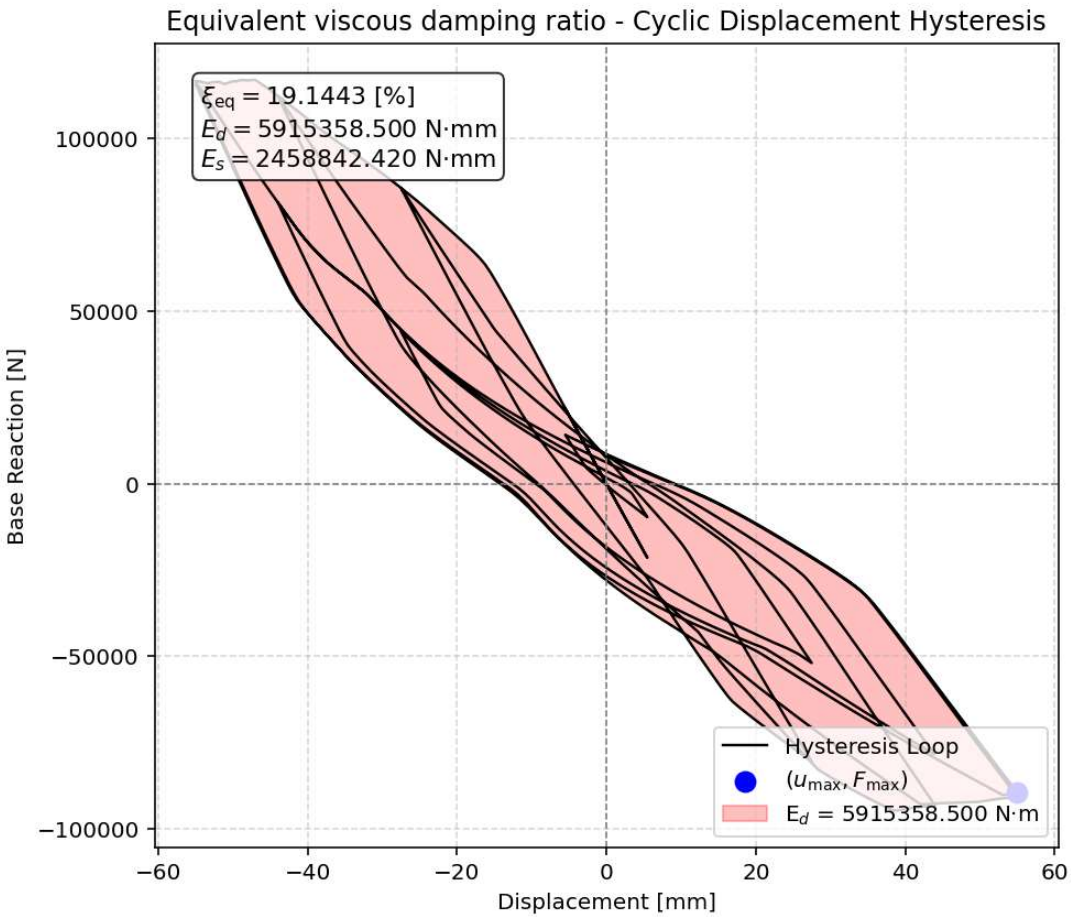

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

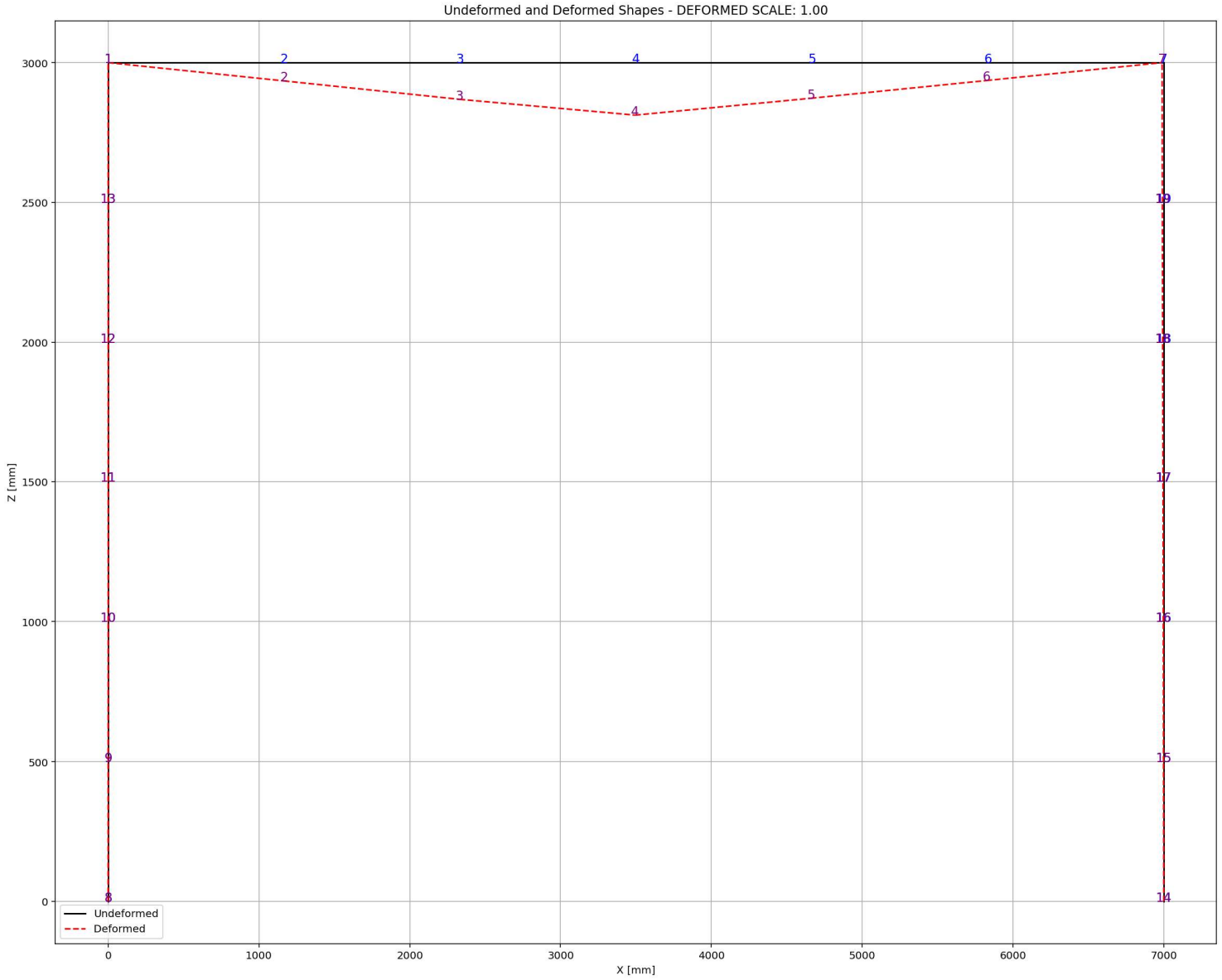
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 47%


Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
100000
50000
0
−50000
−100000
−60
−40
−20
0
20
40
60
Displacement [mm]


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
0
10
20
30
40
−60
−40
−20
0
20
40
60
Displacement [mm]


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 51.821
TOP FIBER - Stress: 0.327
Stress (N/mm^2)
200
100
0
−100
−200
−0.0020
−0.0015
−0.0010
−0.0005
0.0000
0.0005
0.0010
0.0015
0.0020
Strain (mm/mm)


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq}$ = 19.1443 [%]
$E_d$ = 5915358.500 N·mm
$E_s$ = 2458842.420 N·mm
Base Reaction [N]
Displacement [mm]
100000
50000
0
−50000
−100000
−60
−40
−20
0
20
40
60
Hysteresis Loop
($u_{max}$, $F_{max}$)
$E_d$ = 5915358.500 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

# STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

P(t)_STEEL_FRAME_8_ANA.py

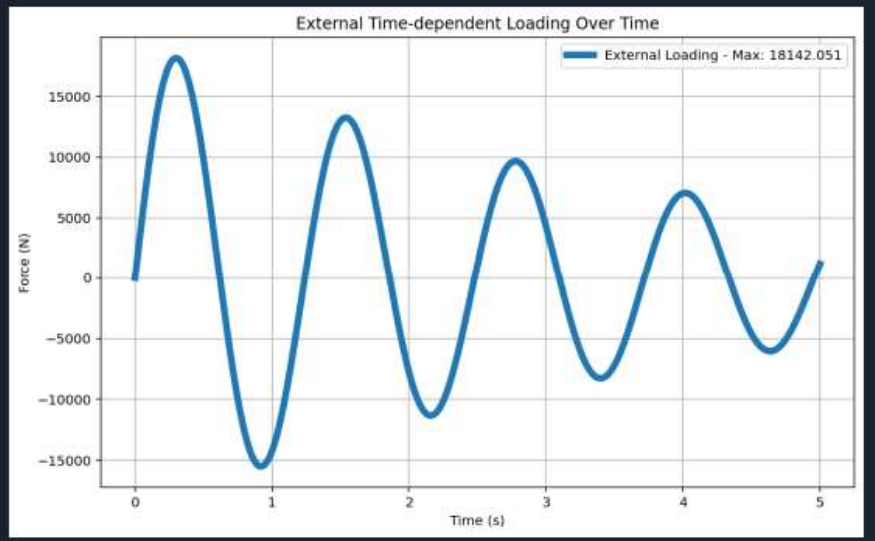
```
# EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
BRACE = False                        # True: WITH BRACE - False: WITHOUT B

DATA = STEEL_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS

(reactionX_ETDLS, reactionY_ETDLS, disp_mid_ETDLS,
 ele_axialforce_ETDLS, ele_shearforce_ETDLS, ele_momentforce_ETDLS,
 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
  dispX_ETDLS, dispY_ETDLS,
 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS,
 STRESSb_ETDLS, STRAINb_ETDLS, STRESSt_ETDLS, STRAINt_ETDLS, CURVATURE_ETDL

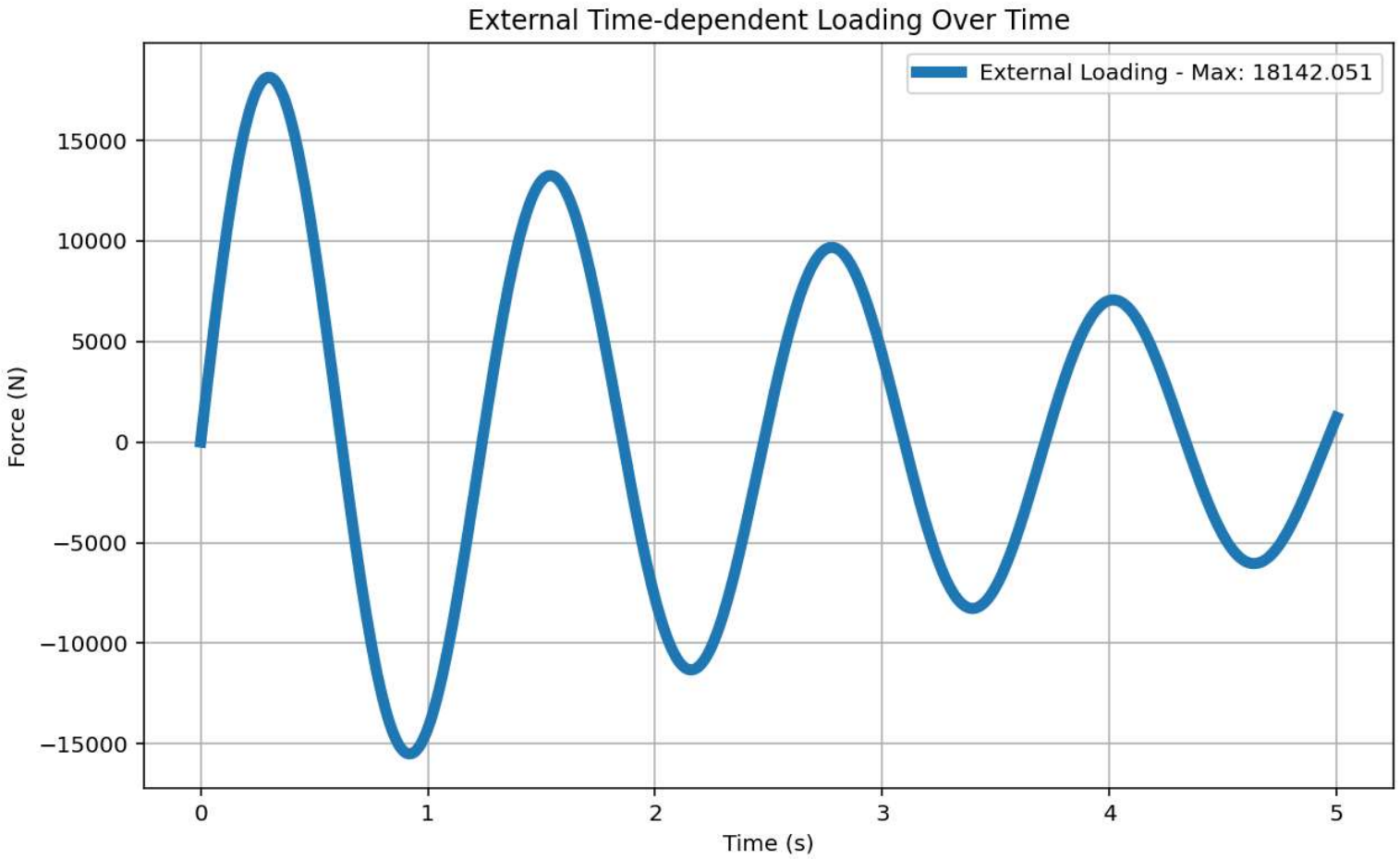

XDATA = disp_mid_ETDLS
YDATA = reactionX_ETDLS
XLABEL = 'Displacement [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Static External
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)

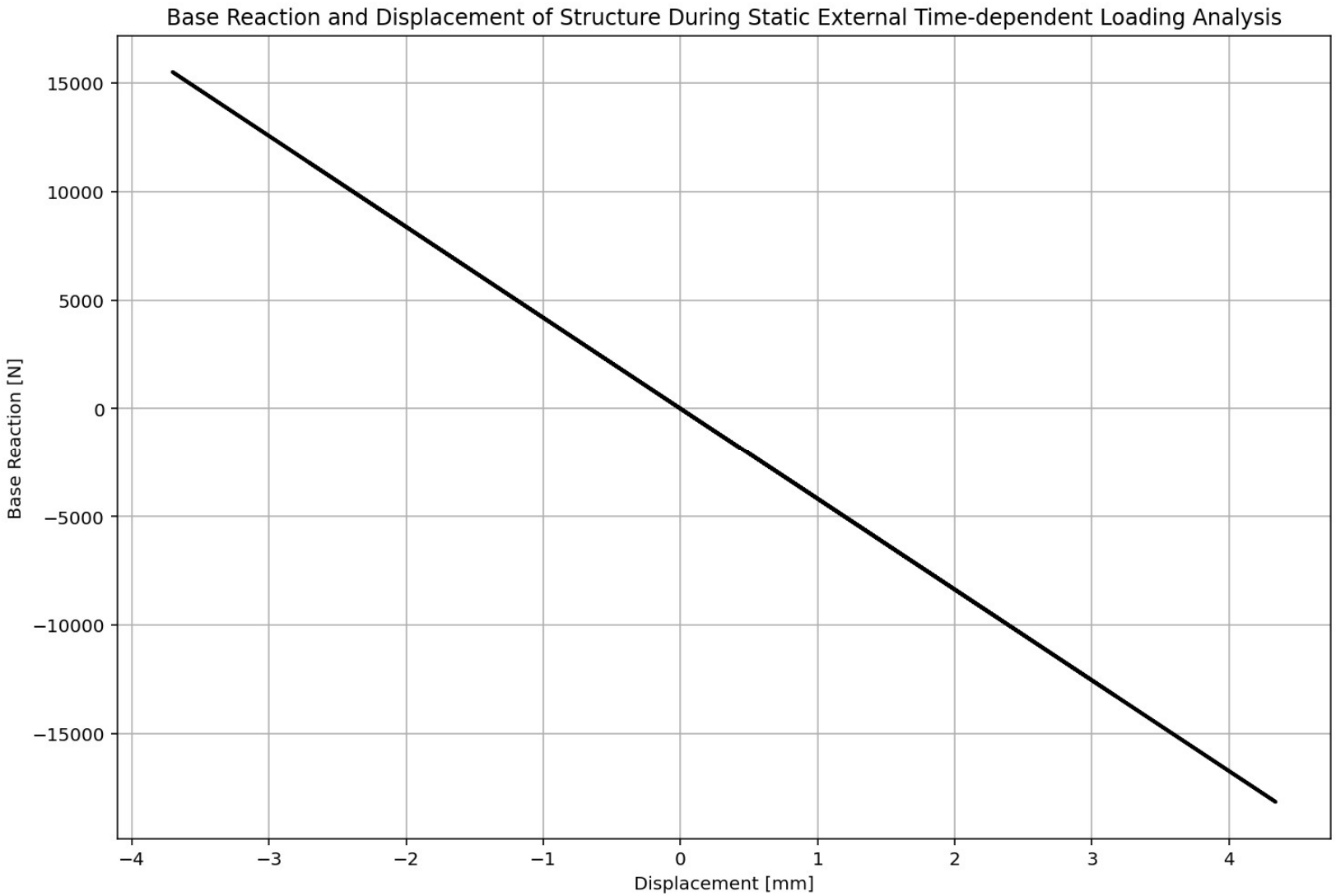
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
plt.title('Period of Structure During Static External Time-dependent Loadin
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
```

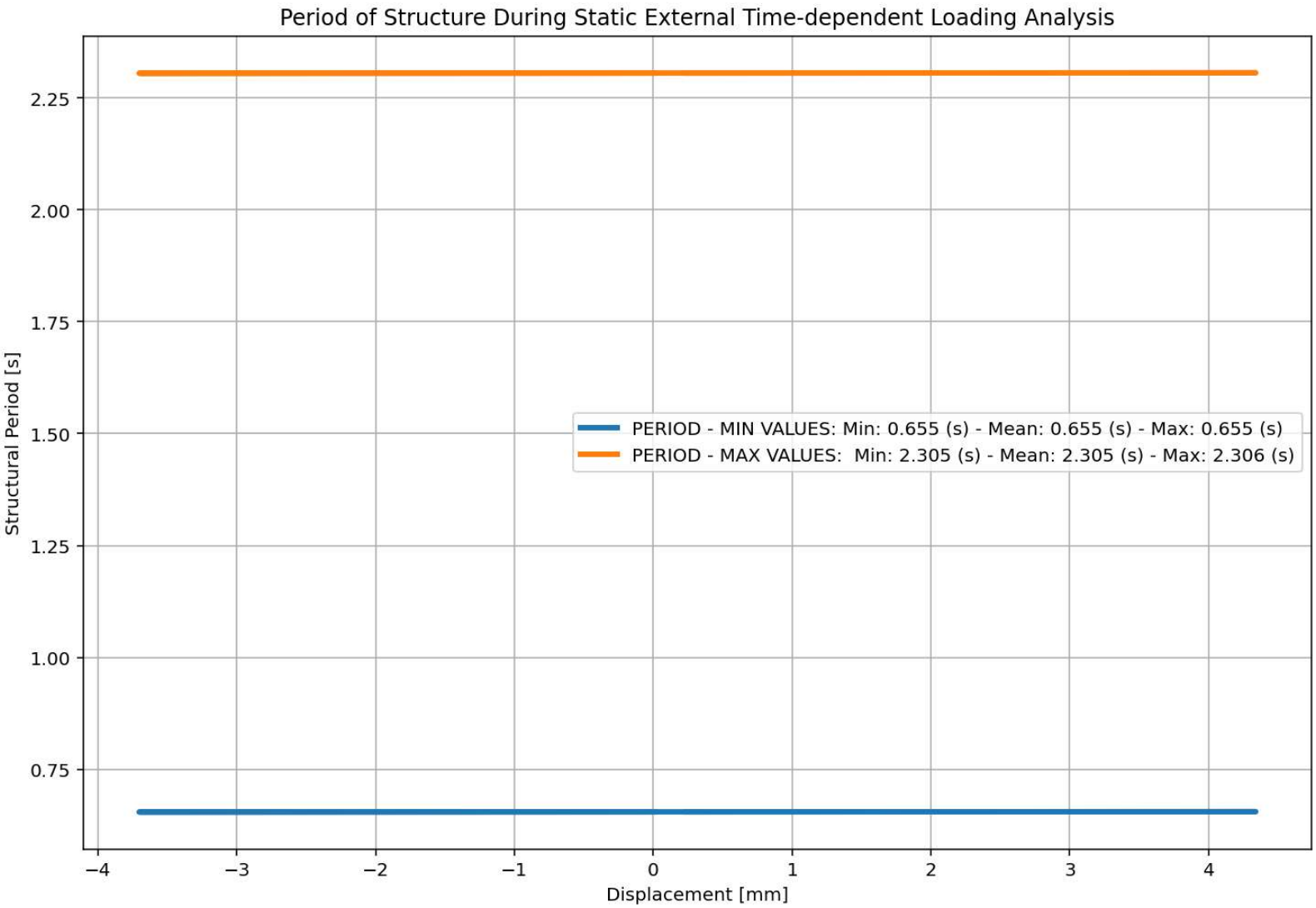


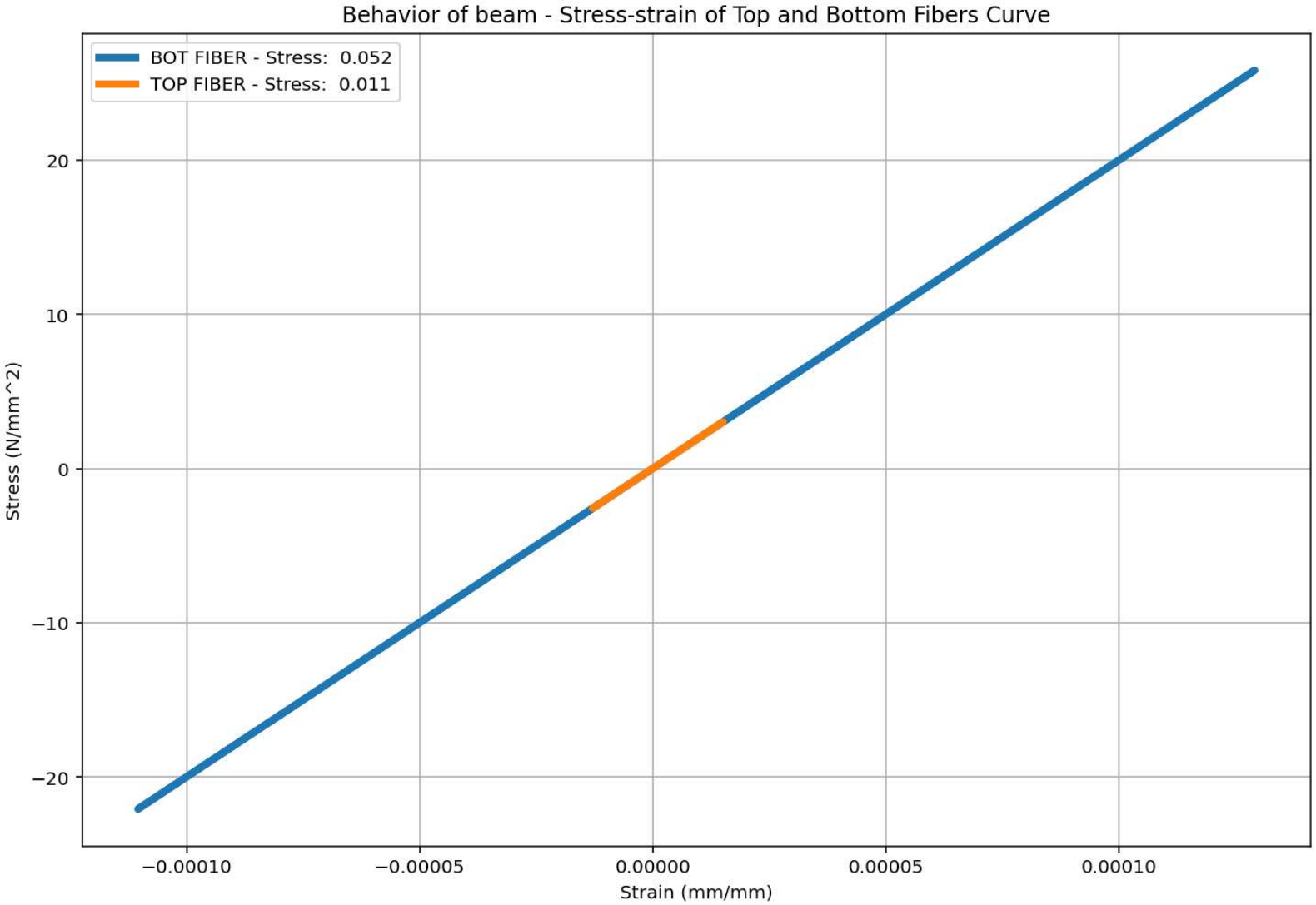

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

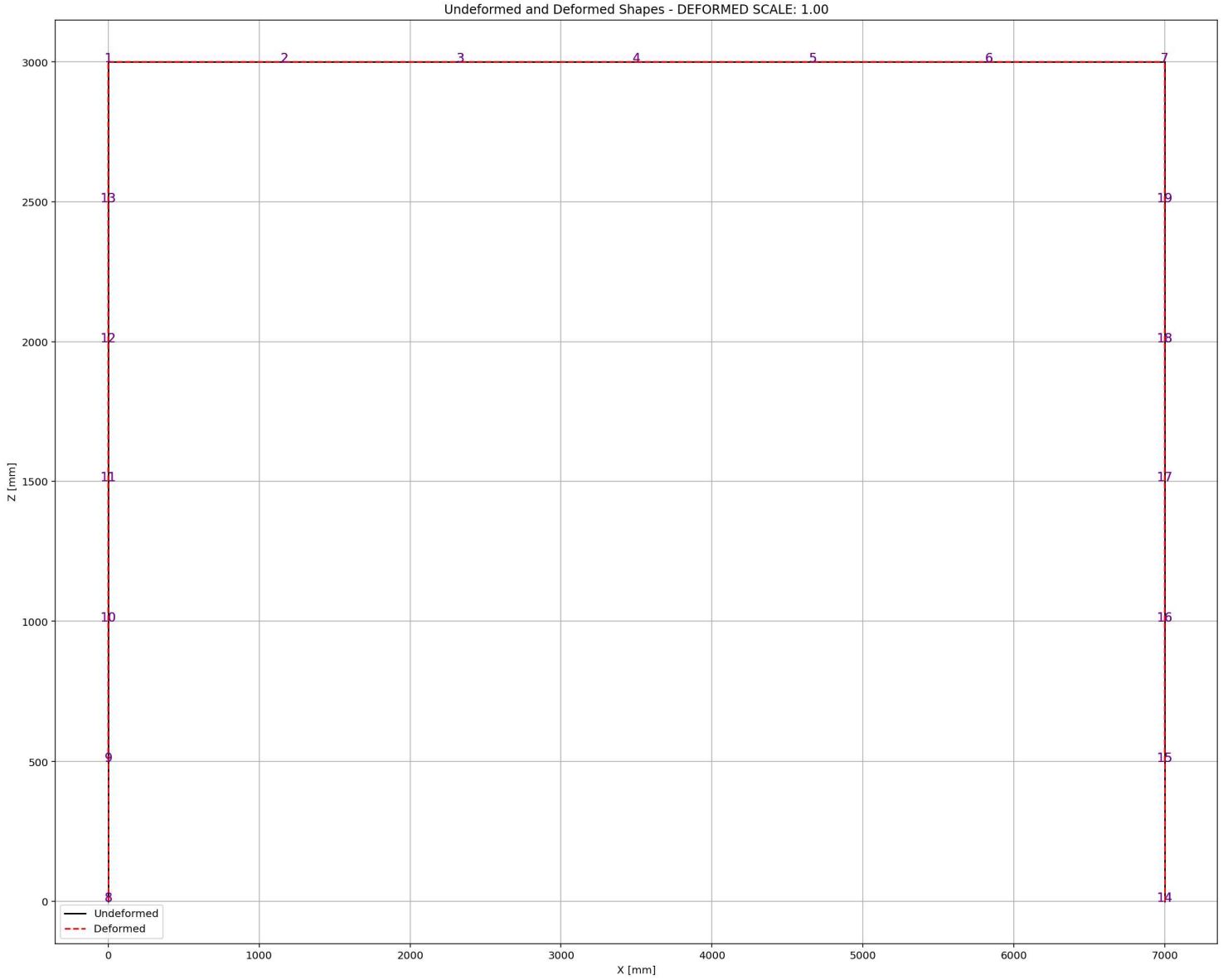
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 47%


External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
15000
10000
5000
0
−5000
−10000
−15000
−4
−3
−2
−1
0
1
2
3
4
Displacement [mm]


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
2.25
2.00
1.75
1.50
1.25
1.00
0.75
−4
−3
−2
−1
0
1
2
3
4
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.655 (s) - Mean: 0.655 (s) - Max: 0.655 (s)
PERIOD - MAX VALUES:  Min: 2.305 (s) - Mean: 2.305 (s) - Max: 2.306 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.052
TOP FIBER - Stress: 0.011
Stress (N/mm^2)
20
10
0
−10
−20
−0.00010
−0.00005
0.00000
0.00005
0.00010
Strain (mm/mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

# DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

P(t)_STEEL_FRAME_8_ANA.py

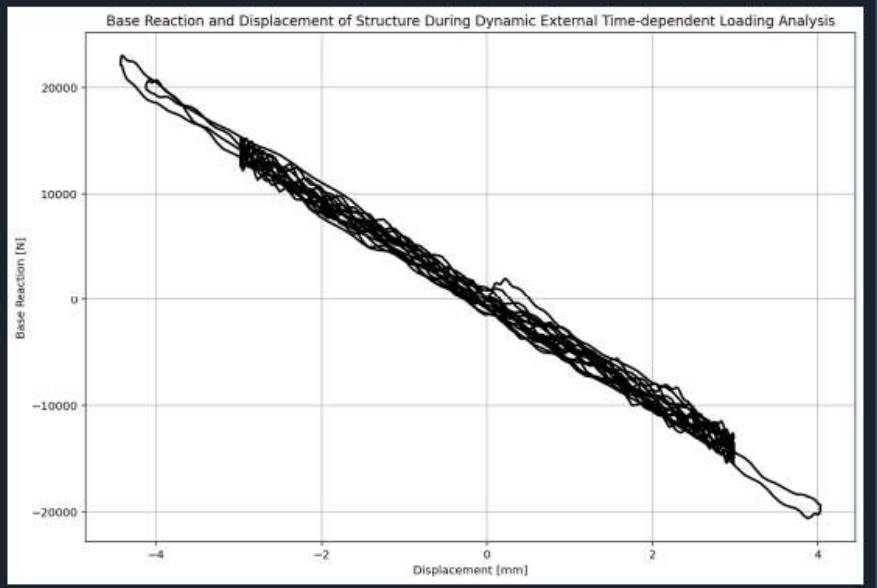
```
# EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
BRACE = False                        # True: WITH BRACE - False: WITHOUT B

DATA = STEEL_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS

(time_ETDLD, reactionX_ETDLD, reactionY_ETDLD, disp_mid_ETDLD,
ele_axialforce_ETDLD, ele_shearforce_ETDLD, ele_momentforce_ETDLD,
node_displacementsX_ETDLD, node_displacementsY_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD,
stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD,
STRESSb_ETDLD, STRAINb_ETDLD, STRESSt_ETDLD, STRAINt_ETDLD, CURVATURE_ETDLD

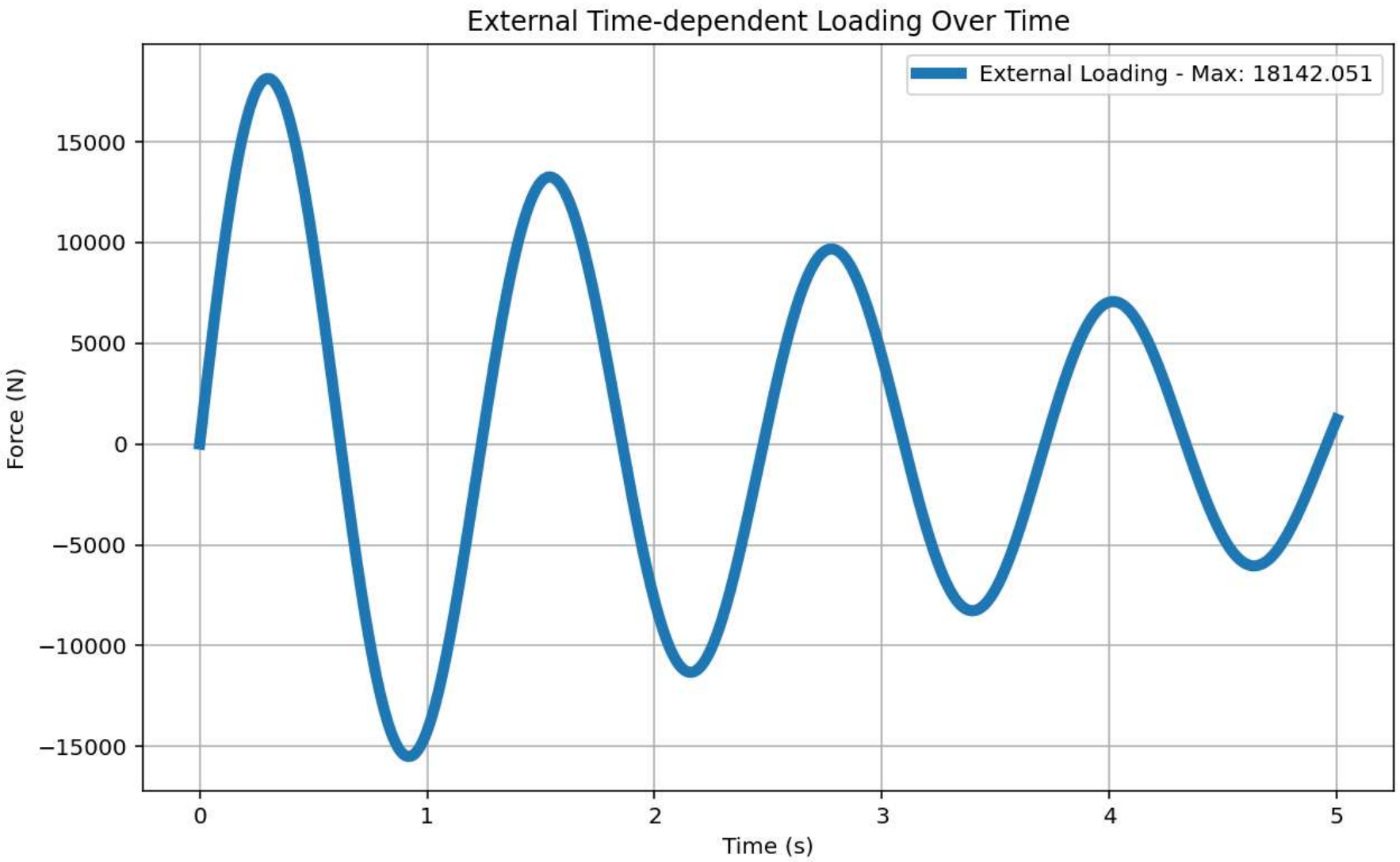

XDATA = disp_mid_ETDLD
YDATA = reactionX_ETDLD
XLABEL = 'Displacement [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Dynamic Externa
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)

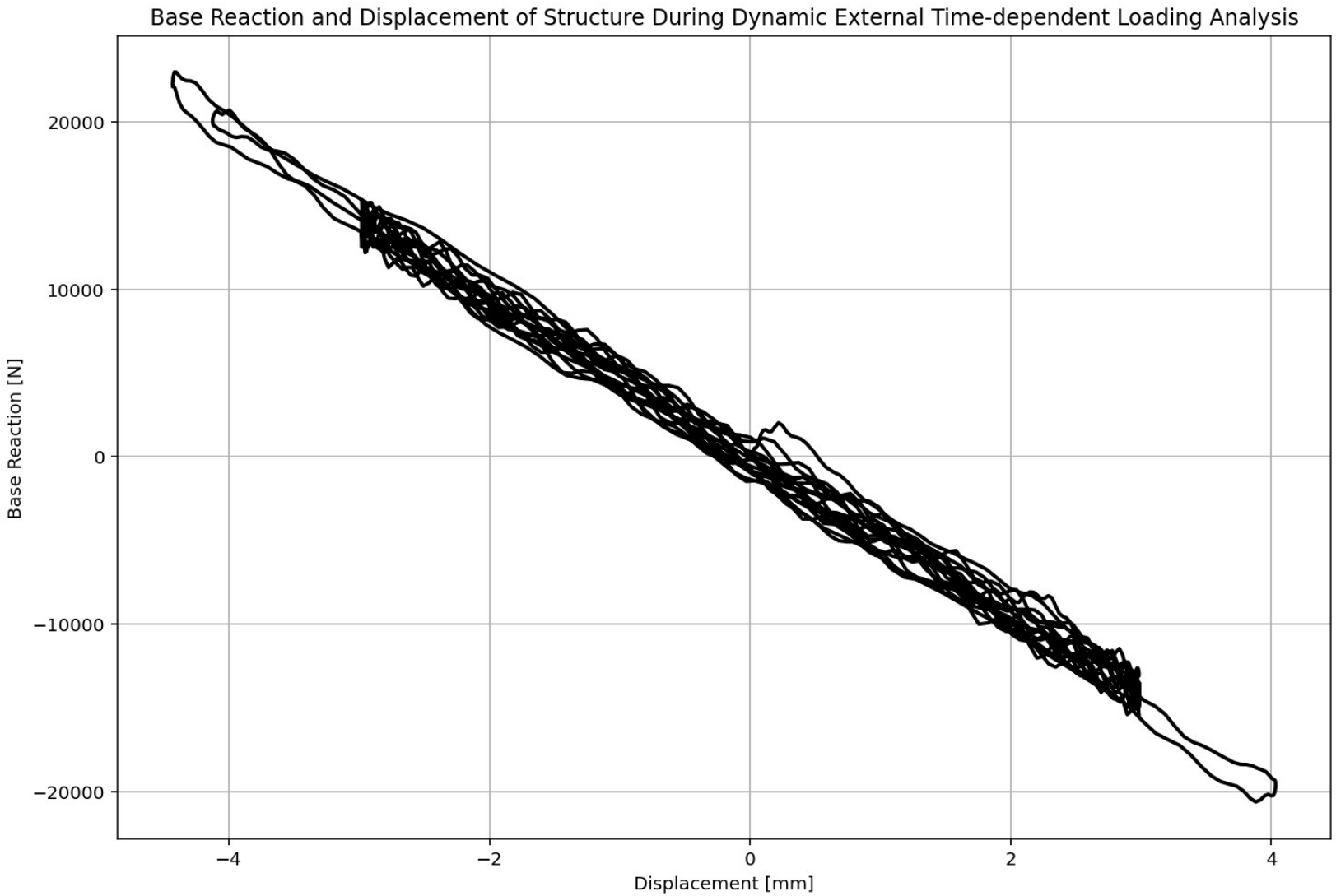
PLOT_TIME_HISTORY(time_ETDLD, reactionY_ETDLD, disp_mid_ETDLD,
                  dispX_ETDLD, dispY_ETDLD,
                  veloX_ETDLD, veloY_ETDLD,
                  accX_ETDLD, accY_ETDLD)
```

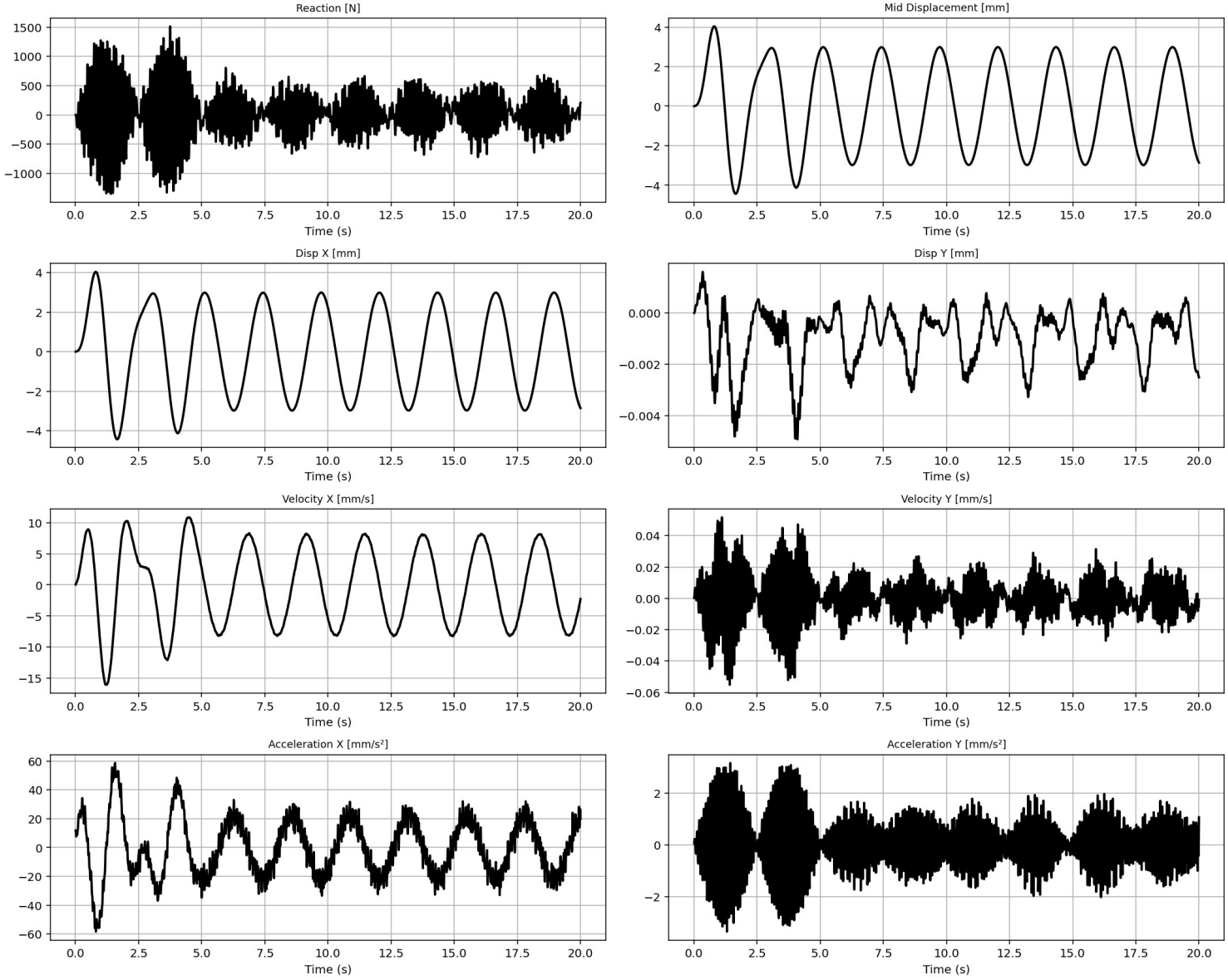


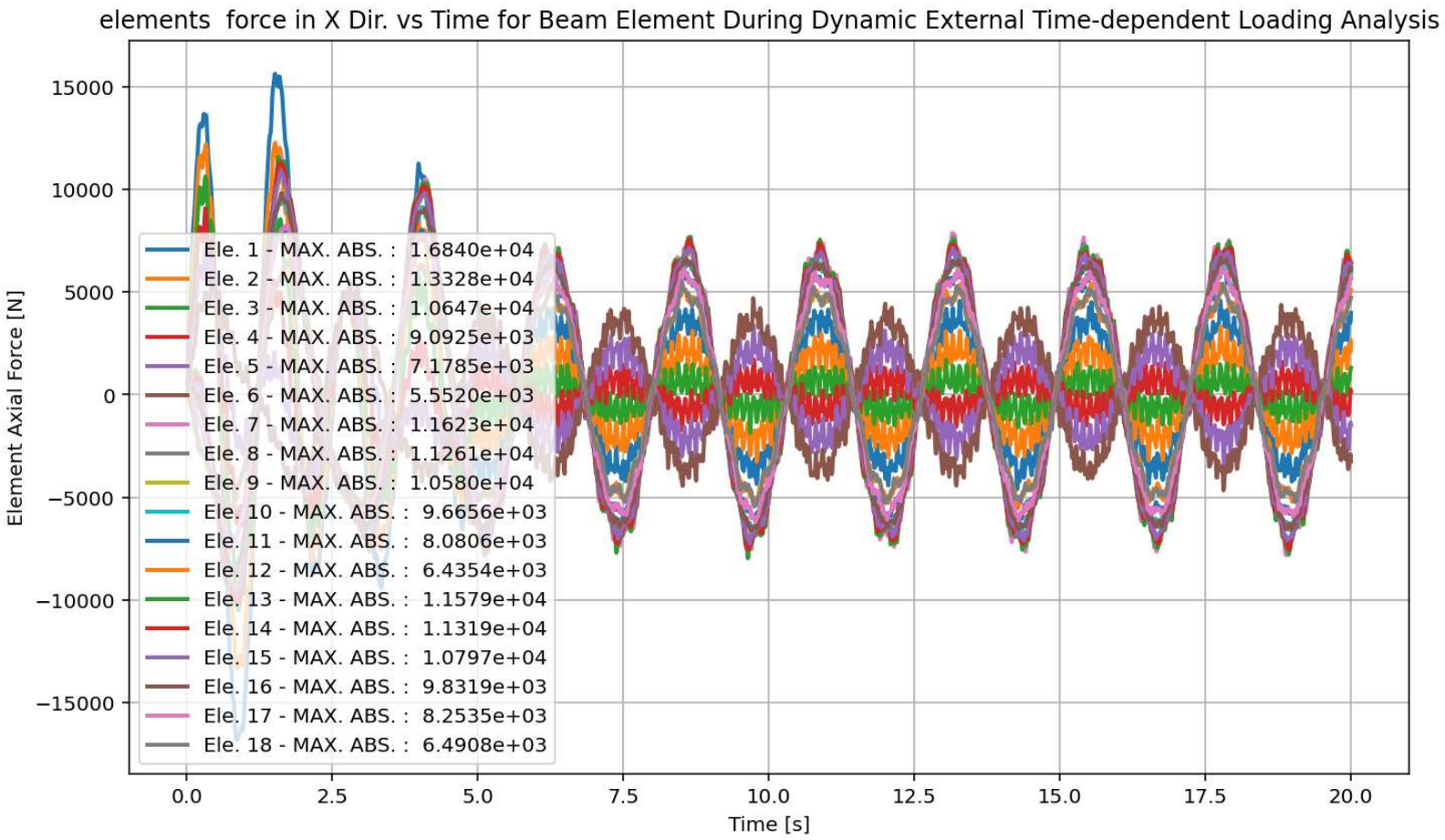

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

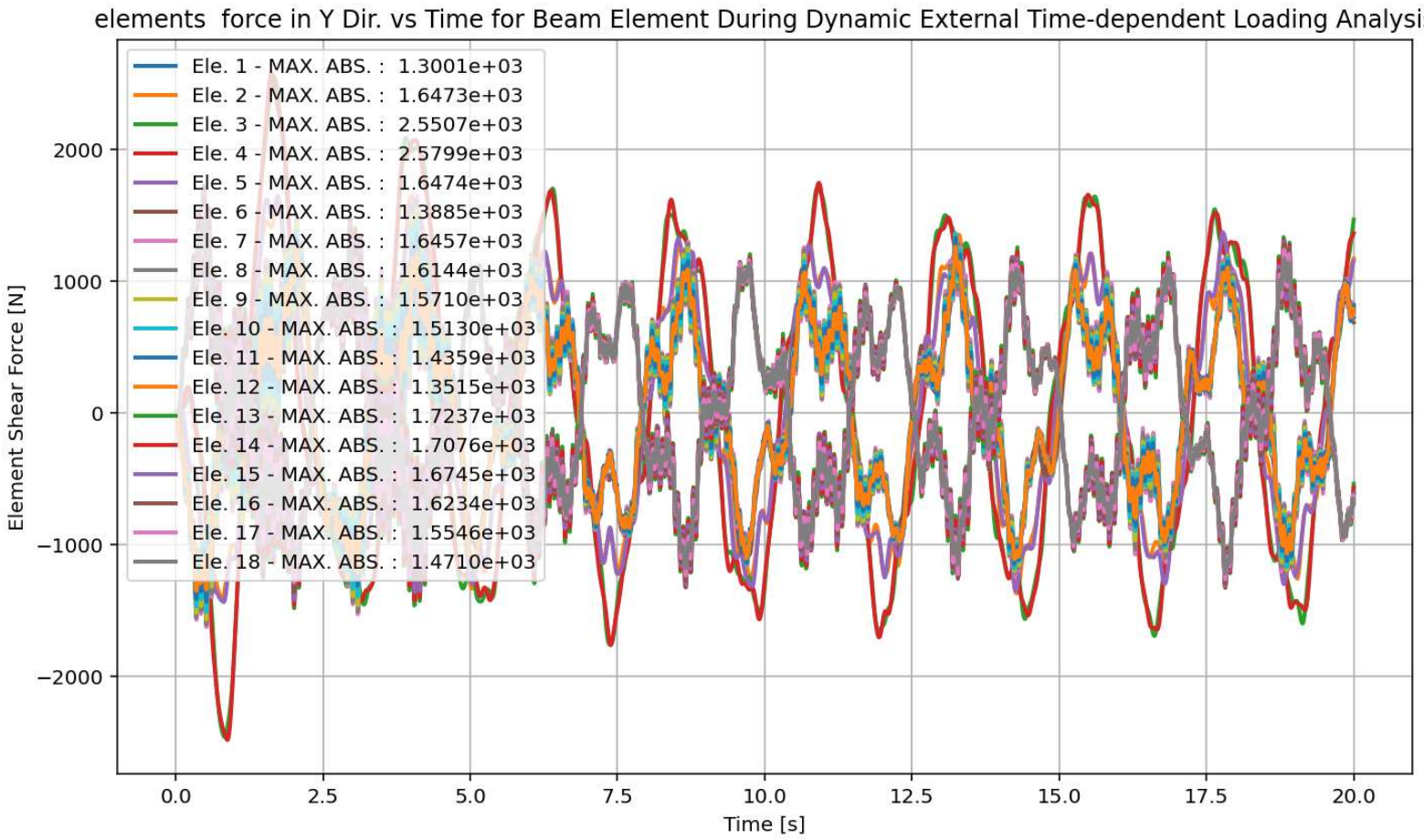
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 12, Col 105 UTF-8 CRLF RW Mem 48%

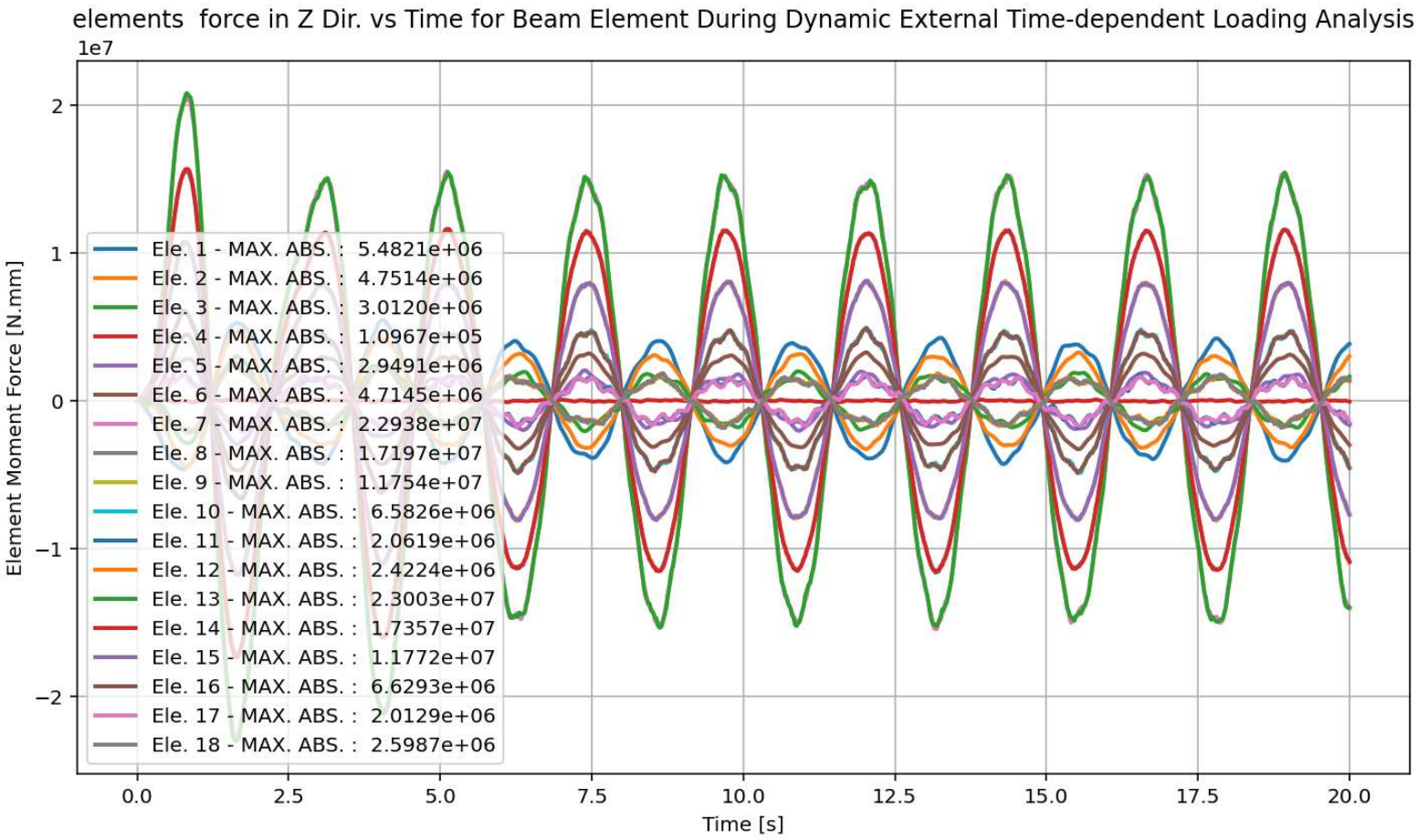

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
15000
10000
5000
0
−5000
−10000
−15000
Force (N)
0
1
2
3
4
5
Time (s)

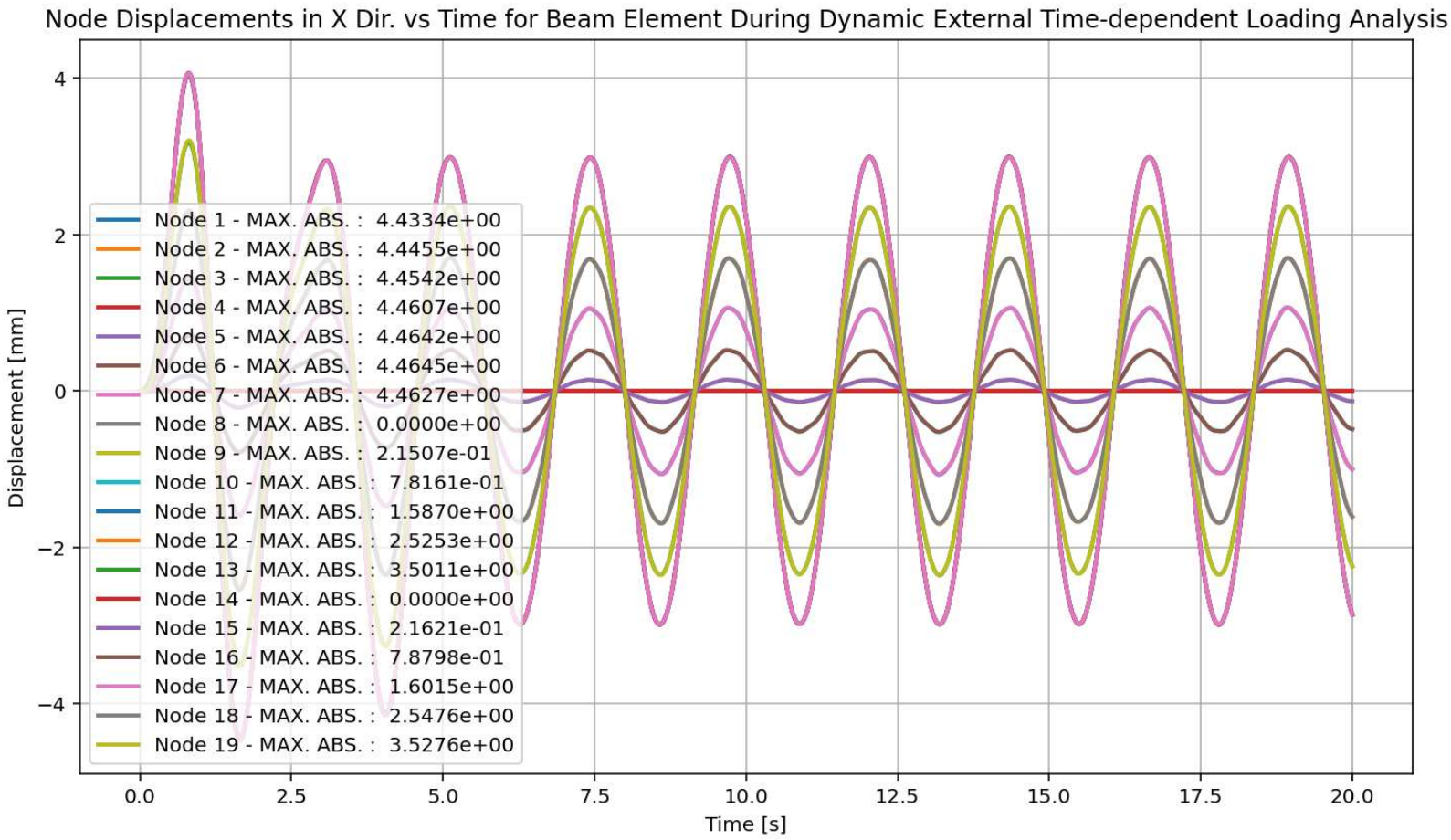

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
20000
10000
0
−10000
−20000
−4
−2
0
2
4
Displacement [mm]

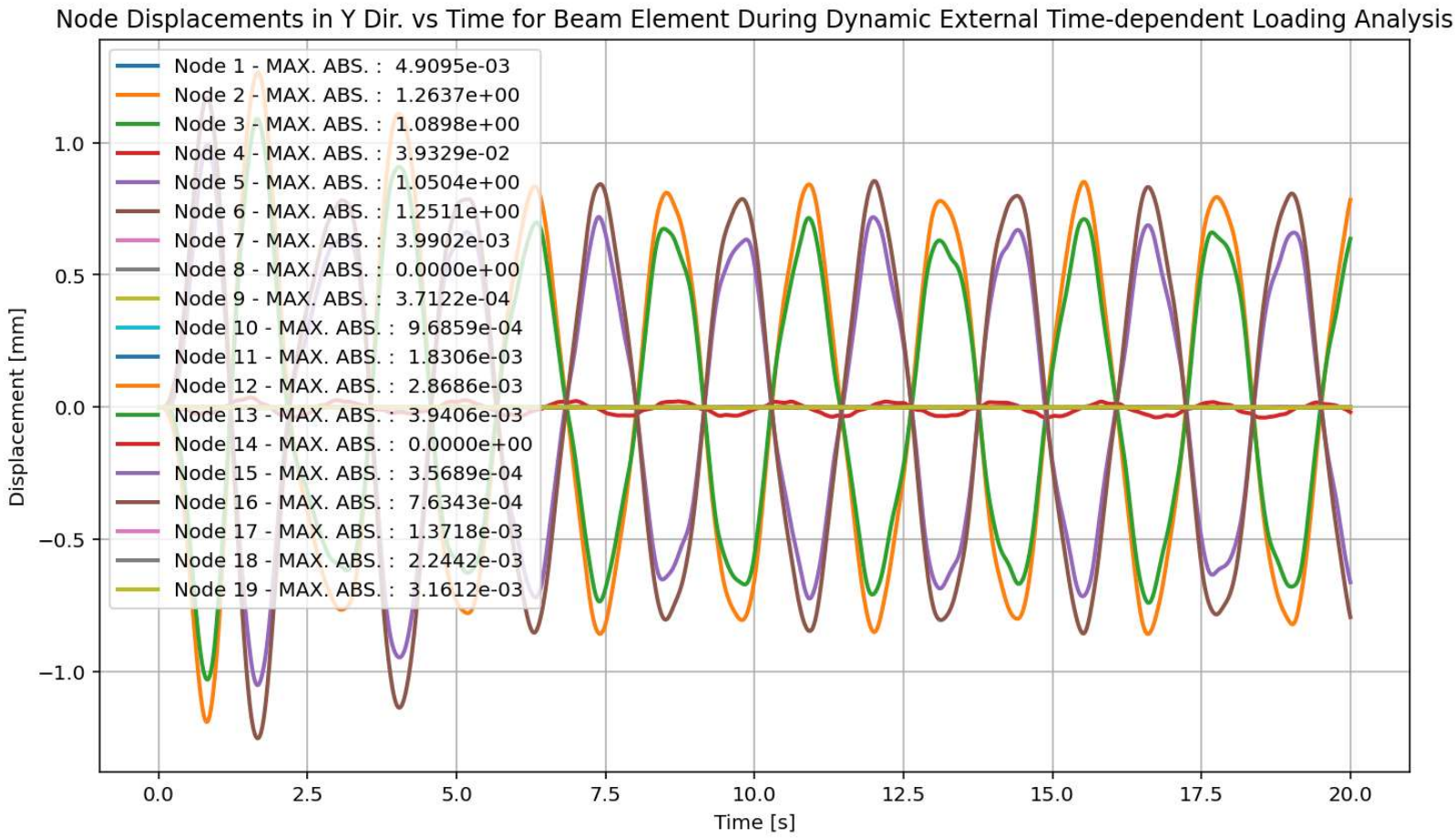

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

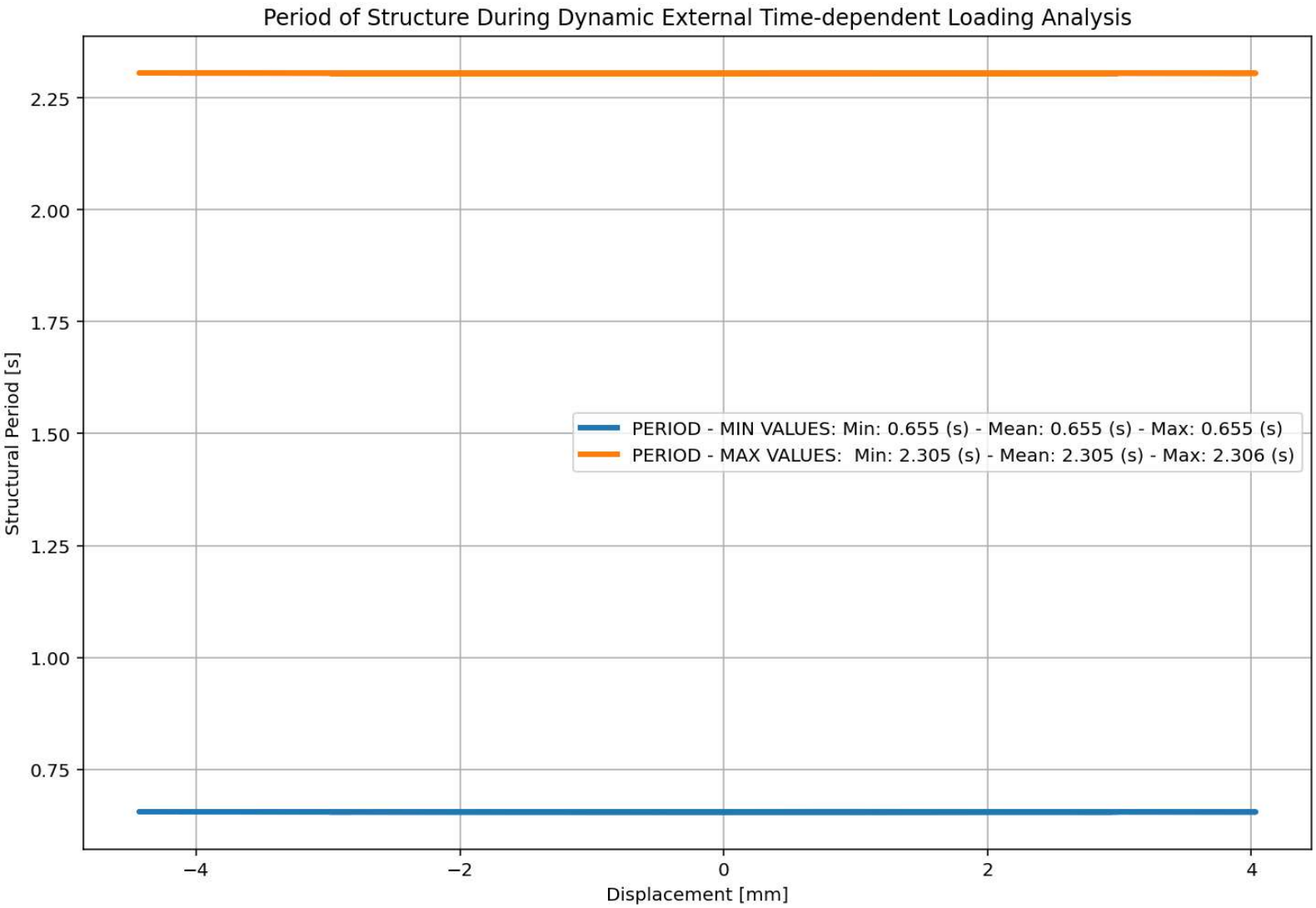

elements  force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Element Axial Force [N]
Time [s]
15000
10000
5000
0
−5000
−10000
−15000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. :  1.6840e+04
Ele. 2 - MAX. ABS. :  1.3328e+04
Ele. 3 - MAX. ABS. :  1.0647e+04
Ele. 4 - MAX. ABS. :  9.0925e+03
Ele. 5 - MAX. ABS. :  7.1785e+03
Ele. 6 - MAX. ABS. :  5.5520e+03
Ele. 7 - MAX. ABS. :  1.1623e+04
Ele. 8 - MAX. ABS. :  1.1261e+04
Ele. 9 - MAX. ABS. :  1.0580e+04
Ele. 10 - MAX. ABS. :  9.6656e+03
Ele. 11 - MAX. ABS. :  8.0806e+03
Ele. 12 - MAX. ABS. :  6.4354e+03
Ele. 13 - MAX. ABS. :  1.1579e+04
Ele. 14 - MAX. ABS. :  1.1319e+04
Ele. 15 - MAX. ABS. :  1.0797e+04
Ele. 16 - MAX. ABS. :  9.8319e+03
Ele. 17 - MAX. ABS. :  8.2535e+03
Ele. 18 - MAX. ABS. :  6.4908e+03

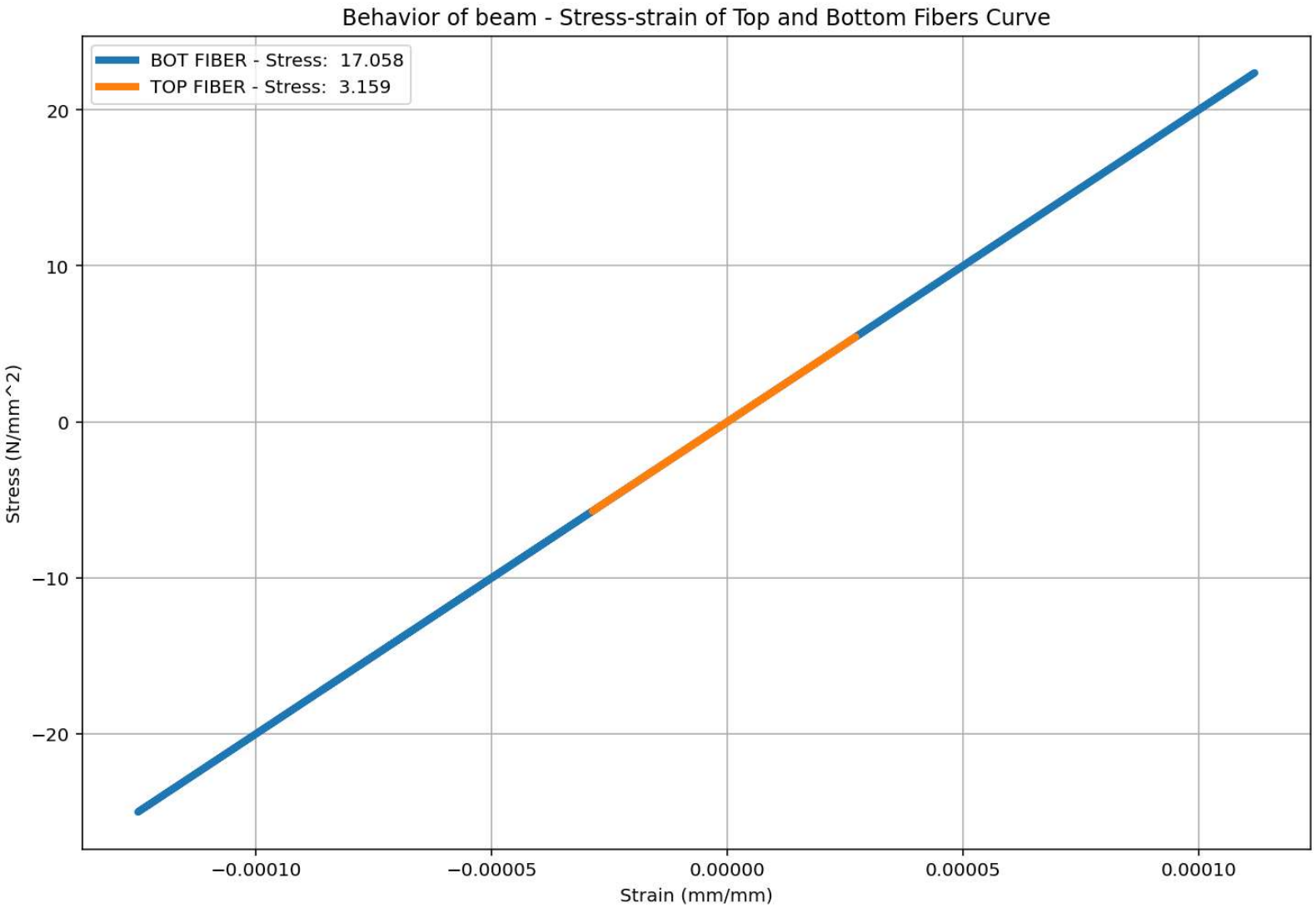

elements  force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. :  1.3001e+03
Ele. 2 - MAX. ABS. :  1.6473e+03
Ele. 3 - MAX. ABS. :  2.5507e+03
Ele. 4 - MAX. ABS. :  2.5799e+03
Ele. 5 - MAX. ABS. :  1.6474e+03
Ele. 6 - MAX. ABS. :  1.3885e+03
Ele. 7 - MAX. ABS. :  1.6457e+03
Ele. 8 - MAX. ABS. :  1.6144e+03
Ele. 9 - MAX. ABS. :  1.5710e+03
Ele. 10 - MAX. ABS. :  1.5130e+03
Ele. 11 - MAX. ABS. :  1.4359e+03
Ele. 12 - MAX. ABS. :  1.3515e+03
Ele. 13 - MAX. ABS. :  1.7237e+03
Ele. 14 - MAX. ABS. :  1.7076e+03
Ele. 15 - MAX. ABS. :  1.6745e+03
Ele. 16 - MAX. ABS. :  1.6234e+03
Ele. 17 - MAX. ABS. :  1.5546e+03
Ele. 18 - MAX. ABS. :  1.4710e+03
Element Shear Force [N]
2000
1000
0
−1000
−2000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

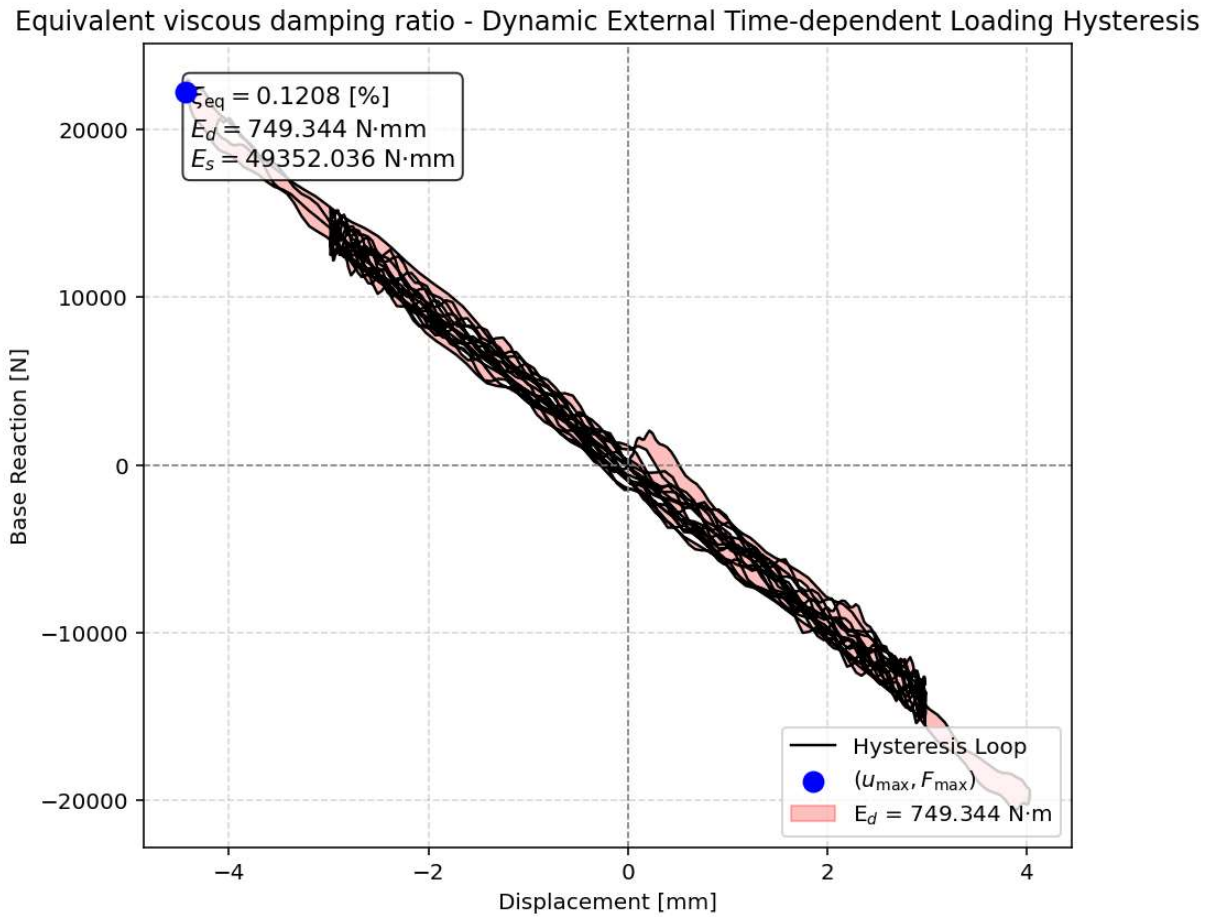

elements  force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
1e7
Element Moment Force [N.mm]
Time [s]
Ele. 1 - MAX. ABS. :  5.4821e+06
Ele. 2 - MAX. ABS. :  4.7514e+06
Ele. 3 - MAX. ABS. :  3.0120e+06
Ele. 4 - MAX. ABS. :  1.0967e+05
Ele. 5 - MAX. ABS. :  2.9491e+06
Ele. 6 - MAX. ABS. :  4.7145e+06
Ele. 7 - MAX. ABS. :  2.2938e+07
Ele. 8 - MAX. ABS. :  1.7197e+07
Ele. 9 - MAX. ABS. :  1.1754e+07
Ele. 10 - MAX. ABS. :  6.5826e+06
Ele. 11 - MAX. ABS. :  2.0619e+06
Ele. 12 - MAX. ABS. :  2.4224e+06
Ele. 13 - MAX. ABS. :  2.3003e+07
Ele. 14 - MAX. ABS. :  1.7357e+07
Ele. 15 - MAX. ABS. :  1.1772e+07
Ele. 16 - MAX. ABS. :  6.6293e+06
Ele. 17 - MAX. ABS. :  2.0129e+06
Ele. 18 - MAX. ABS. :  2.5987e+06

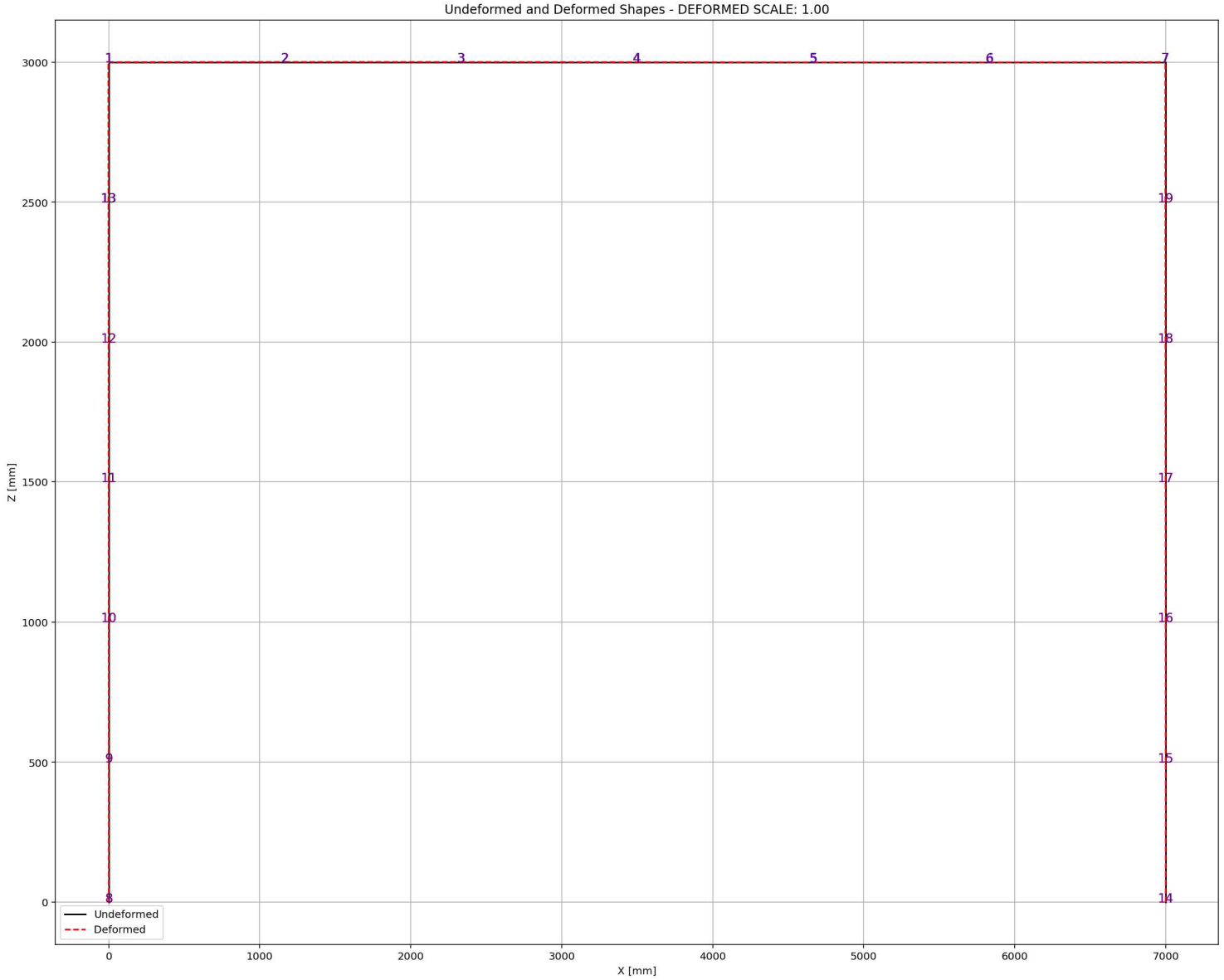

Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
4
2
0
−2
−4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 4.4334e+00
Node 2 - MAX. ABS. : 4.4455e+00
Node 3 - MAX. ABS. : 4.4542e+00
Node 4 - MAX. ABS. : 4.4607e+00
Node 5 - MAX. ABS. : 4.4642e+00
Node 6 - MAX. ABS. : 4.4645e+00
Node 7 - MAX. ABS. : 4.4627e+00
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.1507e-01
Node 10 - MAX. ABS. : 7.8161e-01
Node 11 - MAX. ABS. : 1.5870e+00
Node 12 - MAX. ABS. : 2.5253e+00
Node 13 - MAX. ABS. : 3.5011e+00
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.1621e-01
Node 16 - MAX. ABS. : 7.8798e-01
Node 17 - MAX. ABS. : 1.6015e+00
Node 18 - MAX. ABS. : 2.5476e+00
Node 19 - MAX. ABS. : 3.5276e+00


Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 4.9095e-03
Node 2 - MAX. ABS. : 1.2637e+00
Node 3 - MAX. ABS. : 1.0898e+00
Node 4 - MAX. ABS. : 3.9329e-02
Node 5 - MAX. ABS. : 1.0504e+00
Node 6 - MAX. ABS. : 1.2511e+00
Node 7 - MAX. ABS. : 3.9902e-03
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 3.7122e-04
Node 10 - MAX. ABS. : 9.6859e-04
Node 11 - MAX. ABS. : 1.8306e-03
Node 12 - MAX. ABS. : 2.8686e-03
Node 13 - MAX. ABS. : 3.9406e-03
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 3.5689e-04
Node 16 - MAX. ABS. : 7.6343e-04
Node 17 - MAX. ABS. : 1.3718e-03
Node 18 - MAX. ABS. : 2.2442e-03
Node 19 - MAX. ABS. : 3.1612e-03
Displacement [mm]
1.0
0.5
0.0
−0.5
−1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
2.25
2.00
1.75
1.50
1.25
1.00
0.75
−4
−2
0
2
4
Displacement [mm]
PERIOD - MIN VALUES: Min: 0.655 (s) - Mean: 0.655 (s) - Max: 0.655 (s)
PERIOD - MAX VALUES:  Min: 2.305 (s) - Mean: 2.305 (s) - Max: 2.306 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 17.058
TOP FIBER - Stress: 3.159
Stress (N/mm^2)
20
10
0
−10
−20
−0.00010
−0.00005
0.00000
0.00005
0.00010
Strain (mm/mm)


Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis
$\xi_{eq}$ = 0.1208 [%]
$E_d$ = 749.344 N·mm
$E_s$ = 49352.036 N·mm
Base Reaction [N]
Displacement [mm]
20000
10000
0
−10000
−20000
−4
−2
0
2
4
Hysteresis Loop
($u_{max}$, $F_{max}$)
$E_d$ = 749.344 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

# FREE-VIBRATION ANALYSIS

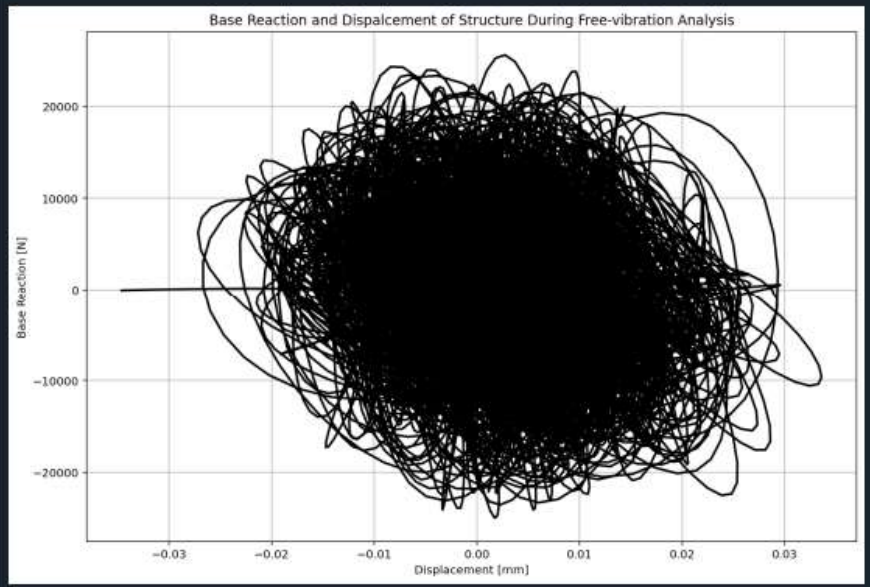
Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

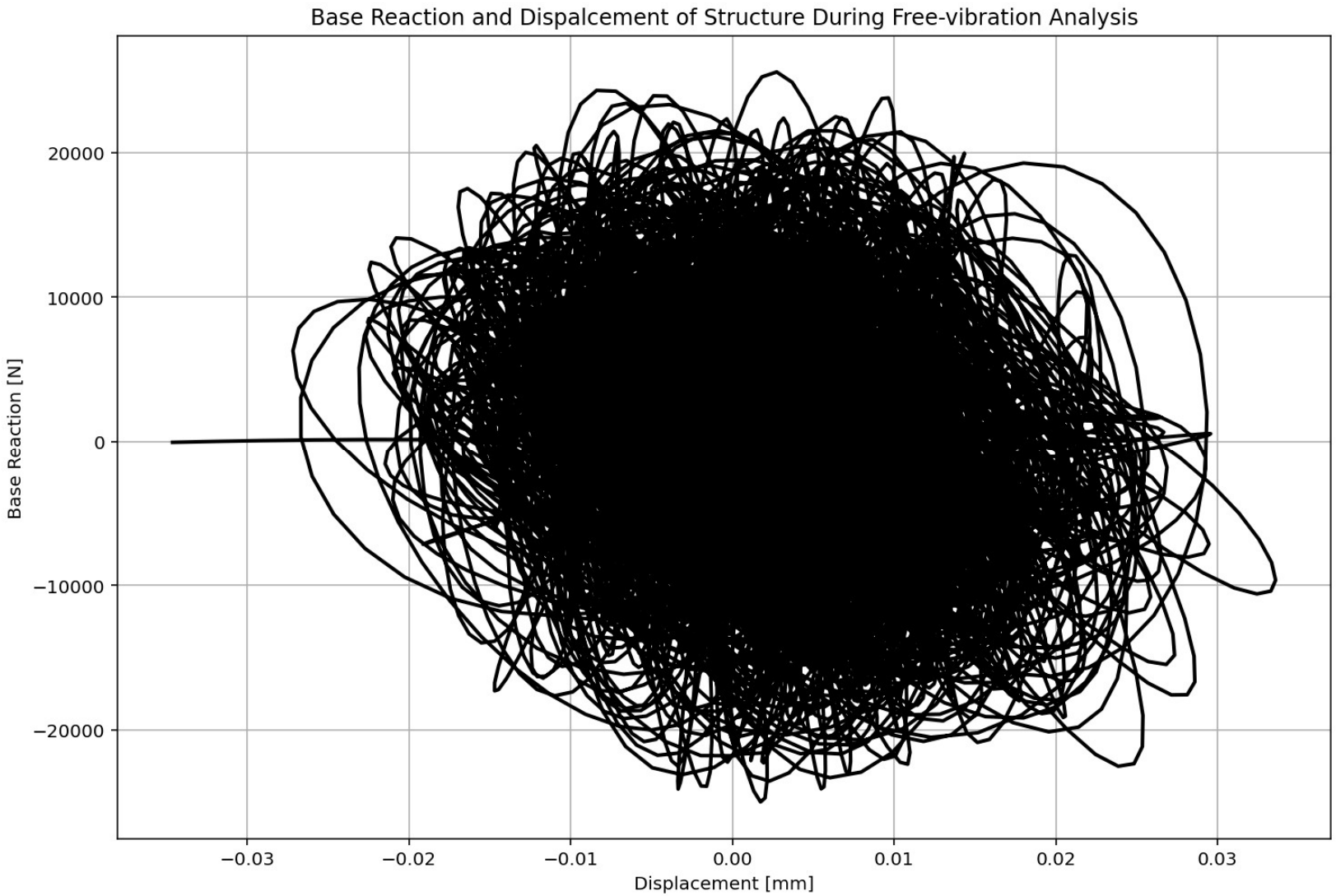
P(t)_STEEL_FRAME_8_ANA.py

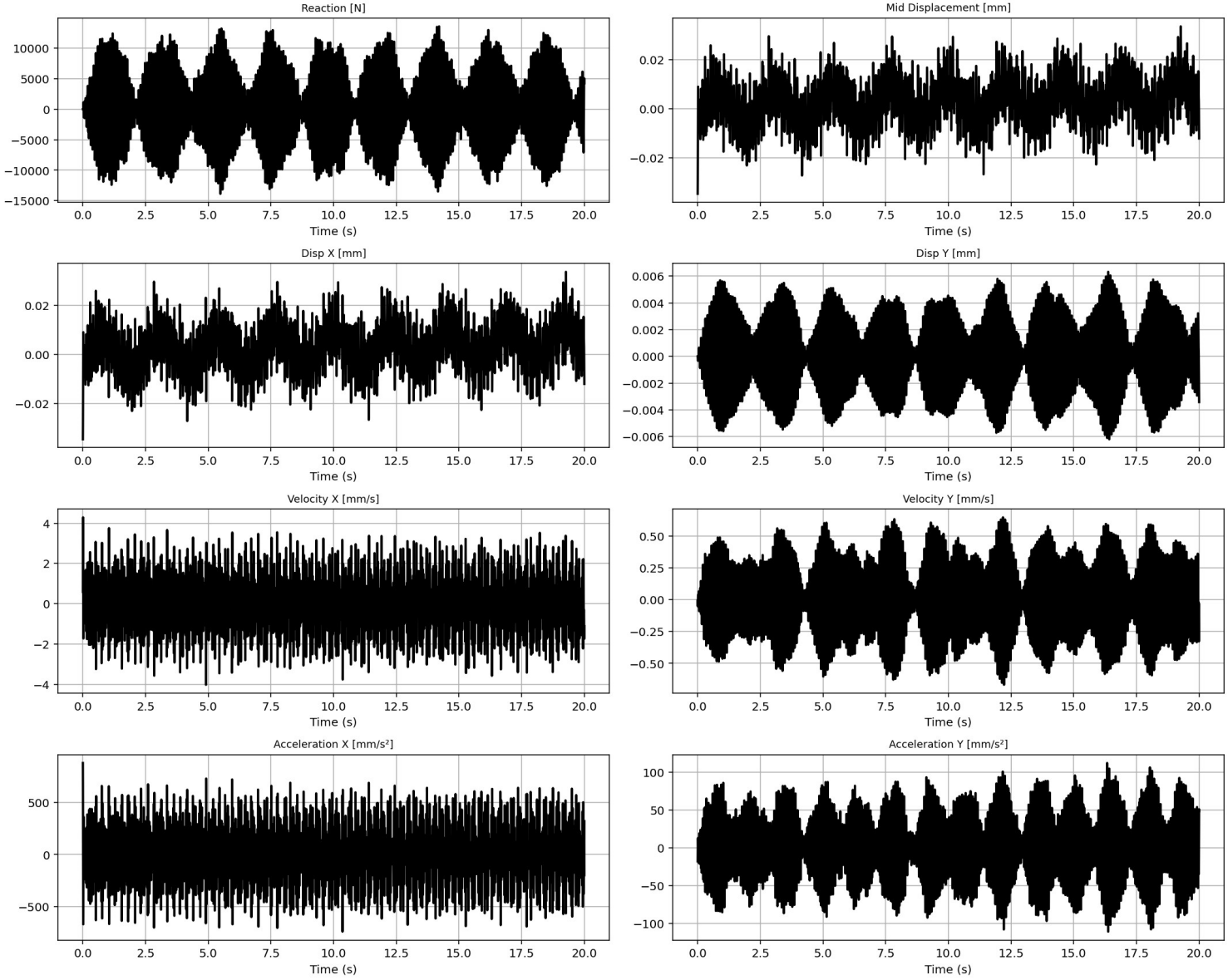
```
1439    # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1440    #ELE_TYPE = 'elasticBeamColumn'      # MATERIAL LINEARITY
1441    ELE_TYPE = 'nonlinearBeamColumn'     # MATERIAL AND GEOMETRIC NONLINEARITIE
1442    MAT_TYPE = 'INELASTIC'               # 'ELASTIC' OR 'INELASTIC'
1443    ANAL_TYPE = 'FREE-VIBRATION'
1444    BRACE = False                         # True: WITH BRACE - False: WITHOUT B
1445
1446    DATA = STEEL_FRAME(LENGTH, HEIGHT, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
1447
1448    (time_FV, reactionX_FV, reactionY_FV, disp_mid_FV,
1449    ele_axialforce_FV, ele_shearforce_FV, ele_momentforce_FV,
1450    node_displacementsX_FV, node_displacementsY_FV,
1451    dispX_FV, dispY_FV,
1452    veloX_FV, veloY_FV,
1453    accX_FV, accY_FV,
1454    stiffness_FV, PERIOD_FV, damping_ratio_FV,
1455    PERIOD_MIN_FV, PERIOD_MAX_FV,
1456    STRESSb_FV, STRAINb_FV, STRESSt_FV, STRAINt_FV, CURVATURE_FV) = DATA
1457
1458    XDATA = disp_mid_FV
1459    YDATA = reactionX_FV
1460    XLABEL = 'Displacement [mm]'
1461    YLABEL = 'Base Reaction [N]'
1462    TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration
1463    COLOR = 'black'
1464    SEMILOGY = False
1465    PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1466
1467    PLOT_TIME_HISTORY(time_FV, reactionY_FV, disp_mid_FV,
1468                      dispX_FV, dispY_FV,
1469                      veloX_FV, veloY_FV,
1470                      accX_FV, accY_FV)
1471
1472    # PLOT ELEMENTS AXIAL FORCE
```

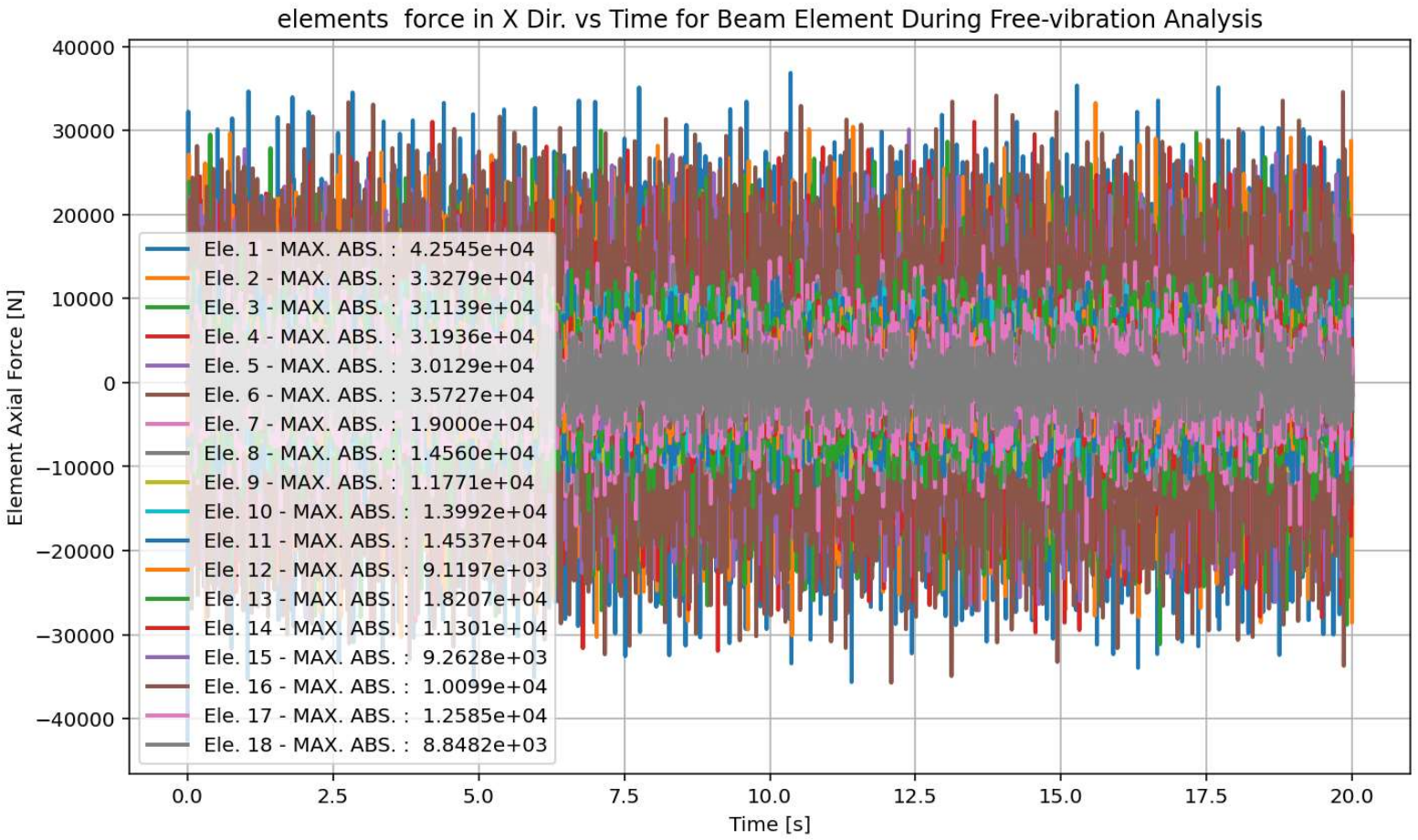


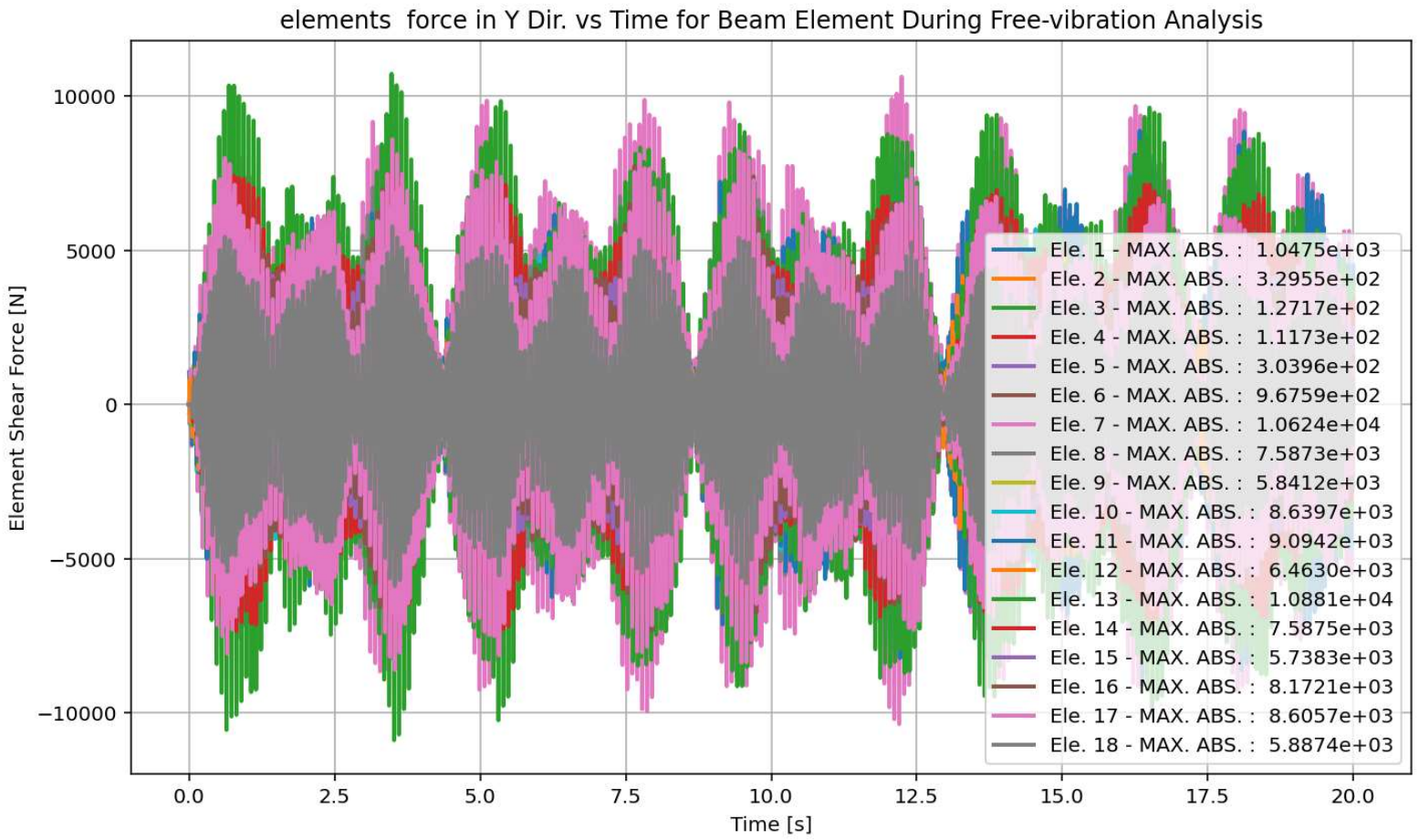

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

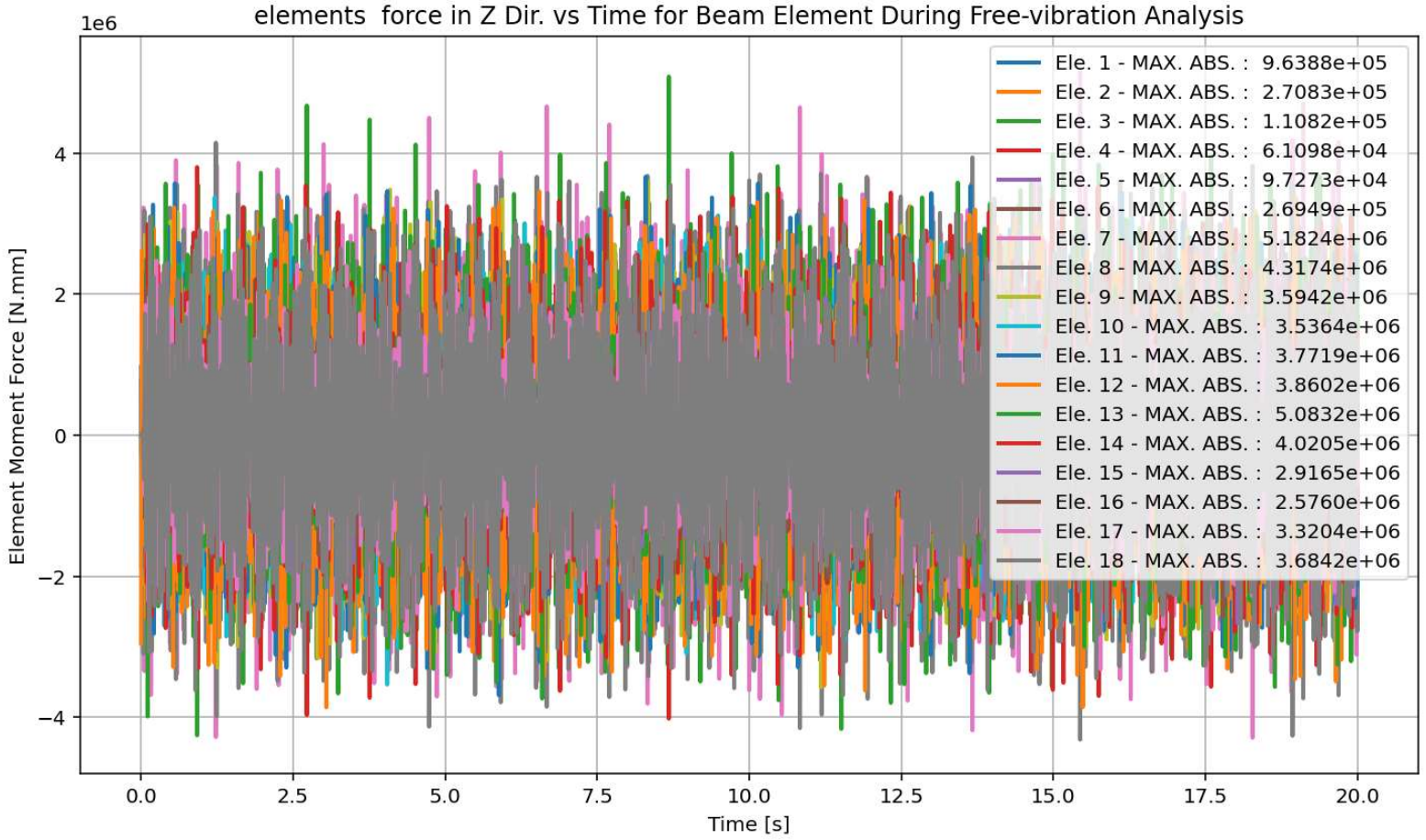
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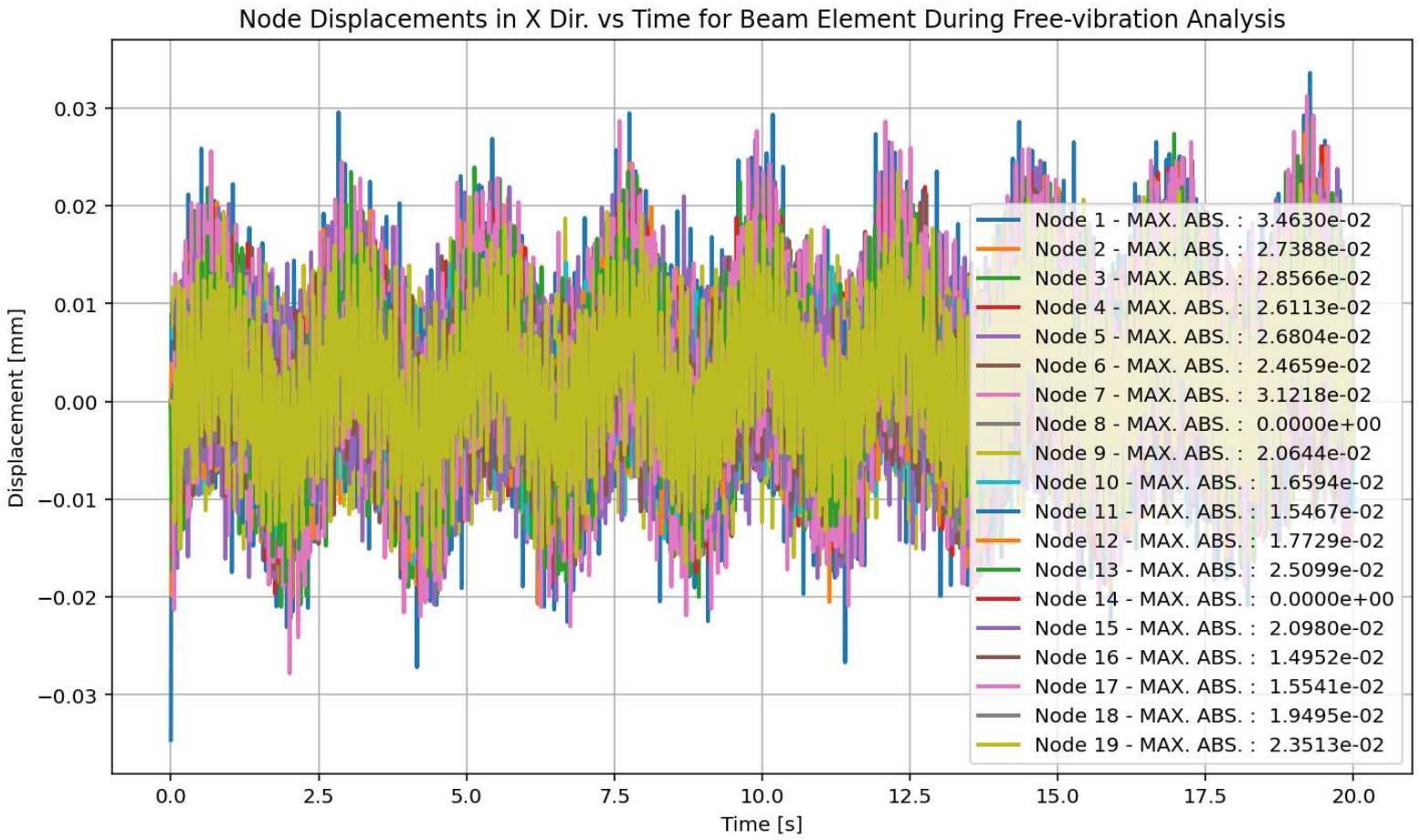

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
20000
10000
0
−10000
−20000
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [mm]

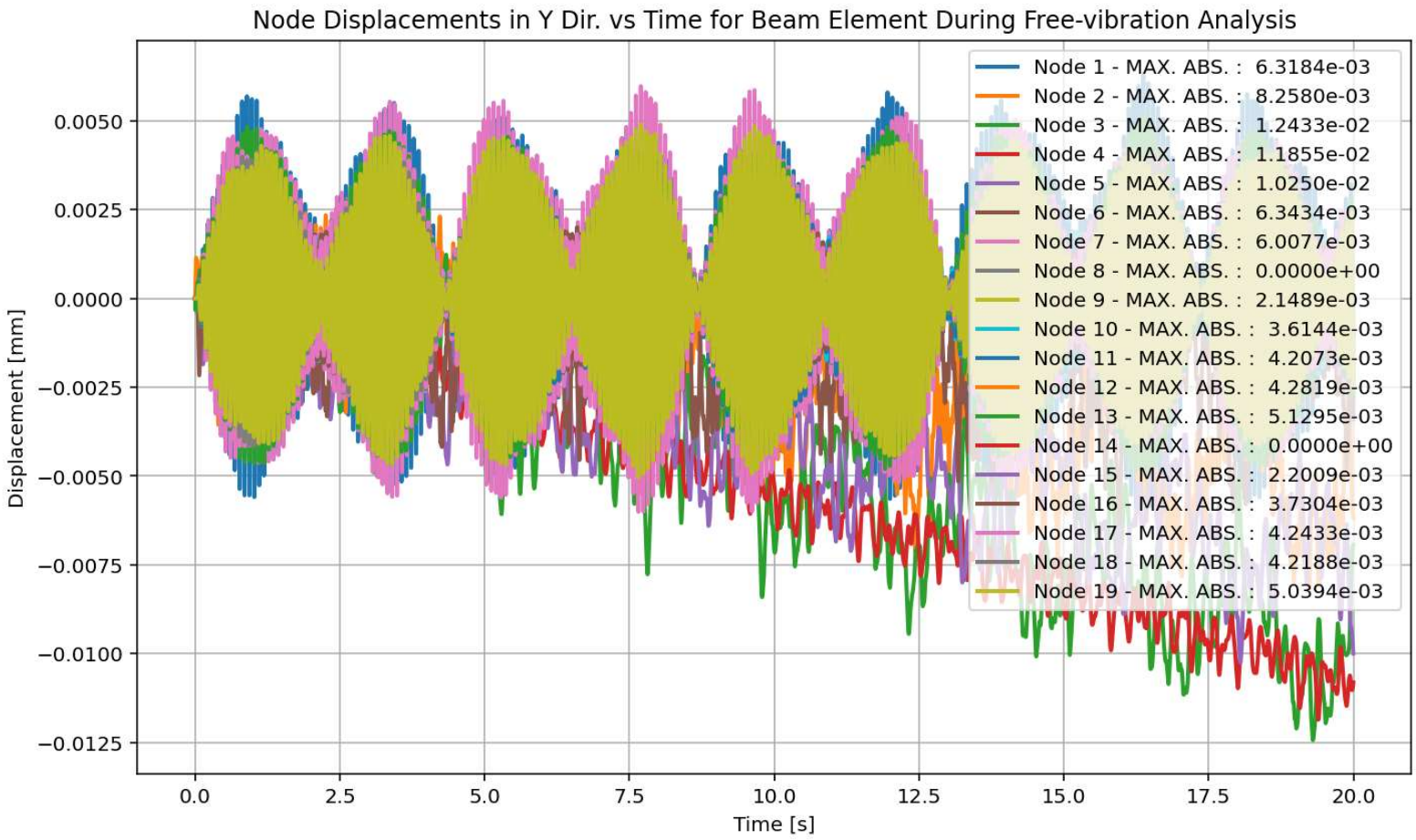

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

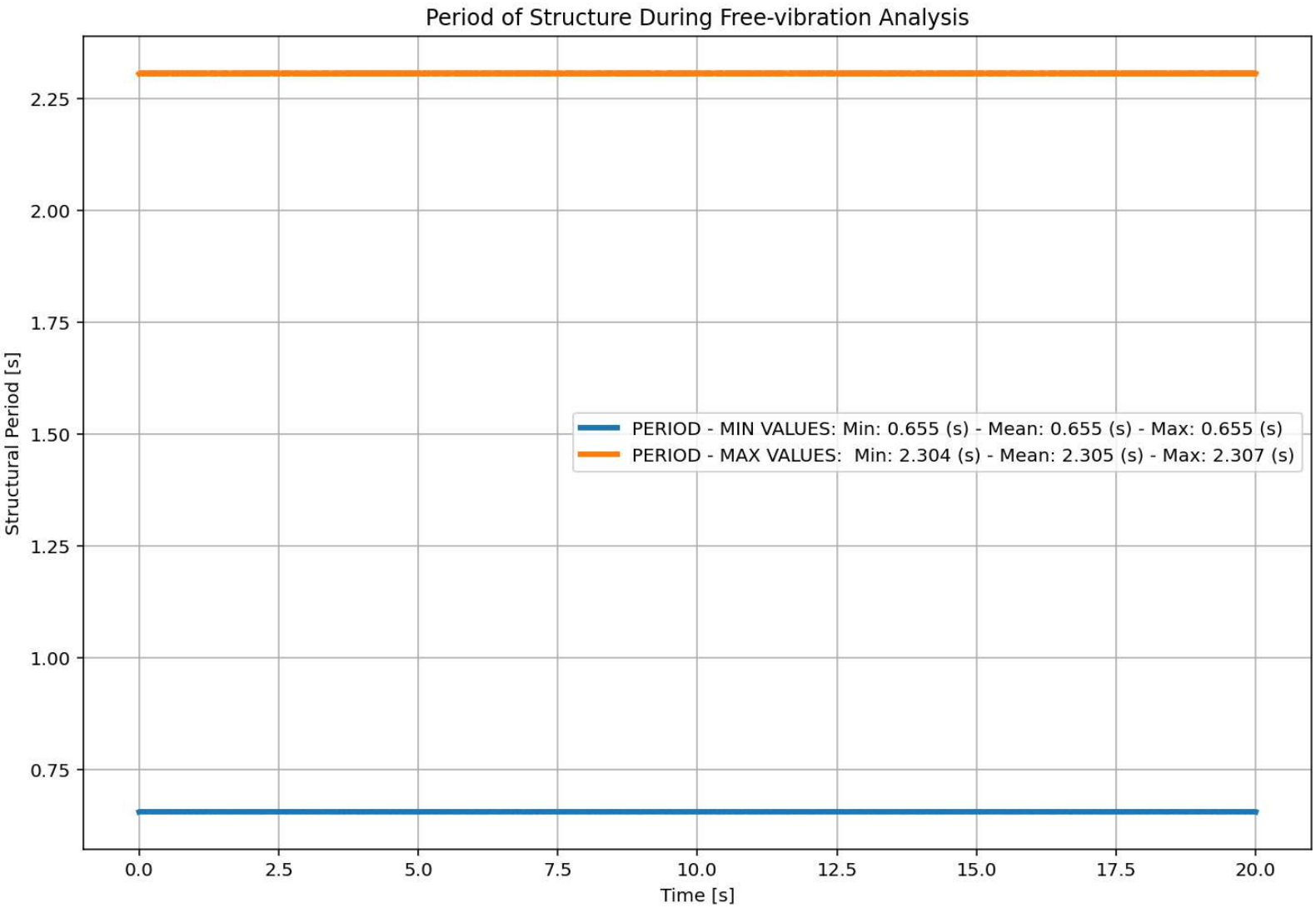

elements  force in X Dir. vs Time for Beam Element During Free-vibration Analysis
Element Axial Force [N]
Time [s]
40000
30000
20000
10000
0
−10000
−20000
−30000
−40000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. :  4.2545e+04
Ele. 2 - MAX. ABS. :  3.3279e+04
Ele. 3 - MAX. ABS. :  3.1139e+04
Ele. 4 - MAX. ABS. :  3.1936e+04
Ele. 5 - MAX. ABS. :  3.0129e+04
Ele. 6 - MAX. ABS. :  3.5727e+04
Ele. 7 - MAX. ABS. :  1.9000e+04
Ele. 8 - MAX. ABS. :  1.4560e+04
Ele. 9 - MAX. ABS. :  1.1771e+04
Ele. 10 - MAX. ABS. :  1.3992e+04
Ele. 11 - MAX. ABS. :  1.4537e+04
Ele. 12 - MAX. ABS. :  9.1197e+03
Ele. 13 - MAX. ABS. :  1.8207e+04
Ele. 14 - MAX. ABS. :  1.1301e+04
Ele. 15 - MAX. ABS. :  9.2628e+03
Ele. 16 - MAX. ABS. :  1.0099e+04
Ele. 17 - MAX. ABS. :  1.2585e+04
Ele. 18 - MAX. ABS. :  8.8482e+03

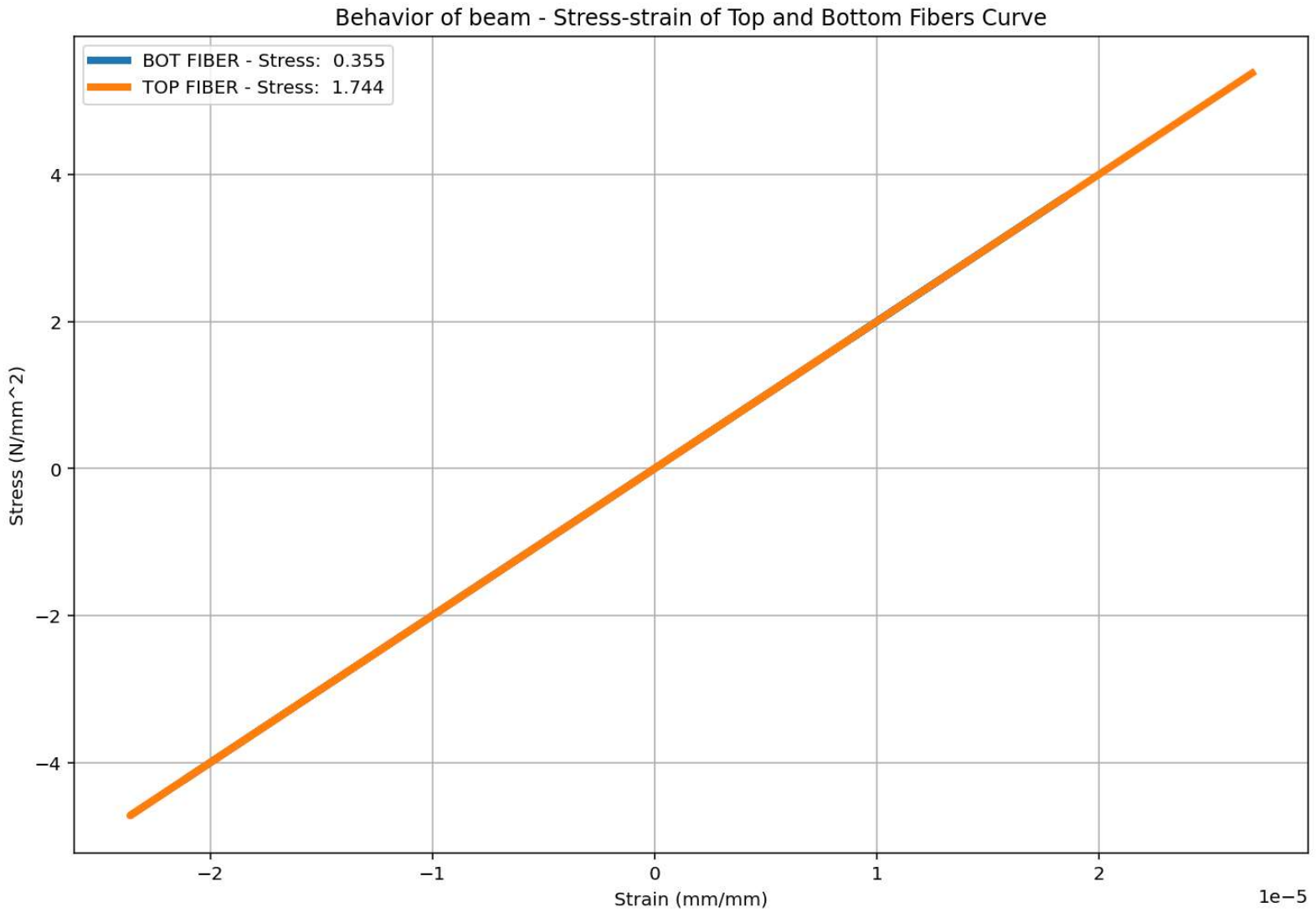

elements  force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Element Shear Force [N]
Time [s]
10000
5000
0
−5000
−10000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. :  1.0475e+03
Ele. 2 - MAX. ABS. :  3.2955e+02
Ele. 3 - MAX. ABS. :  1.2717e+02
Ele. 4 - MAX. ABS. :  1.1173e+02
Ele. 5 - MAX. ABS. :  3.0396e+02
Ele. 6 - MAX. ABS. :  9.6759e+02
Ele. 7 - MAX. ABS. :  1.0624e+04
Ele. 8 - MAX. ABS. :  7.5873e+03
Ele. 9 - MAX. ABS. :  5.8412e+03
Ele. 10 - MAX. ABS. :  8.6397e+03
Ele. 11 - MAX. ABS. :  9.0942e+03
Ele. 12 - MAX. ABS. :  6.4630e+03
Ele. 13 - MAX. ABS. :  1.0881e+04
Ele. 14 - MAX. ABS. :  7.5875e+03
Ele. 15 - MAX. ABS. :  5.7383e+03
Ele. 16 - MAX. ABS. :  8.1721e+03
Ele. 17 - MAX. ABS. :  8.6057e+03
Ele. 18 - MAX. ABS. :  5.8874e+03

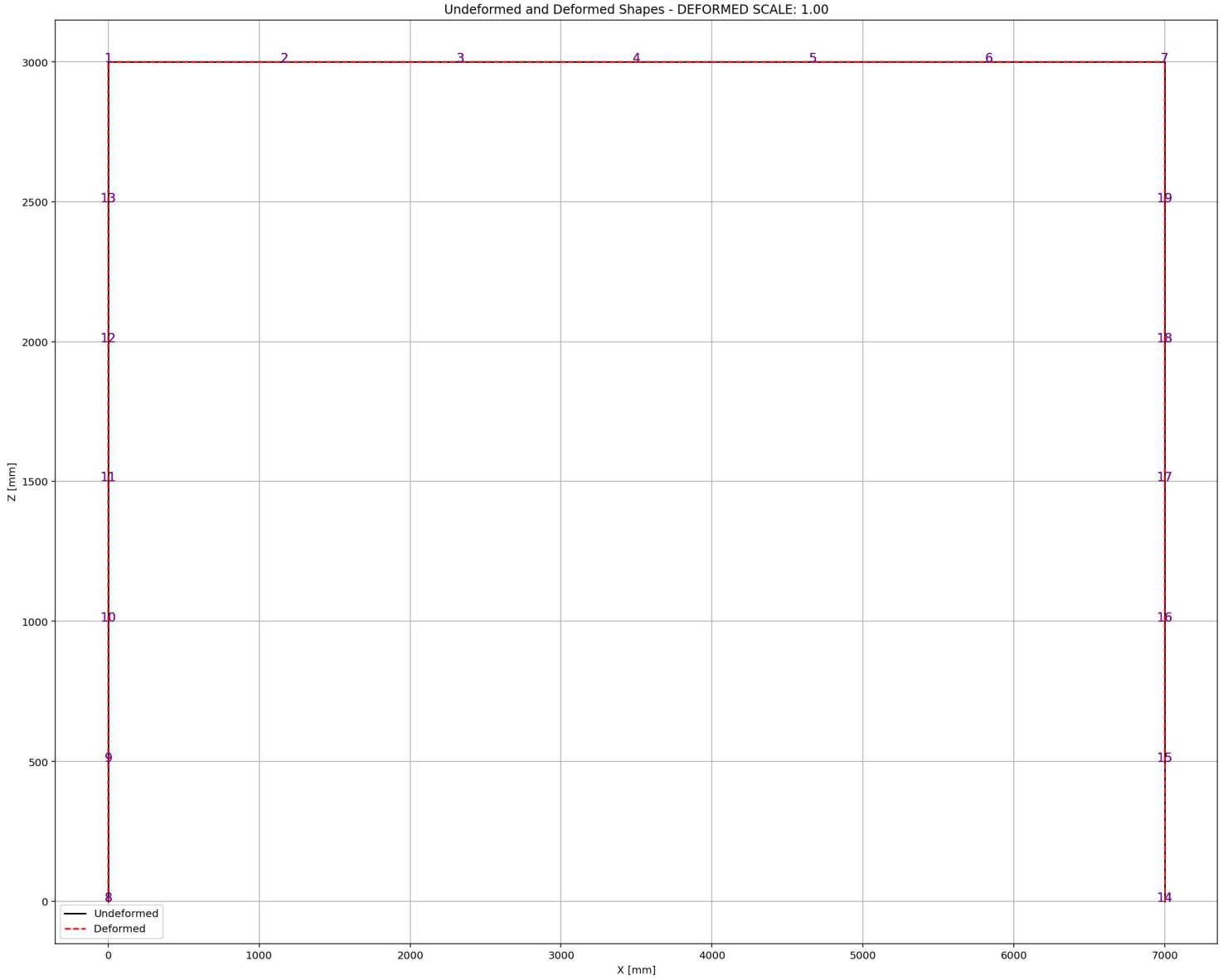

elements  force in Z Dir. vs Time for Beam Element During Free-vibration Analysis
1e6
Element Moment Force [N.mm]
Time [s]
Ele. 1 - MAX. ABS. :  9.6388e+05
Ele. 2 - MAX. ABS. :  2.7083e+05
Ele. 3 - MAX. ABS. :  1.1082e+05
Ele. 4 - MAX. ABS. :  6.1098e+04
Ele. 5 - MAX. ABS. :  9.7273e+04
Ele. 6 - MAX. ABS. :  2.6949e+05
Ele. 7 - MAX. ABS. :  5.1824e+06
Ele. 8 - MAX. ABS. :  4.3174e+06
Ele. 9 - MAX. ABS. :  3.5942e+06
Ele. 10 - MAX. ABS. :  3.5364e+06
Ele. 11 - MAX. ABS. :  3.7719e+06
Ele. 12 - MAX. ABS. :  3.8602e+06
Ele. 13 - MAX. ABS. :  5.0832e+06
Ele. 14 - MAX. ABS. :  4.0205e+06
Ele. 15 - MAX. ABS. :  2.9165e+06
Ele. 16 - MAX. ABS. :  2.5760e+06
Ele. 17 - MAX. ABS. :  3.3204e+06
Ele. 18 - MAX. ABS. :  3.6842e+06


Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 3.4630e-02
Node 2 - MAX. ABS. : 2.7388e-02
Node 3 - MAX. ABS. : 2.8566e-02
Node 4 - MAX. ABS. : 2.6113e-02
Node 5 - MAX. ABS. : 2.6804e-02
Node 6 - MAX. ABS. : 2.4659e-02
Node 7 - MAX. ABS. : 3.1218e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.0644e-02
Node 10 - MAX. ABS. : 1.6594e-02
Node 11 - MAX. ABS. : 1.5467e-02
Node 12 - MAX. ABS. : 1.7729e-02
Node 13 - MAX. ABS. : 2.5099e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.0980e-02
Node 16 - MAX. ABS. : 1.4952e-02
Node 17 - MAX. ABS. : 1.5541e-02
Node 18 - MAX. ABS. : 1.9495e-02
Node 19 - MAX. ABS. : 2.3513e-02


Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Displacement [mm]
Time [s]
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
−0.0125
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 6.3184e-03
Node 2 - MAX. ABS. : 8.2580e-03
Node 3 - MAX. ABS. : 1.2433e-02
Node 4 - MAX. ABS. : 1.1855e-02
Node 5 - MAX. ABS. : 1.0250e-02
Node 6 - MAX. ABS. : 6.3434e-03
Node 7 - MAX. ABS. : 6.0077e-03
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 2.1489e-03
Node 10 - MAX. ABS. : 3.6144e-03
Node 11 - MAX. ABS. : 4.2073e-03
Node 12 - MAX. ABS. : 4.2819e-03
Node 13 - MAX. ABS. : 5.1295e-03
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 2.2009e-03
Node 16 - MAX. ABS. : 3.7304e-03
Node 17 - MAX. ABS. : 4.2433e-03
Node 18 - MAX. ABS. : 4.2188e-03
Node 19 - MAX. ABS. : 5.0394e-03


Period of Structure During Free-vibration Analysis
Structural Period [s]
Time [s]
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
PERIOD - MIN VALUES: Min: 0.655 (s) - Mean: 0.655 (s) - Max: 0.655 (s)
PERIOD - MAX VALUES: Min: 2.304 (s) - Mean: 2.305 (s) - Max: 2.307 (s)


Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.355
TOP FIBER - Stress: 1.744
Stress (N/mm^2)
4
2
0
−2
−4
−2
−1
0
1
2
Strain (mm/mm)
1e−5


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000

# SEISMIC ANALYSIS

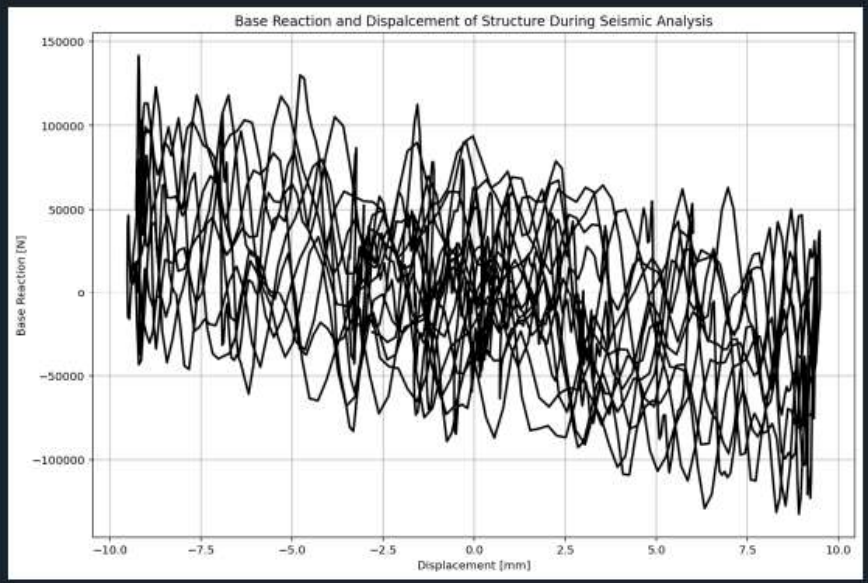
Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...EX\09_STEEL_FRAME_8_ANA\P(t)_STEEL_FRAME_8_ANA.py

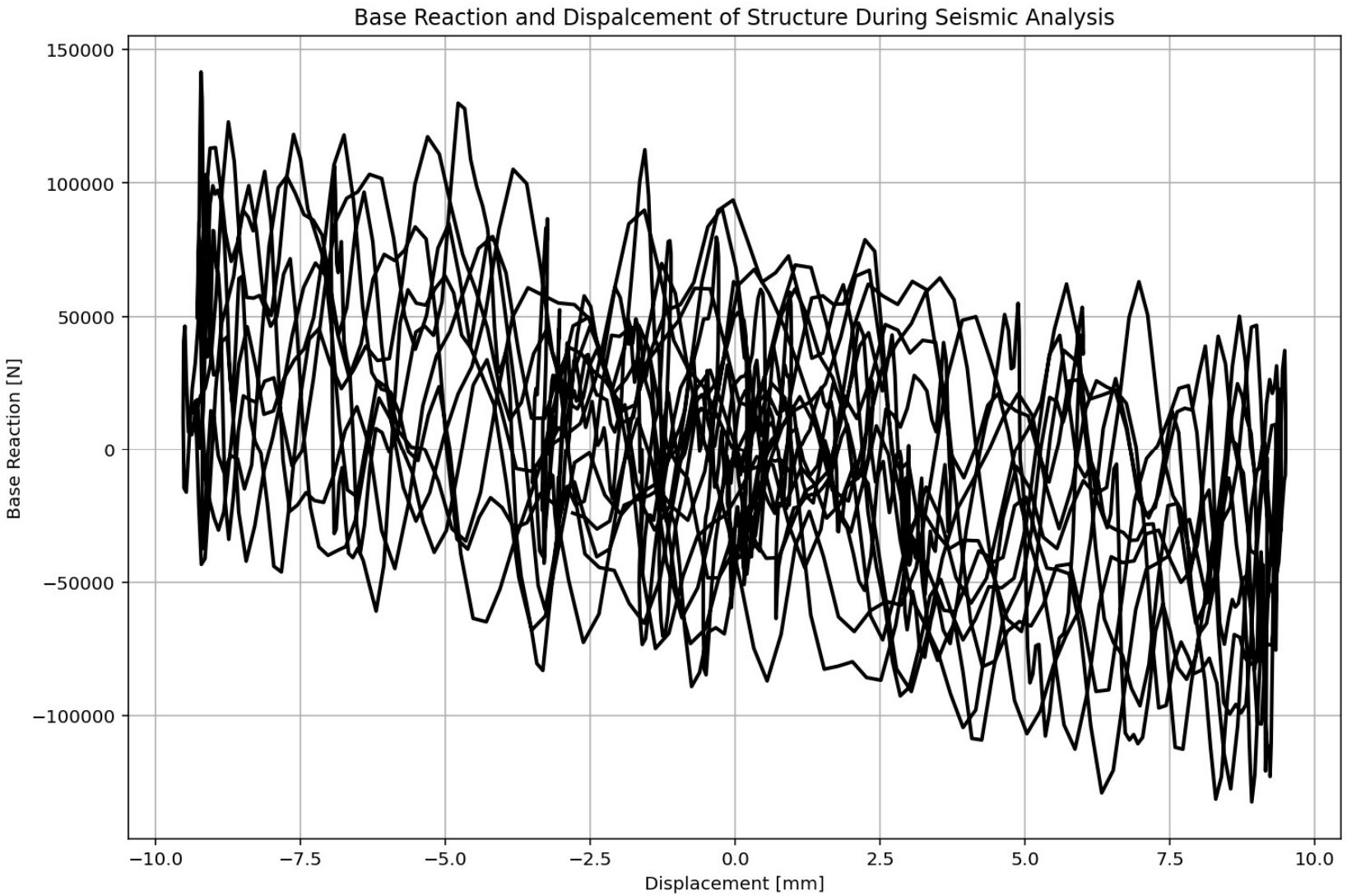
P(t)_STEEL_FRAME_8_ANA.py

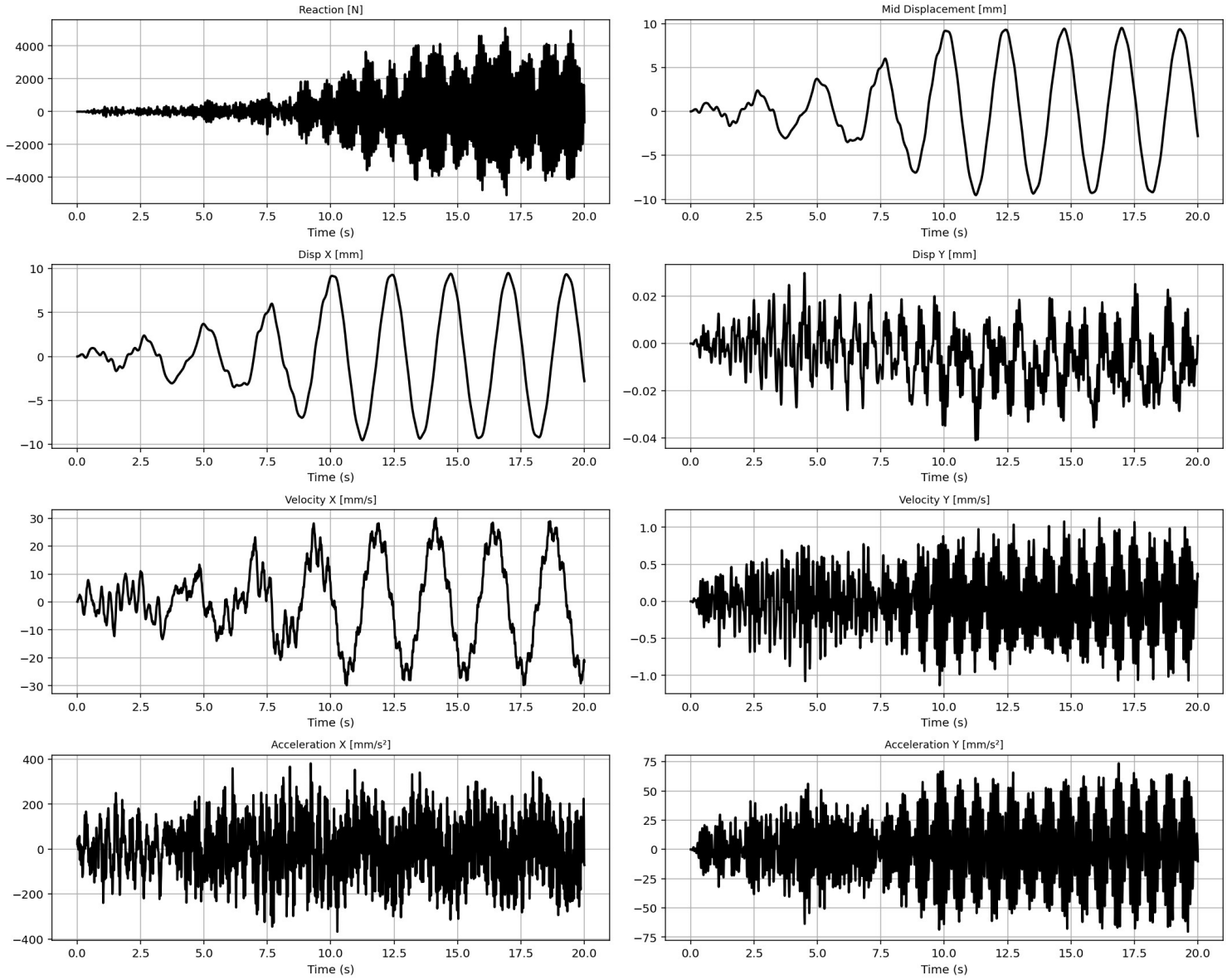
```
944     if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
945         #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
946         duration = 20.0                     # [s] Analysis duration
947         dt = 0.01                           # [s] Time step
948         GMfact = 9810                       # [mm/s²]standard acceleration o
949         SSF_X = 0.50                        # Seismic Acceleration Scale Fac
950         SSF_Y = 0.50                        # Seismic Acceleration Scale Fac
951         iv0_X = 0.0005                      # [mm/s] Initial velocity applie
952         iv0_Y = 0.0005                      # [mm/s] Initial velocity applie
953         st_iv0 = 0.0                        # [s] Initial velocity applied s
954         SEI = 'X'                           # Seismic Direction
955         DR = 0.03                           # Damping ratio
956
957         # Define time series for input motion (Acceleration time history)
958         if SEI == 'X':
959             SEISMIC_TAG_01 = 100
960             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
961             # Define load patterns
962             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
963             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
964         if SEI == 'Y':
965             SEISMIC_TAG_02 = 200
966             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
967             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
968         if SEI == 'XY':
969             SEISMIC_TAG_01 = 100
970             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
971             # Define load patterns
972             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
973             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
974             SEISMIC_TAG_02 = 200
975             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
976             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
977         print('Seismic Defined Done.')
```

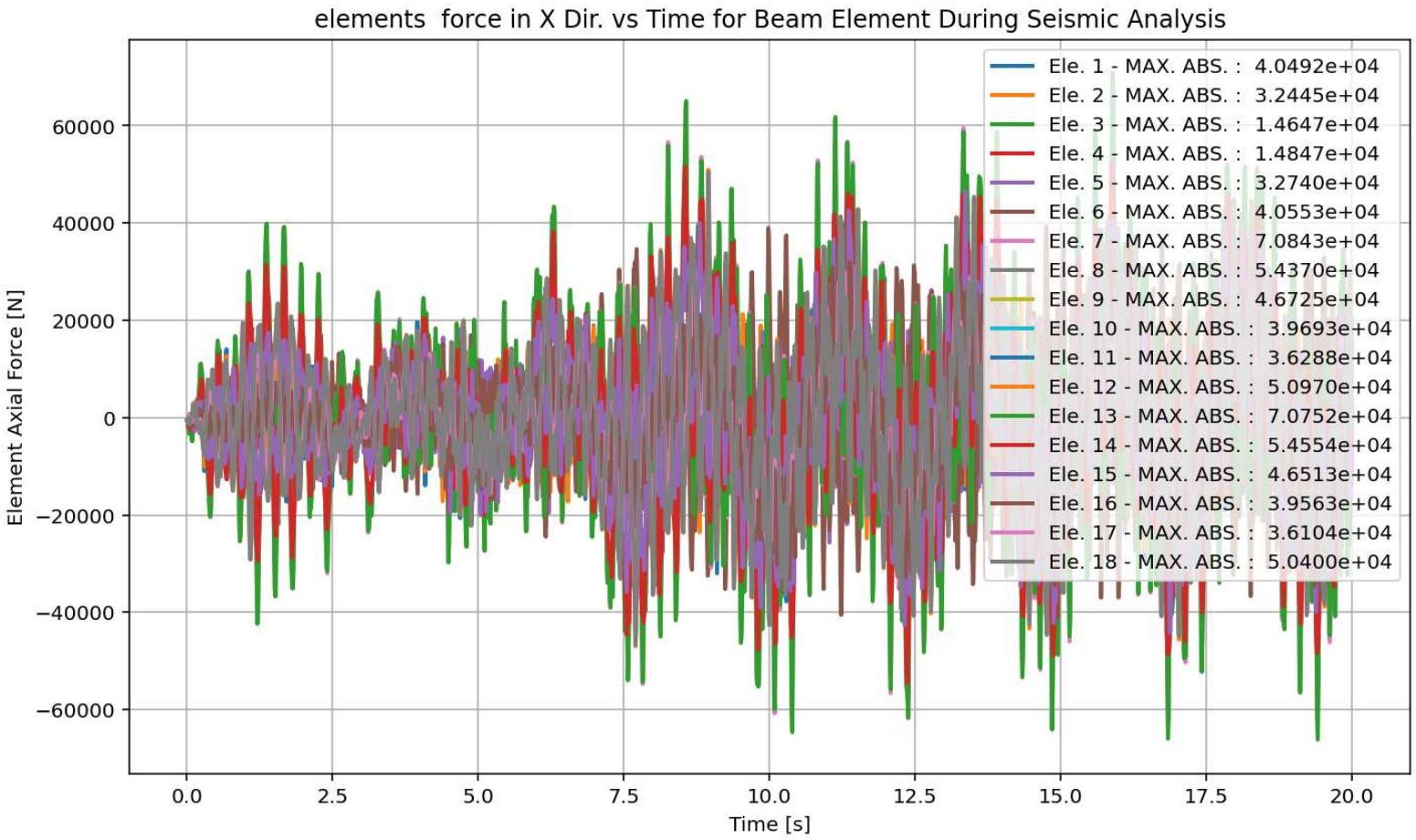


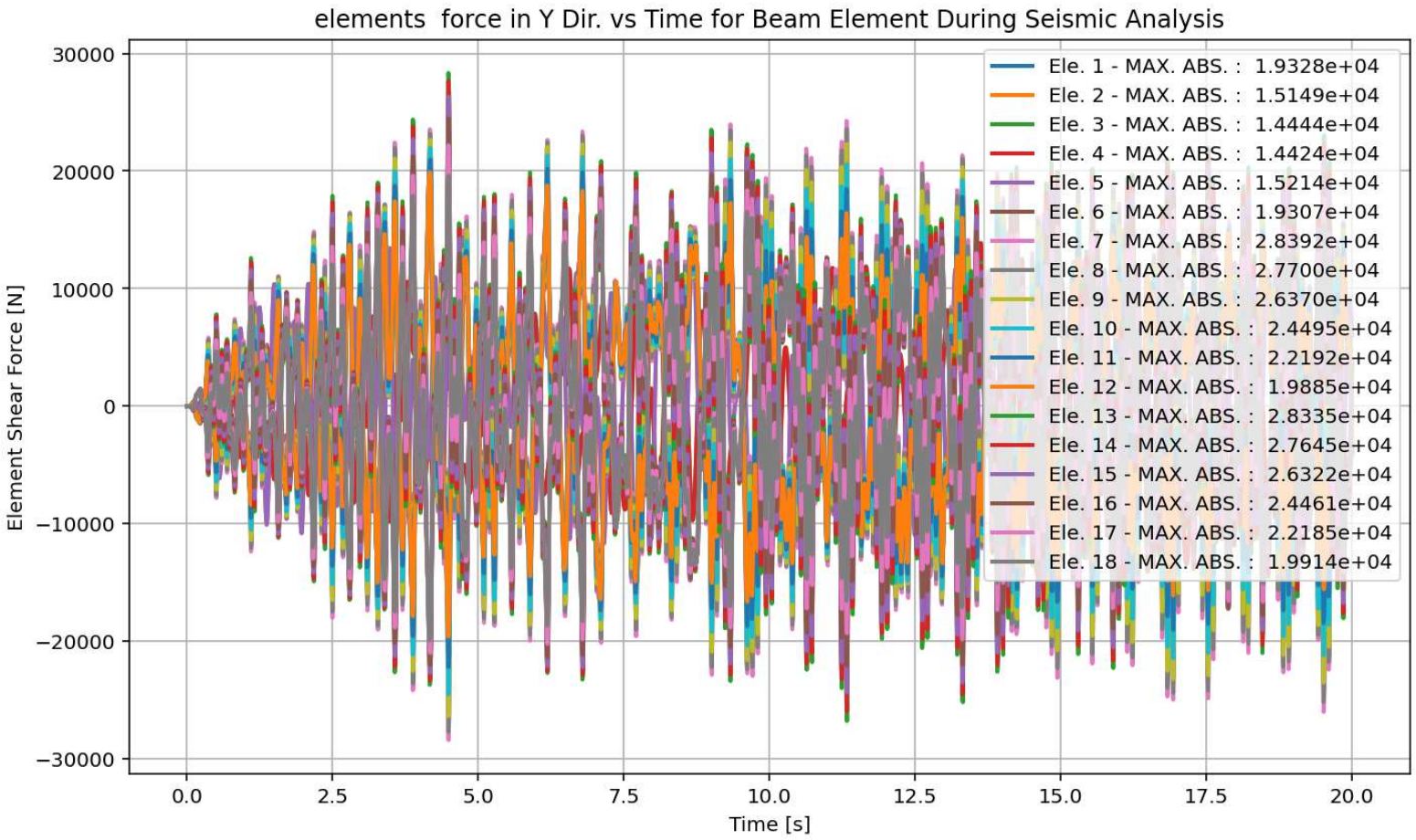

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

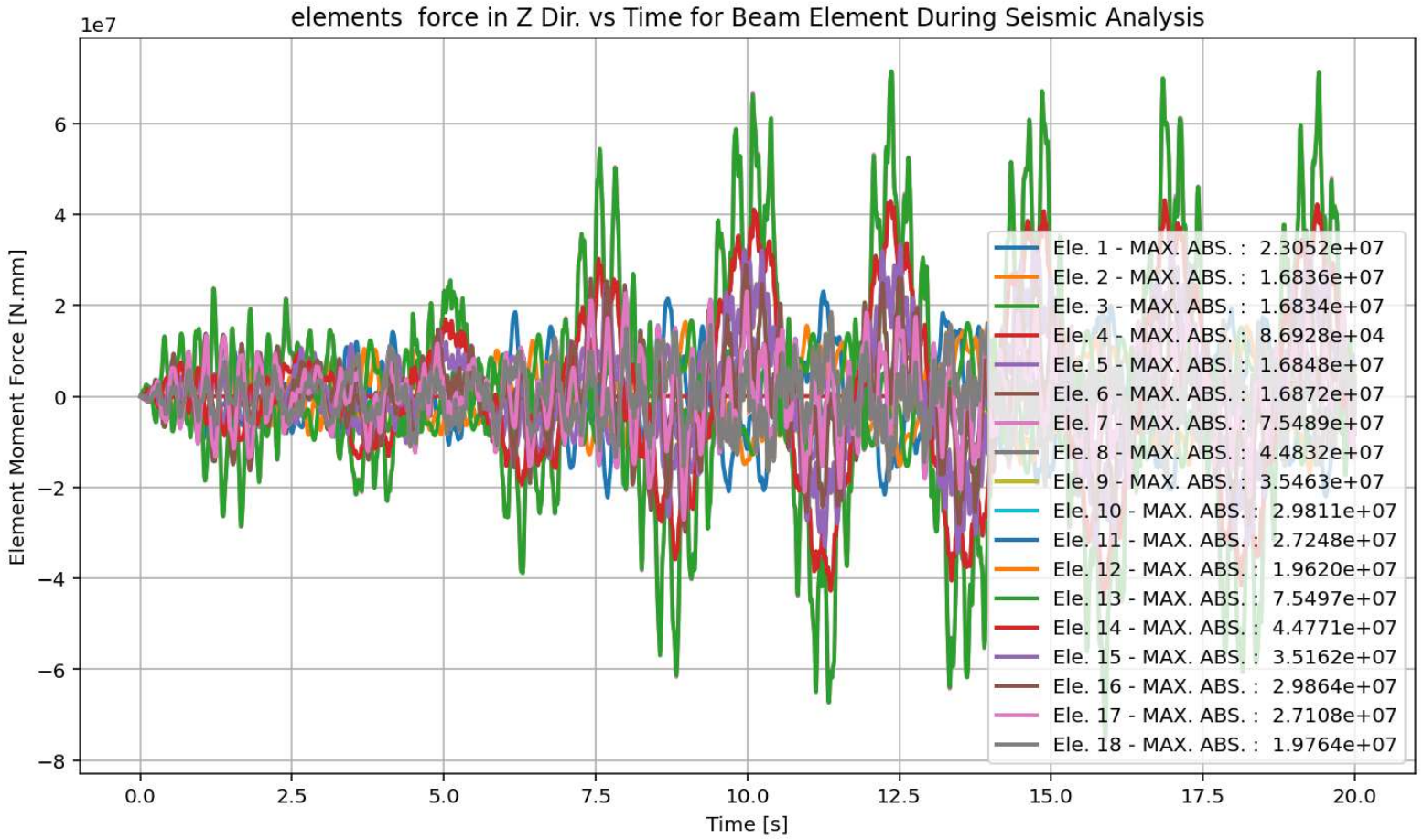
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 943, Col 5 UTF-8 CRLF RW Mem 47%

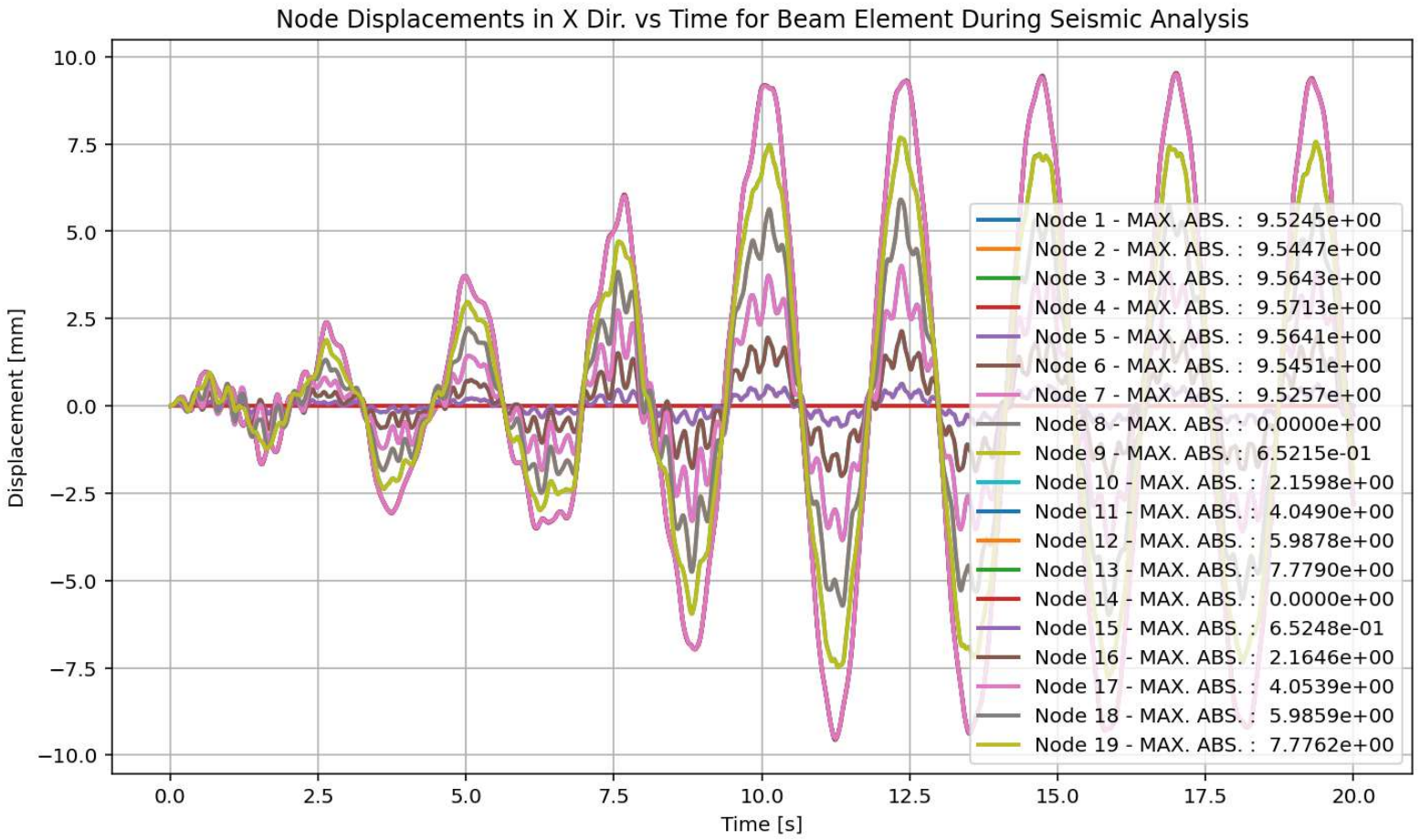

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
150000
100000
50000
0
−50000
−100000
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
Displacement [mm]

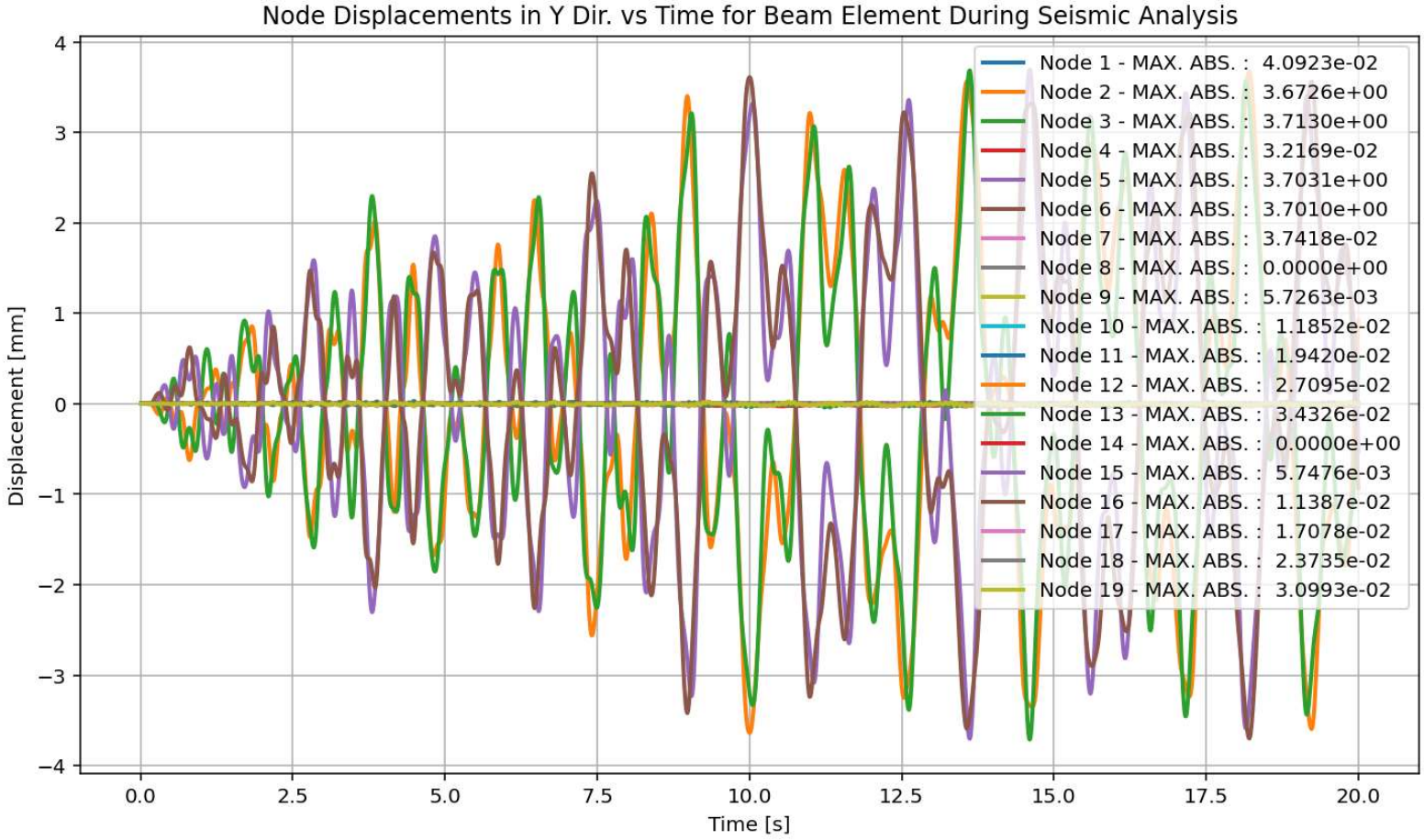

Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

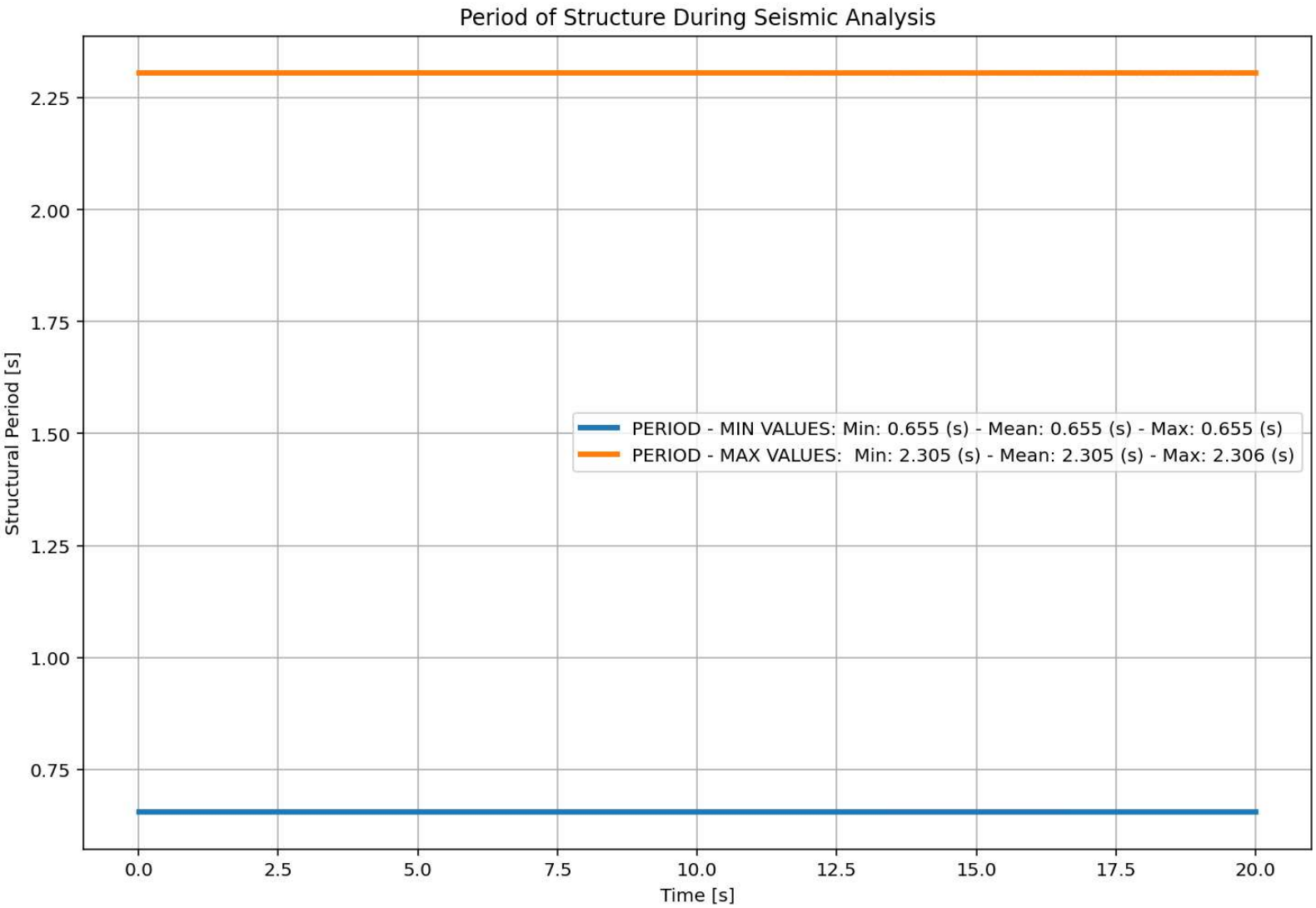

elements  force in X Dir. vs Time for Beam Element During Seismic Analysis
Element Axial Force [N]
Time [s]
60000
40000
20000
0
−20000
−40000
−60000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. :  4.0492e+04
Ele. 2 - MAX. ABS. :  3.2445e+04
Ele. 3 - MAX. ABS. :  1.4647e+04
Ele. 4 - MAX. ABS. :  1.4847e+04
Ele. 5 - MAX. ABS. :  3.2740e+04
Ele. 6 - MAX. ABS. :  4.0553e+04
Ele. 7 - MAX. ABS. :  7.0843e+04
Ele. 8 - MAX. ABS. :  5.4370e+04
Ele. 9 - MAX. ABS. :  4.6725e+04
Ele. 10 - MAX. ABS. :  3.9693e+04
Ele. 11 - MAX. ABS. :  3.6288e+04
Ele. 12 - MAX. ABS. :  5.0970e+04
Ele. 13 - MAX. ABS. :  7.0752e+04
Ele. 14 - MAX. ABS. :  5.4554e+04
Ele. 15 - MAX. ABS. :  4.6513e+04
Ele. 16 - MAX. ABS. :  3.9563e+04
Ele. 17 - MAX. ABS. :  3.6104e+04
Ele. 18 - MAX. ABS. :  5.0400e+04

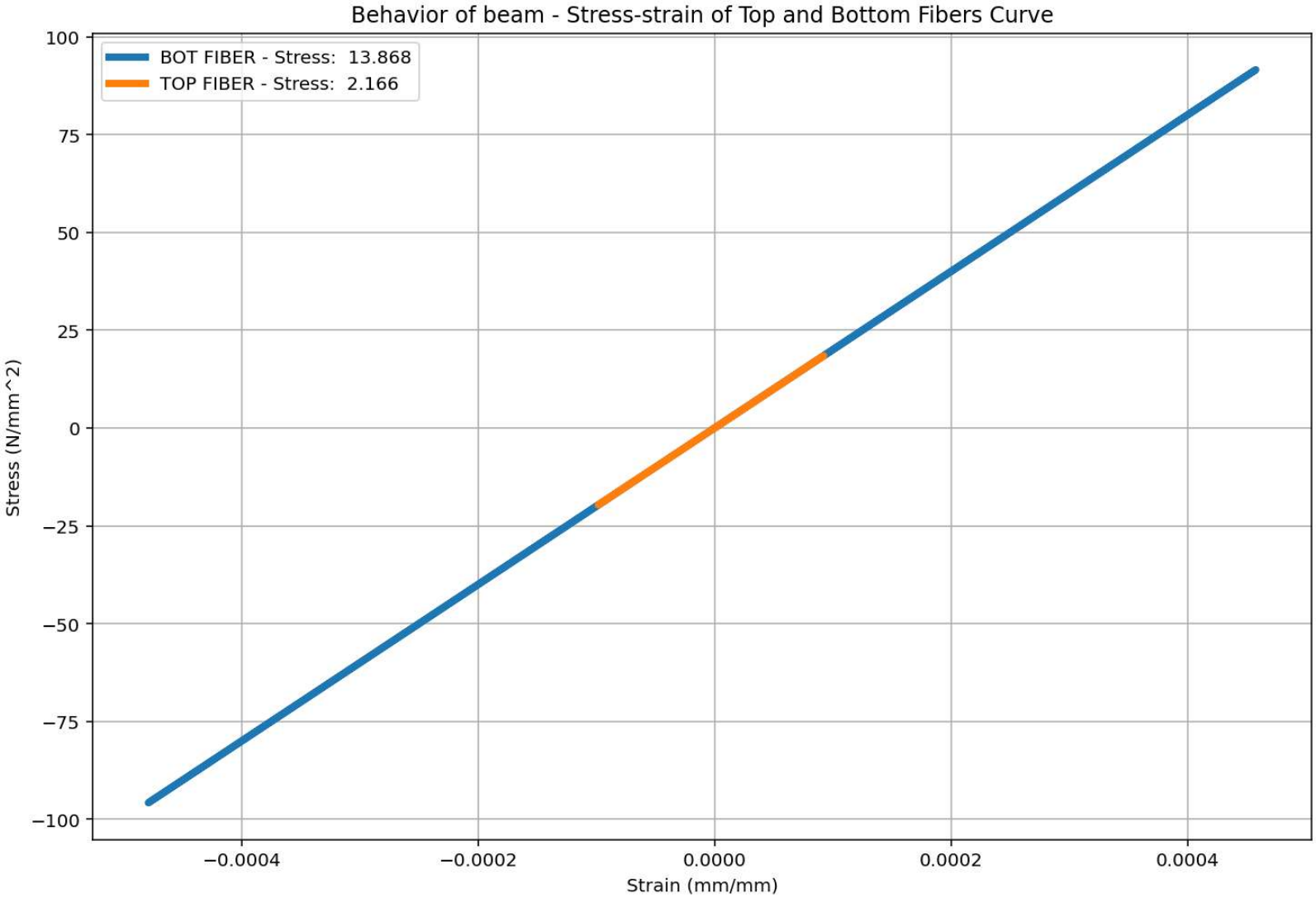

elements  force in Y Dir. vs Time for Beam Element During Seismic Analysis
Element Shear Force [N]
Time [s]
30000
20000
10000
0
−10000
−20000
−30000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. :  1.9328e+04
Ele. 2 - MAX. ABS. :  1.5149e+04
Ele. 3 - MAX. ABS. :  1.4444e+04
Ele. 4 - MAX. ABS. :  1.4424e+04
Ele. 5 - MAX. ABS. :  1.5214e+04
Ele. 6 - MAX. ABS. :  1.9307e+04
Ele. 7 - MAX. ABS. :  2.8392e+04
Ele. 8 - MAX. ABS. :  2.7700e+04
Ele. 9 - MAX. ABS. :  2.6370e+04
Ele. 10 - MAX. ABS. :  2.4495e+04
Ele. 11 - MAX. ABS. :  2.2192e+04
Ele. 12 - MAX. ABS. :  1.9885e+04
Ele. 13 - MAX. ABS. :  2.8335e+04
Ele. 14 - MAX. ABS. :  2.7645e+04
Ele. 15 - MAX. ABS. :  2.6322e+04
Ele. 16 - MAX. ABS. :  2.4461e+04
Ele. 17 - MAX. ABS. :  2.2185e+04
Ele. 18 - MAX. ABS. :  1.9914e+04

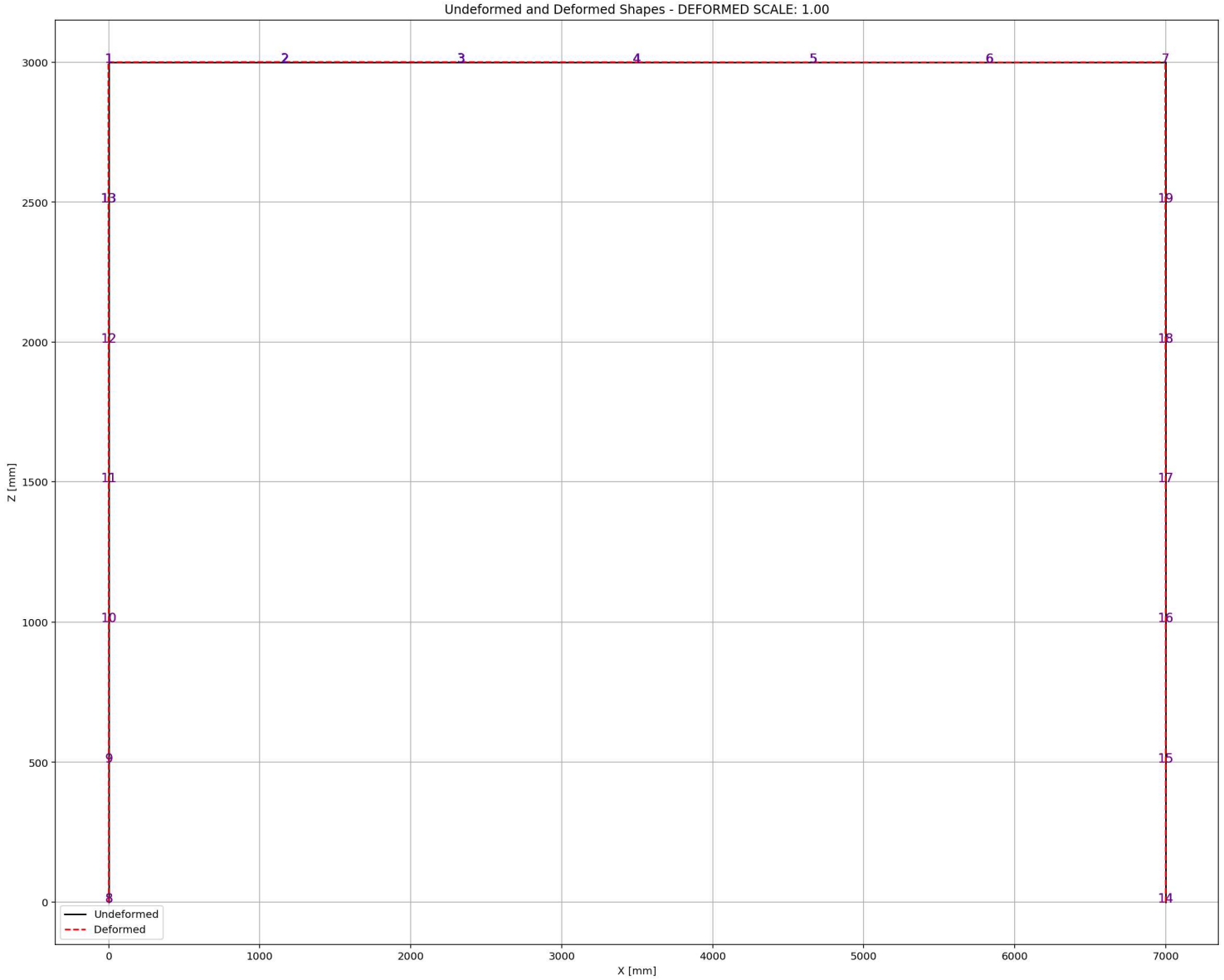

elements  force in Z Dir. vs Time for Beam Element During Seismic Analysis
1e7
Element Moment Force [N.mm]
Time [s]
Ele. 1 - MAX. ABS. :  2.3052e+07
Ele. 2 - MAX. ABS. :  1.6836e+07
Ele. 3 - MAX. ABS. :  1.6834e+07
Ele. 4 - MAX. ABS. :  8.6928e+04
Ele. 5 - MAX. ABS. :  1.6848e+07
Ele. 6 - MAX. ABS. :  1.6872e+07
Ele. 7 - MAX. ABS. :  7.5489e+07
Ele. 8 - MAX. ABS. :  4.4832e+07
Ele. 9 - MAX. ABS. :  3.5463e+07
Ele. 10 - MAX. ABS. :  2.9811e+07
Ele. 11 - MAX. ABS. :  2.7248e+07
Ele. 12 - MAX. ABS. :  1.9620e+07
Ele. 13 - MAX. ABS. :  7.5497e+07
Ele. 14 - MAX. ABS. :  4.4771e+07
Ele. 15 - MAX. ABS. :  3.5162e+07
Ele. 16 - MAX. ABS. :  2.9864e+07
Ele. 17 - MAX. ABS. :  2.7108e+07
Ele. 18 - MAX. ABS. :  1.9764e+07

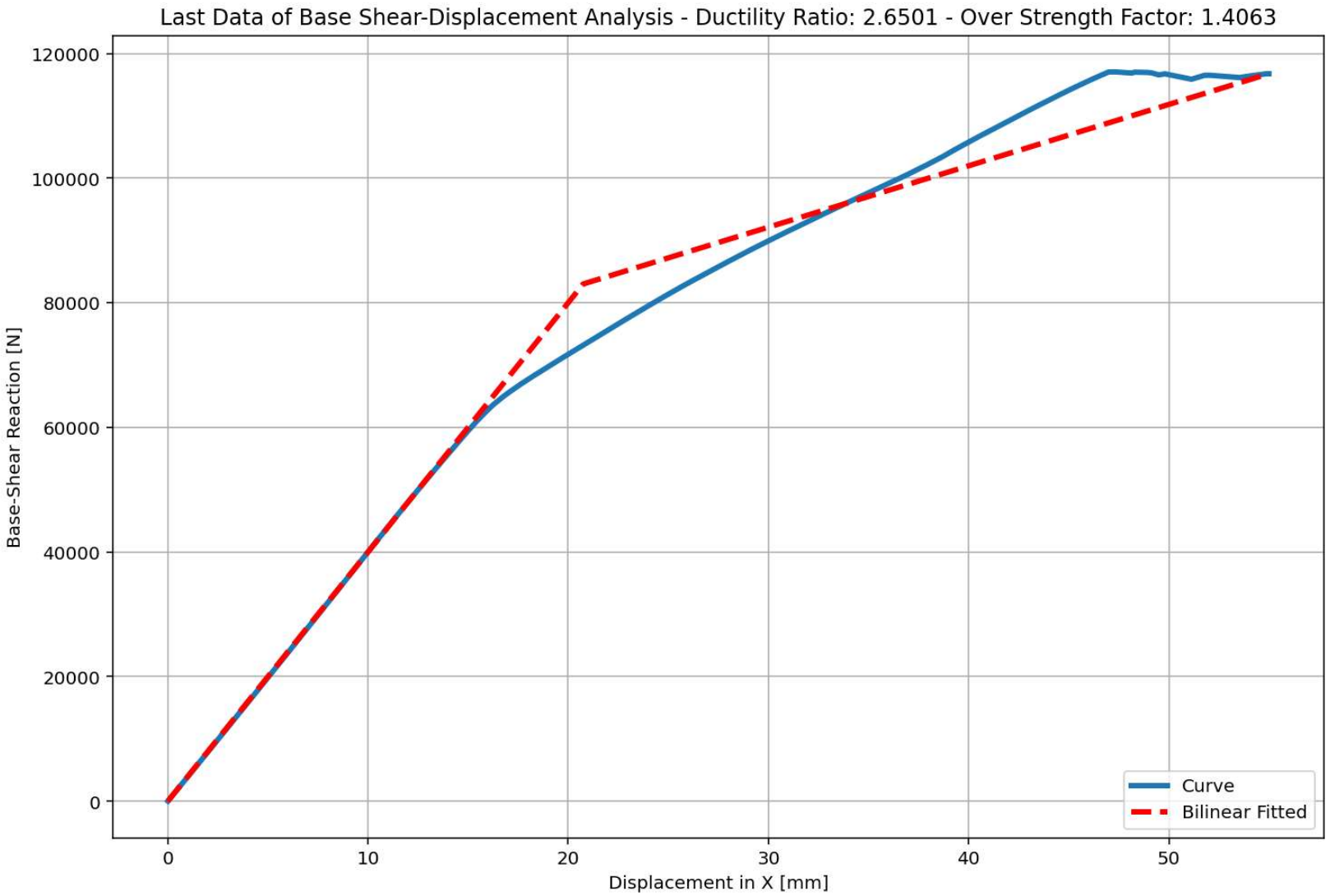

Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 9.5245e+00
Node 2 - MAX. ABS. : 9.5447e+00
Node 3 - MAX. ABS. : 9.5643e+00
Node 4 - MAX. ABS. : 9.5713e+00
Node 5 - MAX. ABS. : 9.5641e+00
Node 6 - MAX. ABS. : 9.5451e+00
Node 7 - MAX. ABS. : 9.5257e+00
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 6.5215e-01
Node 10 - MAX. ABS. : 2.1598e+00
Node 11 - MAX. ABS. : 4.0490e+00
Node 12 - MAX. ABS. : 5.9878e+00
Node 13 - MAX. ABS. : 7.7790e+00
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 6.5248e-01
Node 16 - MAX. ABS. : 2.1646e+00
Node 17 - MAX. ABS. : 4.0539e+00
Node 18 - MAX. ABS. : 5.9859e+00
Node 19 - MAX. ABS. : 7.7762e+00


Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis
Displacement [mm]
Time [s]
Node 1 - MAX. ABS. : 4.0923e-02
Node 2 - MAX. ABS. : 3.6726e+00
Node 3 - MAX. ABS. : 3.7130e+00
Node 4 - MAX. ABS. : 3.2169e-02
Node 5 - MAX. ABS. : 3.7031e+00
Node 6 - MAX. ABS. : 3.7010e+00
Node 7 - MAX. ABS. : 3.7418e-02
Node 8 - MAX. ABS. : 0.0000e+00
Node 9 - MAX. ABS. : 5.7263e-03
Node 10 - MAX. ABS. : 1.1852e-02
Node 11 - MAX. ABS. : 1.9420e-02
Node 12 - MAX. ABS. : 2.7095e-02
Node 13 - MAX. ABS. : 3.4326e-02
Node 14 - MAX. ABS. : 0.0000e+00
Node 15 - MAX. ABS. : 5.7476e-03
Node 16 - MAX. ABS. : 1.1387e-02
Node 17 - MAX. ABS. : 1.7078e-02
Node 18 - MAX. ABS. : 2.3735e-02
Node 19 - MAX. ABS. : 3.0993e-02

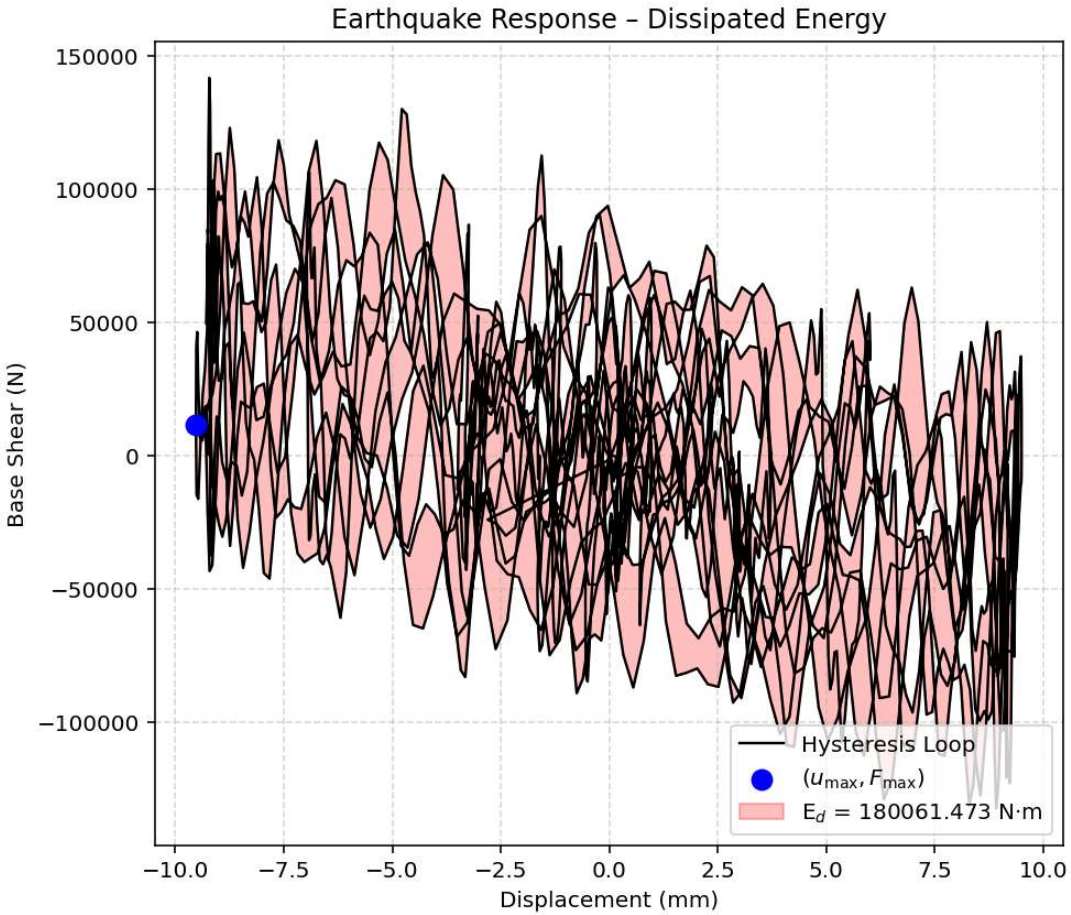

Period of Structure During Seismic Analysis
Structural Period [s]
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.655 (s) - Mean: 0.655 (s) - Max: 0.655 (s)
PERIOD - MAX VALUES: Min: 2.305 (s) - Mean: 2.305 (s) - Max: 2.306 (s)

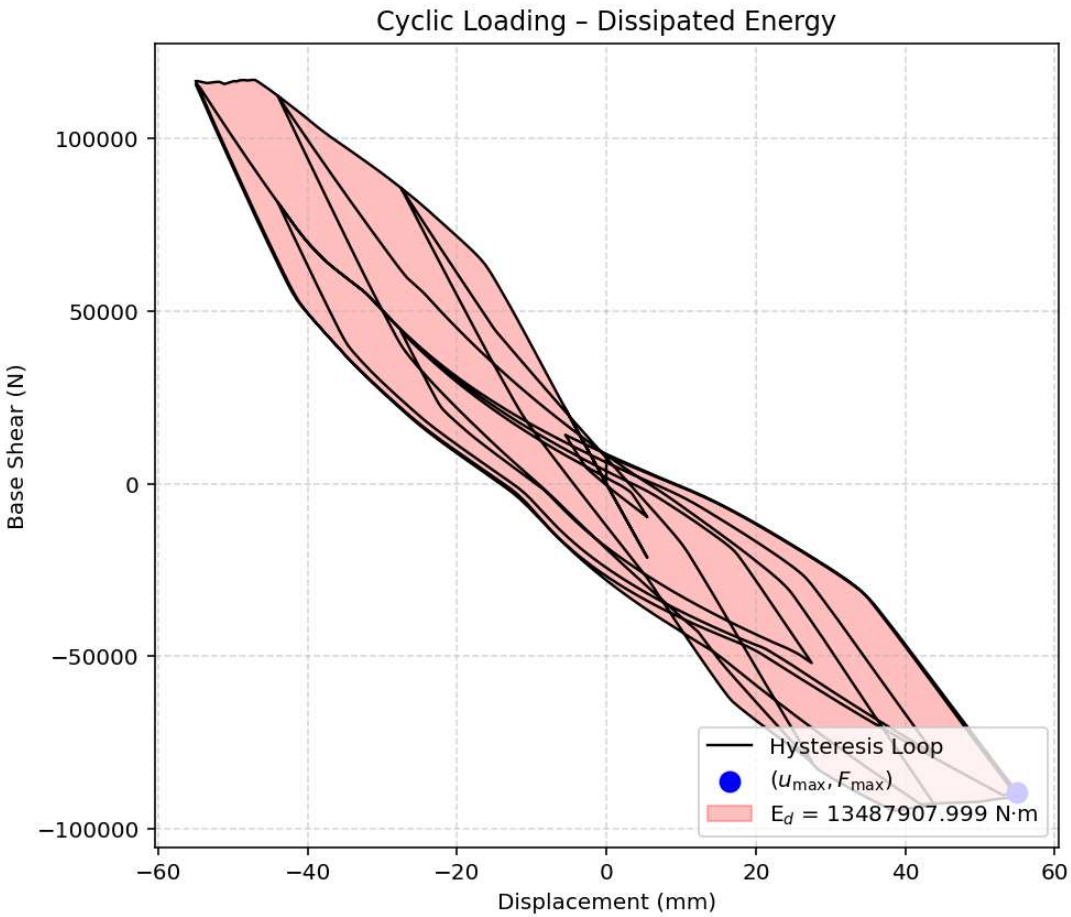

Behavior of beam - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 13.868
TOP FIBER - Stress: 2.166
Stress (N/mm^2)
Strain (mm/mm)
100
75
50
25
0
−25
−50
−75
−100
−0.0004
−0.0002
0.0000
0.0002
0.0004


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000
6000
7000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Undeformed
Deformed


Last Data of Base Shear-Displacement Analysis - Ductility Ratio: 2.6501 - Over Strength Factor: 1.4063
Base-Shear Reaction [N]
Displacement in X [mm]
120000
100000
80000
60000
40000
20000
0
0
10
20
30
40
50
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
--------------------------
[[0.00000000e+00 0.00000000e+00]
 [2.07540189e+01 8.30112638e+04]
 [5.50000000e+01 1.16740465e+05]]
+==========================+
+------------------------------------------------+
 Structure Elastic Stiffness :     3999.77
 Structure Plastic Stiffness :     2122.55
 Structure Tangent Stiffness :     984.91
 Structure Ductility Ratio :      2.65
 Structure Over Strength Factor:   1.41
+------------------------------------------------+
Over Strength Coefficient (Ω0):     1.4063
Displacement Ductility Ratio (μ):    2.6501
Ductility Coefficient (Rμ):         2.6501
Structural Behavior Coefficient (R): 3.7269
Structural Ductility Damage Index in X Direction: -32.7908 (%)
```


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (mm)
150000
100000
50000
0
−50000
−100000
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
Hysteresis Loop
$(u_{max}, F_{max})$
$E_d$ = 180061.473 N·m


Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (mm)
100000
50000
0
−50000
−100000
−60
−40
−20
0
20
40
60
Hysteresis Loop
($u_{max}$, $F_{max}$)
$E_d$ = 13487907.999 N·m

**Dissipated Energy from Earthquake= 180061.47 N·mm**
**Dissipated Energy from Cyclic Displacement= 13487908.00 N·mm**

**DISSIPATED ENERGY CAPACITY INDEX: 1.335 [%]**
**ZONE 1: VERY MINOR DAMAGE**